

INTRODUCTION TO MATRIX ALGEBRA



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Introduction to Matrix Algebra

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1: Introduction

Learning Objectives

After reading this chapter, you should be able to

1. define what a matrix is.
2. identify special types of matrices, and
3. identify when two matrices are equal.

What does a matrix look like?

Matrices are everywhere. If you have used a spreadsheet such as Excel or written numbers in a table, you have used a matrix. Matrices make presentation of numbers clearer and make calculations easier to program. Look at the matrix below about the sale of tires in a Blowoutr'us store – given by quarter and make of tires.

$$\begin{array}{l} \text{Q1. Q2. Q3. Q4} \\ \text{Tirestone} \begin{bmatrix} 25 & 20 & 3 & 2 \end{bmatrix} \\ \text{Michigan} \begin{bmatrix} 5 & 10 & 15 & 25 \end{bmatrix} \\ \text{Copper} \begin{bmatrix} 6 & 16 & 7 & 27 \end{bmatrix} \end{array}$$

If one wants to know how many *Copper* tires were sold in *Quarter 4*, we go along the row *Copper* and column *Q4* and find that it is 27.

So, What is a matrix?

A *matrix* is a rectangular array of elements. The elements can be symbolic expressions or/and numbers. Matrix $[A]$ is denoted by

$$[A] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

Row i of $[A]$ has n elements and is

$$[a_{i1} a_{i2} \dots a_{in}]$$

and column j of $[A]$ has m elements and is

$$\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$$

Each matrix has rows and columns and this defines the size of the matrix. If a matrix $[A]$ has m rows and n columns, the size of the matrix is denoted by $m \times n$. The matrix $[A]$ may also be denoted by $[A]_{m \times n}$ to show that $[A]$ is a matrix with m rows and n columns.

Each entry in the matrix is called the entry or element of the matrix and is denoted by a_{ij} where i is the row number and j is the column number of the element.

The matrix for the tire sales example could be denoted by the matrix $[A]$ as

$$[A] = \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix}$$

There are 3 rows and 4 columns, so the size of the matrix is 3×4 . In the above $[A]$ matrix, $a_{34} = 27$.

What are the special types of matrices?

Vector: A vector is a matrix that has only one row or one column. There are two types of vectors – row vectors and column vectors.

Row Vector:

If a matrix $[B]$ has one row, it is called a row vector $[B] = [b_1 \ b_2 \ \dots \ b_n]$ and n is the dimension of the row vector.

✓ Example 1

Give an example of a row vector.

Solution

$$[B] = [25 \ 20 \ 3 \ 2 \ 0]$$

is an example of a row vector of dimension 5.

Column vector:

If a matrix $[C]$ has one column, it is called a column vector

$$[C] = \begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix}$$

and m is the dimension of the vector.

✓ Example 2

Give an example of a column vector.

Solution

$$[C] = \begin{bmatrix} 25 \\ 5 \\ 6 \end{bmatrix}$$

is an example of a column vector of dimension 3.

Submatrix:

If some row(s) or/and column(s) of a matrix $[A]$ are deleted (no rows or columns may be deleted), the remaining matrix is called a submatrix of $[A]$.

✓ Example 3

Find some of the submatrices of the matrix

$$[A] = \begin{bmatrix} 4 & 6 & 2 \\ 3 & -1 & 2 \end{bmatrix}$$

Solution

$$\begin{bmatrix} 4 & 6 & 2 \\ 3 & -1 & 2 \end{bmatrix}, \begin{bmatrix} 4 & 6 \\ 3 & -1 \end{bmatrix}, [4 \ 6 \ 2], [4], \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

are some of the submatrices of $[A]$. Can you find other submatrices of $[A]$?

Square matrix:

If the number of rows m of a matrix is equal to the number of columns n of a matrix $[A]$, that is, $m = n$, then $[A]$ is called a square matrix. The entries $a_{11}, a_{22}, \dots, a_{nn}$ are called the *diagonal elements* of a square matrix. Sometimes the diagonal of the matrix is also called the *principal* or *main of the matrix*.

✓ Example 4

Give an example of a square matrix.

Solution

$$[A] = \begin{bmatrix} 25 & 20 & 3 \\ 5 & 10 & 15 \\ 6 & 15 & 7 \end{bmatrix}$$

is a square matrix as it has the same number of rows and columns, that is, 3. The diagonal elements of $[A]$ are $a_{11} = 25$, $a_{22} = 10$, $a_{33} = 7$.

Upper triangular matrix:

A $n \times n$ matrix for which $a_{ij} = 0$, $i > j$ for all i, j is called an upper triangular matrix. That is, all the elements below the diagonal entries are zero.

✓ Example 5

Give an example of an upper triangular matrix.

Solution

$$[A] = \begin{bmatrix} 10 & -7 & 0 \\ 0 & -0.001 & 6 \\ 0 & 0 & 15005 \end{bmatrix}$$

is an upper triangular matrix.

Lower triangular matrix:

A $n \times n$ matrix for which $a_{ij} = 0$, $j > i$ for all i, j is called a lower triangular matrix. That is, all the elements above the diagonal entries are zero.

✓ Example 6

Give an example of a lower triangular matrix.

Solution

$$[A] = \begin{bmatrix} 1 & 0 & 0 \\ 0.3 & 1 & 0 \\ 0.6 & 2.5 & 1 \end{bmatrix}$$

is a lower triangular matrix.

Diagonal matrix:

A square matrix with all non-diagonal elements equal to zero is called a diagonal matrix, that is, only the diagonal entries of the square matrix can be non-zero, ($a_{ij} = 0$, $i \neq j$).

✓ Example 7

Give examples of a diagonal matrix.

Solution

$$[A] = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2.1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

is a diagonal matrix.

Any or all the diagonal entries of a diagonal matrix can be zero. For example

$$[A] = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2.1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

is also a diagonal matrix.

Identity matrix:

A diagonal matrix with all diagonal elements equal to 1 is called an identity matrix, ($a_{ij} = 0$, $i \neq j$ for all i, j and $a_{ii} = 1$ for all i).

✓ Example 8

Give an example of an identity matrix.

Solution

$$[A] = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

is an identity matrix.

Zero matrix:

A matrix whose all entries are zero is called a zero matrix, ($a_{ij} = 0$ for all i and j).

✓ Example 9

Give examples of a zero matrix.

Solution

$$[A] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[B] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$[C] = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$[D] = [0 \quad 0 \quad 0]$$

are all examples of a zero matrix.

Tridiagonal matrices:

A tridiagonal matrix is a square matrix in which all elements not on the following are zero - the major diagonal, the diagonal above the major diagonal, and the diagonal below the major diagonal.

✓ Example 10

Give an example of a tridiagonal matrix.

Solution

$$[A] = \begin{bmatrix} 2 & 4 & 0 & 0 \\ 2 & 3 & 9 & 0 \\ 0 & 0 & 5 & 2 \\ 0 & 0 & 3 & 6 \end{bmatrix}$$

is a tridiagonal matrix.

Do non-square matrices have diagonal entries?

Yes, for a $m \times n$ matrix $[A]$, the diagonal entries are $a_{11}, a_{22}, \dots, a_{k-1,k-1}, a_{kk}$ where $k = \min\{m, n\}$.

✓ Example 11

What are the diagonal entries of

$$[A] = \begin{bmatrix} 3.2 & 5 \\ 6 & 7 \\ 2.9 & 3.2 \\ 5.6 & 7.8 \end{bmatrix}$$

Solution

The diagonal elements of $[A]$ are $a_{11} = 3.2$ and $a_{22} = 7$.

Diagonally Dominant Matrix:

A $n \times n$ square matrix $[A]$ is a diagonally dominant matrix if

$$|a_{ii}| \geq \sum_{\substack{j=1 \\ i \neq j}}^n |a_{ij}| \text{ for } i = 1, 2, \dots, n$$

that is, for each row, the absolute value (also called magnitude) of the diagonal element is greater than or equal to the sum of the absolute values of the rest of the elements of that row.

✓ Example 12

Give examples of matrices that are diagonally dominant and those that are not diagonally dominant.

Solution

a. The matrix

$$[A] = \begin{bmatrix} 15 & 6 & 7 \\ 2 & -4.1 & -2 \\ 3 & 2 & 6 \end{bmatrix}$$

is a diagonally dominant matrix.

Why? Because for each and every row, the answer to the question below is Yes.

Row 1: Is $|a_{11}| \geq |a_{12}| + |a_{13}|$? Yes, because

$$|a_{11}| = |15|, |a_{12}| + |a_{13}| = |6| + |7| = 13, 15 \geq 13$$

Row 2: Is $|a_{22}| \geq |a_{21}| + |a_{23}|$? Yes, because

$$|a_{22}| = |-4.1| = 4.1, |a_{21}| + |a_{23}| = |2| + |-2| = 4, 4.1 \geq 4$$

Row 3: Is $|a_{33}| \geq |a_{31}| + |a_{32}|$? Yes, because

$$|a_{33}| = |6|, |a_{31}| + |a_{32}| = |3| + |2| = 5, 6 \geq 5$$

b. The matrix

$$[A] = \begin{bmatrix} -15 & 6 & 9 \\ 2 & -4 & -2 \\ 3 & -2 & 5 \end{bmatrix}$$

is a diagonally dominant matrix.

Why? Because for each and every row, the answer to the question below is Yes.

Row 1: Is $|a_{11}| \geq |a_{12}| + |a_{13}|$? Yes, because

$$|a_{11}| = |15|, |a_{12}| + |a_{13}| = |6| + |9| = 15, 15 \geq 15$$

Row 2: Is $|a_{22}| \geq |a_{21}| + |a_{23}|$? Yes, because

$$|a_{22}| = |-4| = 4, |a_{21}| + |a_{23}| = |2| + |-2| = 4, 4 \geq 4$$

Row 3: Is $|a_{33}| \geq |a_{31}| + |a_{32}|$? Yes, because

$$|a_{33}| = |5|, |a_{31}| + |a_{32}| = |3| + |2| = 5, 5 \geq 5$$

c. The matrix

$$[A] = \begin{bmatrix} -15 & 6 & 9 \\ 2 & -4.1 & -2 \\ 3 & -2 & 5 \end{bmatrix}$$

is a diagonally dominant matrix.

Why? Because for each and every row, the answer to the question below is Yes.

Row 1: Is $|a_{11}| \geq |a_{12}| + |a_{13}|$? Yes, because

$$|a_{11}| = |15|, |a_{12}| + |a_{13}| = |6| + |9| = 15, 15 \geq 15$$

Row 2: Is $|a_{22}| \geq |a_{21}| + |a_{23}|$? Yes, because

$$|a_{22}| = |-4.1| = 4.1, |a_{21}| + |a_{23}| = |2| + |-2| = 4, 4.1 \geq 4$$

Row 3: Is $|a_{33}| \geq |a_{31}| + |a_{32}|$? Yes, because

$$|a_{33}| = |5|, |a_{31}| + |a_{32}| = |3| + |2| = 5, 5 \geq 5$$

d. The matrix

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

is *not* a diagonally dominant matrix.

Why? Because for each and every row, the answer to the question below is not a Yes.

Row 1: Is $|a_{11}| \geq |a_{12}| + |a_{13}|$? Yes, because

$$|a_{11}| = |25|, |a_{12}| + |a_{13}| = |5| + |1| = 6, 25 \geq 6$$

Row 2: Is $|a_{22}| \geq |a_{21}| + |a_{23}|$? No, because

$$|a_{22}| = |8| = 8, |a_{21}| + |a_{23}| = |64| + |1| = 65, 8 < 65$$

Row 3: Is $|a_{33}| \geq |a_{31}| + |a_{32}|$? No, because

$$|a_{33}| = |1|, |a_{31}| + |a_{32}| = |144| + |12| = 156, 1 < 156$$

Weak diagonally dominant matrix:

The answer is simple – the definition of a weak(ly) diagonally dominant matrix is identical to that of a diagonally dominant matrix as the inequality used for the check is a weak inequality of greater than or equal to ($\geq$).

Strictly diagonally dominant matrix:

A $n \times n$ square matrix is a strictly diagonally dominant matrix if

$$|a_{ii}| > \sum_{\substack{j=1 \\ i \neq j}}^n |a_{ij}| \text{ for } i = 1, 2, \dots, n$$

that is, for each row, the absolute value of the diagonal element is strictly greater than the sum of the absolute values of the rest of the elements of that row.

✓ Example 13

Give examples of strictly diagonally dominant matrices and not strictly diagonally dominant matrices.

Solution

a. The matrix

$$[A] = \begin{bmatrix} 15 & 6 & 7 \\ 2 & -4.1 & -2 \\ 3 & 2 & 6 \end{bmatrix}$$

is a strictly diagonally dominant matrix

Why? Because for each and every row, the answer to the question below is Yes.

Row 1: Is $|a_{11}| \geq |a_{12}| + |a_{13}|$? Yes, because

$$|a_{11}| = |15|, |a_{12}| + |a_{13}| = |6| + |7| = 13, 15 > 13.$$

Row 2: Is $|a_{22}| \geq |a_{21}| + |a_{23}|$? Yes, because

$$|a_{22}| = |-4.1| = 4.1, |a_{21}| + |a_{23}| = |2| + |-2| = 4, 4.1 > 4$$

Row 3: Is $|a_{33}| \geq |a_{31}| + |a_{32}|$? Yes, because

$$|a_{33}| = |6|, |a_{31}| + |a_{32}| = |3| + |2| = 5, 6 > 5$$

b) The matrix

$$[A] = \begin{bmatrix} 13 & 6 & 7 \\ 2 & -4.1 & -2 \\ 3 & 2 & 6 \end{bmatrix}$$

is a *not* a strictly diagonally dominant matrix

Why? Because for each and every row, the answer to the question below is *not* a Yes.

Row 1: Is $|a_{11}| \geq |a_{12}| + |a_{13}|$? No, because

$$|a_{11}| = |13|, |a_{12}| + |a_{13}| = |6| + |7| = 13, 13 \not> 13$$

Row 2: Is $|a_{22}| \geq |a_{21}| + |a_{23}|$? Yes, because

$$|a_{22}| = |-4.1| = 4.1, |a_{21}| + |a_{23}| = |2| + |-2| = 4, 4.1 > 4$$

Row 3: Is $|a_{33}| \geq |a_{31}| + |a_{32}|$? Yes, because

$$|a_{33}| = |6|, |a_{31}| + |a_{32}| = |3| + |2| = 5, 6 > 5$$

c. The matrix

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

is *not* a strictly diagonally dominant matrix.

Why? Because for each and every row, the answer to the question below is *not* a Yes.

Row 1: Is $|a_{11}| \geq |a_{12}| + |a_{13}|$? Yes, because

$$|a_{11}| = |25|, |a_{12}| + |a_{13}| = |5| + |1| = 6, 25 > 6$$

Row 2: Is $|a_{22}| \geq |a_{21}| + |a_{23}|$? No, because

$$|a_{22}| = |8| = 8, |a_{21}| + |a_{23}| = |64| + |1| = 65, 8 < 65$$

Row 3: Is $|a_{33}| \geq |a_{31}| + |a_{32}|$? No, because

$$|a_{33}| = |1|, |a_{31}| + |a_{32}| = |144| + |12| = 156, 1 < 156$$

Irreducible diagonally dominant matrix

A $n \times n$ square matrix is an irreducible diagonally dominant matrix if

$[A]$ is irreducible,

$$|a_{ii}| \geq \sum_{\substack{j=1 \\ i \neq j}}^n |a_{ij}| \text{ for } i = 1, 2, \dots, n \text{ and}$$

$$|a_{ii}| > \sum_{\substack{j=1 \\ i \neq j}}^n |a_{ij}| \text{ for at least one row, } i = 1, 2, \dots, n$$

The second condition means that for each row, the absolute value (also called magnitude) of the diagonal element is greater than or equal to the sum of the absolute values of the rest of the elements of that row. The third condition means that for at least one row, the absolute value (also called magnitude) of the diagonal element is greater than the sum of the absolute values of the rest of the elements of that row.

✓ Example 14

Give examples of matrices that are irreducibly diagonally dominant and those that are not irreducibly diagonally dominant.

Solution

a. The matrix

$$[A] = \begin{bmatrix} 15 & 6 & 7 \\ 2 & -4.1 & -2 \\ 3 & 2 & 6 \end{bmatrix}$$

is an irreducible diagonally dominant matrix.

Why? Because the answer to every question below is Yes.

Is $[A]$ irreducible? Yes.

Row 1: Is $|a_{11}| \geq |a_{12}| + |a_{13}|$? Yes, because

$$|a_{11}| = |15|, |a_{12}| + |a_{13}| = |6| + |7| = 13, 15 > 13.$$

Row 2: Is $|a_{22}| \geq |a_{21}| + |a_{23}|$? Yes, because

$$|a_{22}| = |-4.1| = 4.1, |a_{21}| + |a_{23}| = |2| + |-2| = 4, 4.1 > 4$$

Row 3: Is $|a_{33}| \geq |a_{31}| + |a_{32}|$? Yes, because

$$|a_{33}| = |6|, |a_{31}| + |a_{32}| = |3| + |2| = 5, 6 > 5$$

Is the inequality satisfied strictly for at least one row? Yes, it is satisfied for Rows 1, 2 and 3.

b. The matrix

$$[A] = \begin{bmatrix} -15 & 6 & 9 \\ 2 & -4 & -2 \\ 3 & -2 & 5 \end{bmatrix}$$

is a not an irreducible diagonally dominant matrix.

Why? Because the answer to every question below is *not* a Yes.

Is $[A]$ irreducible? Yes.

Row 1: Is $|a_{11}| \geq |a_{12}| + |a_{13}|$? Yes, because

$$|a_{11}| = |15|, |a_{12}| + |a_{13}| = |6| + |9| = 15, 15 \geq 15$$

Row 2: Is $|a_{22}| \geq |a_{21}| + |a_{23}|$? Yes, because

$$|a_{22}| = |-4| = 4, |a_{21}| + |a_{23}| = |2| + |-2| = 4, 4 \geq 4$$

Row 3: Is $|a_{33}| \geq |a_{31}| + |a_{32}|$? Yes, because

$$|a_{33}| = |5|, |a_{31}| + |a_{32}| = |3| + |2| = 5, 5 \geq 5$$

Is the inequality satisfied strictly for at least one row? No.

c. The matrix

$$[A] = \begin{bmatrix} -15 & 6 & 9 \\ 2 & -4.1 & -2 \\ 3 & -2 & 5 \end{bmatrix}$$

is an irreducible diagonally dominant matrix.

Why? Because the answer to every question below is Yes.

Is $[A]$ irreducible? Yes.

Row 1: Is $|a_{11}| \geq |a_{12}| + |a_{13}|$? Yes, because

$$|a_{11}| = |15|, |a_{12}| + |a_{13}| = |6| + |9| = 15, 15 \geq 15$$

Row 2: Is $|a_{22}| \geq |a_{21}| + |a_{23}|$? Yes, because

$$|a_{22}| = |-4.1| = 4.1, |a_{21}| + |a_{23}| = |2| + |-2| = 4, 4.1 \geq 4$$

Row 3: Is $|a_{33}| \geq |a_{31}| + |a_{32}|$? Yes, because

$$|a_{33}| = |5|, |a_{31}| + |a_{32}| = |3| + |2| = 5, 5 \geq 5$$

Is the inequality satisfied strictly for at least one row? Yes, it is satisfied for Row 2.

d. The matrix

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

is *not* an irreducible diagonally dominant matrix.

Why? Because the answer to every question below is *not* a Yes.

Is $[A]$ irreducible? Yes.

Row 1: Is $|a_{11}| \geq |a_{12}| + |a_{13}|$? Yes, because

$$|a_{11}| = |25|, |a_{12}| + |a_{13}| = |5| + |1| = 6, 25 > 6$$

Row 2: Is $|a_{22}| \geq |a_{21}| + |a_{23}|$? No, because

$$|a_{22}| = |8| = 8, |a_{21}| + |a_{23}| = |64| + |1| = 65, 8 < 65$$

Row 3: Is $|a_{33}| \geq |a_{31}| + |a_{32}|$? No, because

$$|a_{33}| = |1|, |a_{31}| + |a_{32}| = |144| + |12| = 156, 1 < 156$$

There is no need to check for strict inequality condition..

Irreducible matrix:

A square matrix is called *reducible* matrix if the following is true. Take the indices $i = 1, 2, \dots, n$ and see if they can be divided into two disjoint nonempty sets $i_1, i_2, \dots, i_\alpha$ and $j_1, j_2, \dots, j_\beta$ such that

$$n = \alpha + \beta, \text{ and}$$

and

$$a_{i_k j_l} = 0, \quad k = 1, 2, \dots, \alpha \text{ and } l = 1, 2, \dots, \beta$$

If the square matrix is not reducible, it is called an *irreducible* matrix.

A square matrix $[A]$ is called *reducible* matrix if and only if for any perturbation matrix $[P]$, the matrix multiplication $[P]^T [A] [P]$ results in a block upper triangular matrix.

✓ Example 15

Give examples of irreducible and reducible matrices.

Solution

a. The matrix

$$\begin{bmatrix} 0 & 5 & 7 \\ 8 & 0 & 0 \\ 10 & 0 & 0 \end{bmatrix}$$

is an irreducible matrix.

b. The matrix

$$\begin{bmatrix} 5 & 0 & 0 \\ 0 & 4 & 6 \\ 10 & 0 & 0 \end{bmatrix}$$

is a reducible matrix. Why? Take the indices $i = 1, 2, 3$ and see that they can be divided into two disjoint nonempty sets 1 and 2,3 such that,

$$\alpha = 1, \beta = 2, \text{ giving } \alpha + \beta = 1 + 2 = 3, \text{ and}$$

$$a_{i_k j_l} = 0, \quad k = 1 \text{ and } l = 1, 2$$

Consequences of diagonally dominant matrices

If a square matrix is strictly diagonally dominant

- then the matrix is non-singular.
- then if the matrix is symmetric with non-negative diagonal entries, the matrix is positive semi-definite.
- then if the matrix is the coefficient matrix for a set of simultaneous linear equations, the iterative Gauss-Seidel numerical method will always converge.
- then if the matrix is the coefficient matrix for a set of simultaneous linear equations, the iterative Jordan numerical method will always converge.
- then if the diagonal entries of the matrix are positive, the real parts of the matrix eigenvalues are positive.
- then if the diagonal entries of the matrix are negative, the real parts of the matrix eigenvalues are negative.
- then if the matrix is column dominant, no pivoting is needed for Gaussian elimination.
- then if the matrix is column dominant, no pivoting is needed for LU factorization.

If a square matrix is irreducible diagonally dominant

- then if the matrix is the coefficient matrix for a set of simultaneous linear equations, the iterative Gauss-Seidel numerical method will always converge.
- then if the matrix is the coefficient matrix for a set of simultaneous linear equations, the iterative Jordan numerical method will always converge.
- the matrix is non-singular.

If a square matrix is diagonally dominant (also called weakly diagonally dominant)

- then if the matrix is column dominant, no pivoting is needed for Gaussian elimination.
- then if the matrix is column dominant, no pivoting is needed for LU factorization.

Equal matrices:

Two matrices $[A]$ and $[B]$ are equal if the size of $[A]$ and $[B]$ is the same (number of rows and columns of $[A]$ are same as that of $[B]$) and $a_{ij} = b_{ij}$ for all i and j .

✓ Example 16

What would make

$$[A] = \begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix}$$

to be equal to

$$[B] = \begin{bmatrix} b_{11} & 3 \\ 6 & b_{22} \end{bmatrix}$$

Solution

The two matrices $[A]$ and $[B]$ would be equal if $b_{11} = 2$ and $b_{22} = 7$.

Key Terms:

Matrix

Vector

Submatrix

Square matrix

Equal matrices

Zero matrix

Identity matrix

Diagonal matrix

Upper triangular matrix

Lower triangular matrix

Tri-diagonal matrix

Introduction Quiz

? Quiz 1

For an $n \times n$ upper triangular matrix $[A]$,

(A) $a_{ij} = 0, i > j$

(B) $a_{ij} = 0, j > i$

(C) $a_{ij} \neq 0, i > j$

(D) $a_{ij} \neq 0, j > i$

? Quiz 2

Which one of these square matrices is strictly diagonally dominant?

(A) $\begin{bmatrix} 5 & 7 & 0 \\ 3 & -6 & 2 \\ 2 & 2 & 9 \end{bmatrix}$

(B) $\begin{bmatrix} 7 & -5 & -2 \\ 6 & -13 & -7 \\ 6 & -7 & -13 \end{bmatrix}$

(C) $\begin{bmatrix} 8 & -5 & -2 \\ 6 & -14 & -7 \\ 6 & -7 & -13 \end{bmatrix}$

(D) $\begin{bmatrix} 8 & 5 & 2 \\ 6 & 14 & 7 \\ 6 & 7.5 & 14 \end{bmatrix}$

? Quiz 3

The order of the following matrix is

$$\begin{bmatrix} 4 & -6 & -7 & 2 \\ 3 & 2 & -5 & 6 \end{bmatrix}$$

(A) 4×2

(B) 2×4

(C) 8×1

(D) not defined

? Quiz 4

To make the following two matrices equal

$$[A] = \begin{bmatrix} 5 & -6 & 7 \\ 3 & 2 & 5 \end{bmatrix}$$

$$[B] = \begin{bmatrix} 5 & p & 7 \\ 3 & 2 & 5 \end{bmatrix}$$

the value of p is

- (A) -6
- (B) 6
- (C) 0
- (D) 7

? Quiz 5

For a square $n \times n$ matrix $[A]$ to be an identity matrix,

- (A) $a_{ij} \neq 0, i \neq j; a_{ij} = 0, i = j$
- (B) $a_{ij} = 0, i \neq j; a_{ij} = 1, i = j$
- (C) $a_{ij} = 0, i \neq j; a_{ij} = i, i = j$
- (D) $a_{ij} = 0, i \neq j; a_{ij} > 0, i = j$

? Quiz 6

To make the following square matrix to be diagonally dominant, the value of p needs to be

$$\begin{bmatrix} 6 & -2 & -4 \\ 7 & 9 & 1 \\ 8 & -5 & p \end{bmatrix}$$

- (A) greater than or equal to 13
- (B) greater than 3
- (C) greater than or equal to 3
- (D) greater than 13

Introduction Exercise

? Exercise 1

Write an example of a row vector of dimension 4.

Answer

$$[5 \quad 6 \quad 2 \quad 3]$$

? Exercise 2

Write an example of a column vector of dimension 4.

Answer

$$\begin{bmatrix} 5 \\ -7 \\ 3 \\ 2.5 \end{bmatrix}$$

? Exercise 3

Write an example of a square matrix of order 4×4 .

Answer

$$\begin{bmatrix} 9 & 0 & -2 & 3 \\ -2 & 3 & 5 & 1 \\ 1.5 & 6 & 7 & 8 \\ 1.1 & 2 & 3 & 4 \end{bmatrix}$$

? Exercise 4

Write an example of a tri-diagonal matrix of order 4×4 .

Answer

$$\begin{bmatrix} 6 & 3 & 0 & 0 \\ 2.1 & 2 & 2.2 & 0 \\ 0 & 6.2 & -3 & 3.5 \\ 0 & 0 & 2.1 & 4.1 \end{bmatrix}$$

? Exercise 5

Write an example of a identity matrix of order 5×5 .

Answer

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

? Exercise 6

Write an example of a upper triangular matrix of order 4×4 .

Answer

$$\begin{bmatrix} 6 & 2 & 3 & 9 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 4 & 5 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

? Exercise 7

Write an example of a lower triangular matrix of order 4×4 .

Answer

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 \\ 4 & 2 & 4 & 0 \\ 5 & 3 & 5 & 6 \end{bmatrix}$$

? Exercise 8

Which of these matrices are strictly diagonally dominant?

$$\begin{aligned} \text{A. } [A] &= \begin{bmatrix} 15 & 6 & 7 \\ 2 & -4 & 2 \\ 3 & 2 & 6 \end{bmatrix} \\ \text{B. } [A] &= \begin{bmatrix} 5 & 6 & 7 \\ 2 & -4 & 2 \\ 3 & 2 & -5 \end{bmatrix} \\ \text{C. } [A] &= \begin{bmatrix} 5 & 3 & 2 \\ 6 & -8 & 2 \\ 7 & -5 & 12 \end{bmatrix} \end{aligned}$$

Answer

(A) Yes (B) No (C) No

? Exercise 9

Find all the submatrices of

$$[A] = \begin{bmatrix} 10 & -7 & 0 \\ 0 & -0.001 & 6 \end{bmatrix}$$

Answer

$$\begin{aligned} &[10] \ [-7] \ , \ [0], \ [-0.001] \ [6] \ \begin{bmatrix} 10 \\ 0 \end{bmatrix}, \ \begin{bmatrix} -7 \\ -0.001 \end{bmatrix}, \ \begin{bmatrix} 0 \\ 6 \end{bmatrix}, \ [10 \ -7 \ 0], \ [0 \ -0.001 \ 6], \ \begin{bmatrix} 10 & -7 \\ 0 & -0.001 \end{bmatrix}, \ \begin{bmatrix} 10 & 0 \\ 0 & 6 \end{bmatrix}, \\ &\begin{bmatrix} -7 & 0 \\ -0.001 & 6 \end{bmatrix}, \ [10, -7], \ [10, 0], \ [-7, 0], \ [0, 6], \ [0, -0.001] \ [-0.001, 6] \end{aligned}$$

? Exercise 10

If

$$[A] = \begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix},$$

what are b_{11} and b_{12} in

$$[B] = \begin{bmatrix} b_{11} & b_{12} \\ 0 & 4 \end{bmatrix}$$

if $[B] = 2[A]$.

Answer

$8, -2$

? Exercise 11

Are matrix

$$[A] = \begin{bmatrix} 10 & -7 & 0 \\ 0 & -0.001 & 6 \end{bmatrix}$$

and matrix

$$[B] = \begin{bmatrix} 10 & 0 \\ -7 & -0.001 \\ 0 & 6 \end{bmatrix}$$

equal?

Answer

No

? Exercise 12

A square matrix $[A]$ is lower triangular if

- A. $a_{ij} = 0$ for $i > j$
- B. $a_{ij} = 0$ for $j > i$
- C. $a_{ij} = 0$ for $i = j$
- D. $a_{ij} = 0$ for $i + j = \text{odd integer}$

Answer

B

? Exercise 13

A square matrix $[A]$ is upper triangular if

- A. $a_{ij} = 0$ for $i > j$
- B. $a_{ij} = 0$ for $j > i$
- C. $a_{ij} = 0$ for $i = j$
- D. $a_{ij} = 0$ for $i + j = \text{odd integer}$

Answer

A

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2: Vectors

Learning Objectives

After reading this chapter, you should be able to:

1. define a vector
2. add and subtract vectors,
3. find linear combinations of vectors and their relationship to a set of equations,
4. explain what it means to have a linearly independent set of vectors, and
5. find the rank of a set of vectors.

What is a vector?

A vector is a collection of numbers in a definite order. If it is a collection of n numbers, it is called a n -dimensional vector. So, the vector $\vec{A}$ given by

$$\vec{A} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

is a n -dimensional column vector with n components, $a_1, a_2, \dots, a_n$. The above is a column vector. A row vector $[B]$ is of the form $\vec{B} = [b_1, b_2, \dots, b_n]$ where $\vec{B}$ is a n -dimensional row vector with n components $b_1, b_2, \dots, b_n$.

✓ Example 1

Give an example of a 3-dimensional column vector.

Solution

Assume a point in space is given by its (x, y, z) coordinates. Then if the value of $x = 3$, $y = 2$, $z = 5$, the column vector corresponding to the location of the points is

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

When are two vectors equal?

Two vectors $\vec{A}$ and $\vec{B}$ are equal if they are of the same dimension and if their corresponding components are equal.

Given

$$\vec{A} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$$

and

$$\vec{B} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

then $\vec{A} = \vec{B}$ if $a_i = b_i, i = 1, 2, \dots, n$.

✓ Example 2

What are the values of the unknown components in $\vec{B}$ if

$$\vec{A} = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix}$$

and

$$\vec{B} = \begin{bmatrix} b_1 \\ 3 \\ 4 \\ b_4 \end{bmatrix}$$

and $\vec{A} = \vec{B}$.

Solution

$$b_1 = 2, b_4 = 1$$

How do you add two vectors?

Two vectors can be added only if they are of the same dimension and the addition is given by

$$\begin{aligned} [A] + [B] &= \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \\ &= \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{bmatrix} \end{aligned}$$

✓ Example 3

Add the two vectors

$$\vec{A} = \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix}$$

and

$$\vec{B} = \begin{bmatrix} 5 \\ -2 \\ 3 \\ 7 \end{bmatrix}$$

Solution

$$\begin{aligned}\vec{A} + \vec{B} &= \begin{bmatrix} 2 \\ 3 \\ 4 \\ 1 \end{bmatrix} + \begin{bmatrix} 5 \\ -2 \\ 3 \\ 7 \end{bmatrix} \\ &= \begin{bmatrix} 2+5 \\ 3-2 \\ 4+3 \\ 1+7 \end{bmatrix} \\ &= \begin{bmatrix} 7 \\ 1 \\ 7 \\ 8 \end{bmatrix}\end{aligned}$$

✓ Example 4

A store sells three brands of tires: Tirestone, Michigan and Copper. In quarter 1, the sales are given by the column vector

$$\vec{A}_1 = \begin{bmatrix} 25 \\ 5 \\ 6 \end{bmatrix}$$

where the rows represent the three brands of tires sold – Tirestone, Michigan and Copper respectively. In quarter 2, the sales are given by

$$\vec{A}_2 = \begin{bmatrix} 20 \\ 10 \\ 6 \end{bmatrix}$$

What is the total sale of each brand of tire in the first half of the year?

Solution

The total sales would be given by

$$\begin{aligned}\vec{C} &= \vec{A}_1 + \vec{A}_2 \\ &= \begin{bmatrix} 25 \\ 5 \\ 6 \end{bmatrix} + \begin{bmatrix} 20 \\ 10 \\ 6 \end{bmatrix} \\ &= \begin{bmatrix} 25+20 \\ 5+10 \\ 6+6 \end{bmatrix} \\ &= \begin{bmatrix} 45 \\ 15 \\ 12 \end{bmatrix}\end{aligned}$$

So, the number of Tirestone tires sold is 45, Michigan is 15 and Copper is 12 in the first half of the year.

What is a null vector?

A null vector (also called zero vector) is where all the components of the vector are zero.

✓ Example 5

Give an example of a null vector or zero vector.

Solution

The vector

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

is an example of a zero or null vector.

What is a unit vector?

A unit vector $\vec{U}$ is defined as

$$\vec{U} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$$

where

$$\sqrt{u_1^2 + u_2^2 + u_3^2 + \dots + u_n^2} = 1$$

✓ Example 6

Give examples of 3-dimensional unit column vectors.

Solution

Examples include

$$\begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \text{ etc.}$$

How do you multiply a vector by a scalar?

If k is a scalar and $\vec{A}$ is a n -dimensional vector, then

$$\begin{aligned} k\vec{A} &= k \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \\ &= \begin{bmatrix} ka_1 \\ ka_2 \\ \vdots \\ ka_n \end{bmatrix} \end{aligned}$$

✓ Example 7

What is $2\vec{A}$ if

$$\vec{A} = \begin{bmatrix} 25 \\ 20 \\ 5 \end{bmatrix}$$

Solution

$$\begin{aligned} 2\vec{A} &= 2 \begin{bmatrix} 25 \\ 20 \\ 5 \end{bmatrix} \\ &= \begin{bmatrix} 2 \times 25 \\ 2 \times 20 \\ 2 \times 5 \end{bmatrix} \\ &= \begin{bmatrix} 50 \\ 40 \\ 10 \end{bmatrix} \end{aligned}$$

✓ Example 8

A store sells three brands of tires: Tirestone, Michigan and Copper. In quarter 1, the sales are given by the column vector

$$\vec{A} = \begin{bmatrix} 25 \\ 25 \\ 6 \end{bmatrix}$$

If the goal is to increase the sales of all tires by at least 25% in the next quarter, how many of each brand should be sold?

Solution

Since the goal is to increase the sales by 25%, one would multiply the $\vec{A}$ vector by 1.25,

$$\begin{aligned} \vec{B} &= 1.25 \begin{bmatrix} 25 \\ 25 \\ 6 \end{bmatrix} \\ &= \begin{bmatrix} 31.25 \\ 31.25 \\ 7.5 \end{bmatrix} \end{aligned}$$

Since the number of tires must be an integer, we can say that the goal of sales is

$$\vec{B} = \begin{bmatrix} 32 \\ 32 \\ 8 \end{bmatrix}$$

What do you mean by a linear combination of vectors?

Given

$$\vec{A}_1, \vec{A}_2, \dots, \vec{A}_m$$

as m vectors of same dimension n , and if $k_1, k_2, \dots, k_m$ are scalars, then

$$k_1 \vec{A}_1 + k_2 \vec{A}_2 + \dots + k_m \vec{A}_m$$

is a linear combination of the m vectors.

✓ Example 9

Find the linear combinations

(a) $\vec{A} - \vec{B}$ and

(b) $\vec{A} + \vec{B} - 3\vec{C}$

where

$$\vec{A} = \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix}, \vec{B} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \vec{C} = \begin{bmatrix} 10 \\ 1 \\ 2 \end{bmatrix}$$

Solution

(a)

$$\begin{aligned} \vec{A} - \vec{B} &= \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 2-1 \\ 3-1 \\ 6-2 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix} \end{aligned}$$

(b)

$$\begin{aligned} \vec{A} + \vec{B} - 3\vec{C} &= \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - 3 \begin{bmatrix} 10 \\ 1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 2+1-30 \\ 3+1-3 \\ 6+2-6 \end{bmatrix} \\ &= \begin{bmatrix} -27 \\ 1 \\ 2 \end{bmatrix} \end{aligned}$$

What do you mean by vectors being linearly independent?

A set of vectors $\vec{A}_1, \vec{A}_2, \dots, \vec{A}_m$ are considered to be linearly independent if

$$k_1 \vec{A}_1 + k_2 \vec{A}_2 + \dots + k_m \vec{A}_m = \vec{0}$$

has only one solution of

$$k_1 = k_2 = \dots = k_m = 0$$

✓ Example 10

Are the three vectors

$$\vec{A}_1 = \begin{bmatrix} 25 \\ 64 \\ 144 \end{bmatrix}, \vec{A}_2 = \begin{bmatrix} 5 \\ 8 \\ 12 \end{bmatrix}, \vec{A}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

linearly independent?

Solution

Writing the linear combination of the three vectors

$$k_1 \begin{bmatrix} 25 \\ 64 \\ 144 \end{bmatrix} + k_2 \begin{bmatrix} 5 \\ 8 \\ 12 \end{bmatrix} + k_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

gives

$$\begin{bmatrix} 25k_1 + 5k_2 + k_3 \\ 64k_1 + 8k_2 + k_3 \\ 144k_1 + 12k_2 + k_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The above equations have only one solution, $k_1 = k_2 = k_3 = 0$. However, how do we show that this is the only solution? This is shown below.

The above equations are

$$25k_1 + 5k_2 + k_3 = 0 \quad (1)$$

$$64k_1 + 8k_2 + k_3 = 0 \quad (2)$$

$$144k_1 + 12k_2 + k_3 = 0 \quad (3)$$

Subtracting Eqn (1) from Eqn (2) gives

$$39k_1 + 3k_2 = 0$$

$$k_2 = -13k_1 \quad (4)$$

Multiplying Eqn (1) by 8 and subtracting it from Eqn (2) that is first multiplied by 5 gives

$$120k_1 - 3k_3 = 0$$

$$k_3 = 40k_1 \quad (5)$$

Remember we found Eqn (4) and Eqn (5) just from Eqns (1) and (2).

Substitution of Eqns (4) and (5) in Eqn (3) for k_2 and k_3 gives

$$144k_1 + 12(-13k_1) + 40k_1 = 0$$

$$28k_1 = 0$$

$$k_1 = 0$$

This means that k_1 has to be zero, and coupled with (4) and (5), k_2 and k_3 are also zero. So the only solution is $k_1 = k_2 = k_3 = 0$. The three vectors hence are linearly independent.

✓ Example 11

Are the three vectors

$$\vec{A}_1 = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}, \vec{A}_2 = \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix}, \vec{A}_3 = \begin{bmatrix} 6 \\ 14 \\ 24 \end{bmatrix}$$

linearly independent?

Solution

By inspection,

$$\vec{A}_3 = 2\vec{A}_1 + 2\vec{A}_2$$

or

$$-2\vec{A}_1 - 2\vec{A}_2 + \vec{A}_3 = \vec{0}$$

So the linear combination

$$k_1\vec{A}_1 + k_2\vec{A}_2 + k_3\vec{A}_3 = \vec{0}$$

has a non-zero solution

$$k_1 = -2, k_2 = -2, k_3 = 1$$

Hence, the set of vectors is linearly dependent.

What if I cannot prove by inspection, what do I do? Put the linear combination of three vectors equal to the zero vector,

$$k_1 \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix} + k_2 \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix} + k_3 \begin{bmatrix} 6 \\ 14 \\ 24 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

to give

$$k_1 + 2k_2 + 6k_3 = 0 \quad (1)$$

$$2k_1 + 5k_2 + 14k_3 = 0 \quad (2)$$

$$5k_1 + 7k_2 + 24k_3 = 0 \quad (3)$$

Multiplying Eqn (1) by 2 and subtracting from Eqn (2) gives

$$k_2 + 2k_3 = 0$$

$$k_2 = -2k_3 \quad (4)$$

Multiplying Eqn (1) by 2.5 and subtracting from Eqn (2) gives

$$-0.5k_1 - k_3 = 0$$

$$k_1 = -2k_3 \quad (5)$$

Remember we found Eqn (4) and Eqn (5) just from Eqns (1) and (2).

Substitute Eqn (4) and (5) in Eqn (3) for k_1 and k_2 gives

$$5(-2k_3) + 7(-2k_3) + 24k_3 = 0$$

$$-10k_3 - 14k_3 + 24k_3 = 0$$

$$0 = 0$$

This means any values satisfying Eqns (4) and (5) will satisfy Eqns (1), (2) and (3) simultaneously.

For example, chose

$$k_3 = 6, \text{ then}$$

$$k_2 = -12 \text{ from Eqn (4), and}$$

$$k_1 = -12 \text{ from Eqn (5).}$$

Hence we have a nontrivial solution of $[k_1 \ k_2 \ k_3] = [-12 \ -12 \ 6]$. This implies the three given vectors are linearly dependent. Can you find another nontrivial solution?

What about the following three vectors?

$$\begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \\ 7 \end{bmatrix}, \begin{bmatrix} 6 \\ 14 \\ 25 \end{bmatrix}$$

Are they linearly dependent or linearly independent?

Note that the only difference between this set of vectors and the previous one is the third entry in the third vector. Hence, equations (4) and (5) are still valid. What conclusion do you draw when you plug in equations (4) and (5) in the third equation: $5k_1 + 7k_2 + 25k_3 = 0$? What has changed?

✓ Example 12

Are the three vectors

$$\vec{A}_1 = \begin{bmatrix} 25 \\ 64 \\ 89 \end{bmatrix}, \vec{A}_2 = \begin{bmatrix} 5 \\ 8 \\ 13 \end{bmatrix}, \vec{A}_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

linearly independent?

Solution

Writing the linear combination of the three vectors and equating to zero vector

$$k_1 \begin{bmatrix} 25 \\ 64 \\ 89 \end{bmatrix} + k_2 \begin{bmatrix} 5 \\ 8 \\ 13 \end{bmatrix} + k_3 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

gives

$$\begin{bmatrix} 25k_1 + 5k_2 + k_3 \\ 64k_1 + 8k_2 + k_3 \\ 89k_1 + 13k_2 + 2k_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

In addition to $k_1 = k_2 = k_3 = 0$, one can find other solutions for which k_1, k_2, k_3 are not equal to zero. For example, $k_1 = 1, k_2 = -13, k_3 = 40$ is also a solution as

$$1 \begin{bmatrix} 25 \\ 64 \\ 89 \end{bmatrix} - 13 \begin{bmatrix} 5 \\ 8 \\ 13 \end{bmatrix} + 40 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Hence $\vec{A}_1, \vec{A}_2, \vec{A}_3$ are linearly dependent.

What do you mean by the rank of a set of vectors?

From a set of n -dimensional vectors, the maximum number of linearly independent vectors in the set is called the rank of the set of vectors. *Note that the rank of the vectors can never be greater than the vectors dimension.*

✓ Example 13

What is the rank of

$$\vec{A}_1 = \begin{bmatrix} 25 \\ 64 \\ 144 \end{bmatrix}, \vec{A}_2 = \begin{bmatrix} 5 \\ 8 \\ 12 \end{bmatrix}, \vec{A}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}?$$

Solution

Since we found in Example 2.10 that $\vec{A}_1, \vec{A}_2, \vec{A}_3$ are linearly independent, the rank of the set of vectors $\vec{A}_1, \vec{A}_2, \vec{A}_3$ is 3. If we were given another vector $\vec{A}_4$, the rank of the set of the vectors $\vec{A}_1, \vec{A}_2, \vec{A}_3, \vec{A}_4$ would still be 3 as the rank of a set of vectors is always less than or equal to the dimension of the vectors and that at least $\vec{A}_1, \vec{A}_2, \vec{A}_3$ are linearly independent.

✓ Example 14

What is the rank of

$$\vec{A}_1 = \begin{bmatrix} 25 \\ 64 \\ 89 \end{bmatrix}, \vec{A}_2 = \begin{bmatrix} 5 \\ 8 \\ 13 \end{bmatrix}, \vec{A}_3 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}?$$

Solution

In Example 2.12, we found that $\vec{A}_1, \vec{A}_2, \vec{A}_3$ are linearly dependent, the rank of $\vec{A}_1, \vec{A}_2, \vec{A}_3$ is hence not 3, and is less than 3. Is it 2? Let us choose two of the three vectors

$$\vec{A}_1 = \begin{bmatrix} 25 \\ 64 \\ 89 \end{bmatrix}, \vec{A}_2 = \begin{bmatrix} 5 \\ 8 \\ 13 \end{bmatrix}$$

Linear combination of $\vec{A}_1$ and $\vec{A}_2$ equal to zero has only one solution – the trivial solution. Therefore, the rank is 2.

✓ Example 15

What is the rank of

$$\vec{A}_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \vec{A}_2 = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}, \vec{A}_3 = \begin{bmatrix} 3 \\ 3 \\ 5 \end{bmatrix}?$$

Solution

From inspection,

$$\vec{A}_2 = 2\vec{A}_1,$$

that implies

$$2\vec{A}_1 - \vec{A}_2 + 0\vec{A}_3 = \vec{0}$$

Hence

$$k_1\vec{A}_1 + k_2\vec{A}_2 + k_3\vec{A}_3 = \vec{0}$$

has a nontrivial solution.

So $\vec{A}_1, \vec{A}_2, \vec{A}_3$ are linearly dependent, and hence the rank of the three vectors is not 3. Since

$$\vec{A}_2 = 2\vec{A}_1,$$

$\vec{A}_1$ and $\vec{A}_2$ are linearly dependent, but

$$k_1\vec{A}_1 + k_3\vec{A}_3 = \vec{0}.$$

has trivial solution as the only solution. So $\vec{A}_1$ and $\vec{A}_3$ are linearly independent. The rank of the above three vectors is 2.

Prove that if a set of vectors contains the null vector, the set of vectors is linearly dependent.

Let $\vec{A}_1, \vec{A}_2, \dots, \vec{A}_m$ be a set of n -dimensional vectors, then

$$k_1\vec{A}_1 + k_2\vec{A}_2 + \dots + k_m\vec{A}_m = \vec{0}$$

is a linear combination of the m vectors. Then assuming if $\vec{A}_1$ is the zero or null vector, any value of k_1 coupled with $k_2 = k_3 = \dots = k_m = 0$ will satisfy the above equation. Hence, the set of vectors is linearly dependent as more than one solution exists.

Prove that if a set of m vectors is linearly independent, then a subset of the m vectors also has to be linearly independent.

Let this subset of vectors be

$$\vec{A}_{a1}, \vec{A}_{a2}, \dots, \vec{A}_{ap}$$

where $p < m$.

Then if this subset of vectors is linearly dependent, the linear combination

$$k_1 \vec{A}_{a1} + k_2 \vec{A}_{a2} + \dots + k_p \vec{A}_{ap} = \vec{0}$$

has a non-trivial solution.

So

$$k_1 \vec{A}_{a1} + k_2 \vec{A}_{a2} + \dots + k_p \vec{A}_{ap} + 0 \vec{A}_{a(p+1)} + \dots + 0 \vec{A}_{am} = \vec{0}$$

also has a non-trivial solution too, where $\vec{A}_{a(p+1)}, \dots, \vec{A}_{am}$ are the rest of the $(m-p)$ vectors. However, this is a contradiction. Therefore, a subset of linearly independent vectors cannot be linearly dependent.

Prove that if a set of vectors is linearly dependent, then at least one vector can be written as a linear combination of others.

Let $\vec{A}_1, \vec{A}_2, \dots, \vec{A}_m$ be linearly dependent set of vectors, then there exists a set of scalars

$k_1, \dots, k_m$ not all of which are zero for the linear combination equation

$$k_1 \vec{A}_1 + k_2 \vec{A}_2 + \dots + k_m \vec{A}_m = \vec{0}.$$

Let k_p be one of the non-zero values of k_i , $i = 1, \dots, m$, that is, $k_p \neq 0$, then

$$\vec{A}_p = -\frac{k_2}{k_p} \vec{A}_2 - \dots - \frac{k_{p-1}}{k_p} \vec{A}_{p-1} - \frac{k_{p+1}}{k_p} \vec{A}_{p+1} - \dots - \frac{k_m}{k_p} \vec{A}_m$$

and that proves the theorem.

Prove that if the dimension of a set of vectors is less than the number of vectors in the set, then the set of vectors is linearly dependent.

Can you prove it?

How can vectors be used to write simultaneous linear equations?

If a set of m simultaneous linear equations with n unknowns is written as

$$a_{11}x_1 + \dots + a_{1n}x_n = c_1$$

$$a_{21}x_1 + \dots + a_{2n}x_n = c_2$$

$$\vdots \quad \vdots$$

$$a_{m1}x_1 + \dots + a_{mn}x_n = c_n$$

where

$x_1, x_2, \dots, x_n$ are the unknowns, then in the vector notation they can be written as

$$x_1 \vec{A}_1 + x_2 \vec{A}_2 + \dots + x_n \vec{A}_n = \vec{C}$$

where

$$\vec{A}_1 = \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix}$$

where

$$\vec{A}_1 = \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix}$$

$$\vec{A}_2 = \begin{bmatrix} a_{12} \\ \vdots \\ a_{m2} \end{bmatrix}$$

$$\vec{A}_n = \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

$$\vec{C}_1 = \begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix}$$

The problem now becomes whether you can find the scalars $x_1, x_2, \dots, x_n$ such that the linear combination

$$x_1 \vec{A}_1 + \dots + x_n \vec{A}_n$$

is equal to the $\vec{C}$, that is

$$x_1 \vec{A}_1 + \dots + x_n \vec{A}_n = \vec{C}$$

✓ Example 16

Write

$$25x_1 + 5x_2 + x_3 = 106.8$$

$$64x_1 + 8x_2 + x_3 = 177.2$$

$$144x_1 + 12x_2 + x_3 = 279.2$$

as a linear combination of set of vectors equal to another vector.

Solution

$$\begin{bmatrix} 25x_1 & +5x_2 & +x_3 \\ 64x_1 & +8x_2 & +x_3 \\ 144x_1 & +12x_2 & +x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

$$x_1 \begin{bmatrix} 25 \\ 64 \\ 144 \end{bmatrix} + x_2 \begin{bmatrix} 5 \\ 8 \\ 12 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

What is the definition of the dot product of two vectors?

Let $\vec{A} = [a_1, a_2, \dots, a_n]$ and $\vec{B} = [b_1, b_2, \dots, b_n]$ be two n -dimensional vectors. Then the dot product of the two vectors $\vec{A}$ and $\vec{B}$ is defined as

$$\vec{A} \cdot \vec{B} = a_1 b_1 + a_2 b_2 + \dots + a_n b_n = \sum_{i=1}^n a_i b_i$$

A dot product is also called an inner product.

✓ Example 17

Find the dot product of the two vectors $\vec{A} = [4, 1, 2, 3]$ and $\vec{B} = [3, 1, 7, 2]$.

Solution

$$\begin{aligned} \vec{A} \cdot \vec{B} &= [4, 1, 2, 3] \cdot [3, 1, 7, 2] \\ &= (4)(3) + (1)(1) + (2)(7) + (3)(2) \\ &= 33 \end{aligned}$$

✓ Example 18

A product line needs three types of rubber as given in the table below.

Rubber type		Weight (lbs)
A		200
B		250
C		310

Use the definition of a dot product to find the total price of the rubber needed.

Solution

The weight vector is given by

$$\vec{W} = [200, 250, 310]$$

and the cost vector is given by

$$\vec{C} = [20.23, 30.56, 29.12]$$

The total cost of the rubber would be the dot product of $\vec{W}$ and $\vec{C}$.

$$\begin{aligned}\vec{W} \cdot \vec{C} &= [200, 250, 310] \cdot [20.23, 30.56, 29.12] \\ &= (200)(20.23) + (250)(30.56) + (310)(29.12) \\ &= 4046 + 7640 + 9027.2 \\ &= \$20713.20\end{aligned}$$

Vectors Quiz

? Quiz 1

A set of equations

$$4x_1 + 7x_2 + 11x_3 = 13$$

$$17x_1 + 39x_2 + 23x_3 = 31$$

$$13x_1 + 67x_2 + 59x_3 = 37$$

can also be written as

$$(A) \ x_1 \begin{bmatrix} 4 \\ 17 \\ 13 \end{bmatrix} + x_2 \begin{bmatrix} 7 \\ 39 \\ 23 \end{bmatrix} + x_3 \begin{bmatrix} 11 \\ 23 \\ 59 \end{bmatrix} = \begin{bmatrix} 13 \\ 31 \\ 37 \end{bmatrix}$$

$$(B) \ 4 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + 39 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + 59 \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 13 \\ 31 \\ 37 \end{bmatrix}$$

$$(C) \ x_1 \begin{bmatrix} 4 \\ 7 \\ 11 \end{bmatrix} + x_2 \begin{bmatrix} 17 \\ 39 \\ 23 \end{bmatrix} + x_3 \begin{bmatrix} 13 \\ 67 \\ 59 \end{bmatrix} = \begin{bmatrix} 13 \\ 31 \\ 37 \end{bmatrix}$$

$$(D) \ x_1 \begin{bmatrix} 13 \\ 17 \\ 4 \end{bmatrix} + x_2 \begin{bmatrix} 67 \\ 39 \\ 7 \end{bmatrix} + x_3 \begin{bmatrix} 59 \\ 23 \\ 11 \end{bmatrix} = \begin{bmatrix} 57 \\ 13 \\ 31 \end{bmatrix}$$

? Quiz 2

The magnitude of the vector, $V = (5, -3, 2)$ is

(A) 4

(B) 10

(C) $\sqrt{38}$

(D) $\sqrt{20}$

? Quiz 3

The rank of the vector

$$\vec{A} \begin{bmatrix} 2 \\ 3 \\ 7 \end{bmatrix}, \begin{bmatrix} 6 \\ 9 \\ 21 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \\ 7 \end{bmatrix}$$

is

- (A) 1
- (B) 2
- (C) 3
- (D) 4

? Quiz 4

If $\vec{A} = (5, 2, 3)$ and $\vec{B} = (6, -7, 3)$, then $4\vec{A} + 5\vec{B}$ is

- (A) $(50, -5, 6)$
- (B) $(50, -27, 27)$
- (C) $(11, -5, 6)$
- (D) $(20, 8, 12)$

? Quiz 5

The dot product of two vectors $\vec{A}$ and $\vec{B}$

$$\begin{aligned}\vec{A} &= 3i + 5j + 7k \\ \vec{B} &= 11i + 13j + 17k\end{aligned}$$

most nearly is

- (A) 14.80
- (B) 33.00
- (C) 56.00
- (D) 217.0

? Quiz 6

The angle in degrees between two vectors $\vec{u}$ and $\vec{v}$

$$\begin{aligned}\vec{u} &= 3i + 5j + 7k \\ \vec{v} &= 11i + 13j + 17k\end{aligned}$$

most nearly is

- (A) 8.124
- (B) 11.47
- (C) 78.52
- (D) 81.88

Vectors Exercise

? Exercise 1

For

$$\vec{A} = \begin{bmatrix} 2 \\ 9 \\ -7 \end{bmatrix}, \vec{B} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}, \vec{C} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

find $\vec{A} + \vec{B}$ and $2\vec{A} - 3\vec{B} + \vec{C}$.

Answer

$$\begin{bmatrix} 5 \\ 11 \\ -2 \end{bmatrix}; \begin{bmatrix} -4 \\ 13 \\ -28 \end{bmatrix}$$

? Exercise 2

Are

$$\vec{A} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \vec{B} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}, \vec{C} = \begin{bmatrix} 1 \\ 4 \\ 25 \end{bmatrix}$$

linearly independent?.

What is the rank of the above set of vectors?

Answer

3

? Exercise 3

Add exercises text here.

Are

$$\vec{A} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \vec{B} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}, \vec{C} = \begin{bmatrix} 3 \\ 5 \\ 7 \end{bmatrix}$$

linearly independent?.

What is the rank of the above set of vectors?

Answer

3

? Exercise 4

Are

$$\vec{A} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}, \vec{B} = \begin{bmatrix} 2 \\ 4 \\ 10 \end{bmatrix}, \vec{C} = \begin{bmatrix} 1.1 \\ 2.2 \\ 5.5 \end{bmatrix}$$

linearly independent?

What is the rank of the above set of vectors?

Answer

No;1

? Exercise 5

If a set of vectors contains the null vector, the set of vectors is linearly

- A. Independent
- B. Dependent?

Answer

B

? Exercise 6

If a set of vectors is linearly independent, a subset of the vectors is linearly

- A. Independent.
- B. Dependent.

Answer

A

? Exercise 7

If a set of vectors is linearly dependent, then

- A. At least one vector can be written as a linear combination of others.
- B. At least one vector is a null vector.

Answer

A

? Exercise 8

If the dimension of a set of vectors is less than the number of vectors in the set, then the set of vectors is linearly

- A. Dependent.
- B. Independent.

Answer

A

? Exercise 9

Find the dot product of $\vec{A} = (2, 1, 2.5, 3)$ and $\vec{B} = (-3, 2, 1, 2.5)$

Answer

6

? Exercise 10

If $\vec{u}$, $\vec{v}$, $\vec{w}$ are three nonzero vector of 2-dimensions, then

- A. $\vec{u}$, $\vec{v}$, $\vec{w}$ are linearly independent
- B. $\vec{u}$, $\vec{v}$, $\vec{w}$ are linearly dependent
- C. $\vec{u}$, $\vec{v}$, $\vec{w}$ are unit vectors
- D. $k_1 \vec{u} + k_2 \vec{v} + k_3 \vec{w} = \vec{0}$ has a unique solution.

Answer

B

? Exercise 11

$\vec{u}$ and $\vec{v}$ are two non-zero vectors of dimension n . Prove that if $\vec{u}$ and $\vec{v}$ are linearly dependent, there is a scalar q such that $\vec{v} = q \vec{u}$.

Answer

Hint :

Start with $k_1 \vec{u} + k_2 \vec{v} = \vec{0}$

Show that $k_1 \neq 0$ and $k_2 \neq 0$ because $\vec{u}$ and $\vec{v}$ are both nonzero.

Hence

$$\begin{aligned} \vec{v} &= -\frac{k_1}{k_2} \vec{u} \\ &= q \vec{u} \quad q = -\frac{k_1}{k_2} \end{aligned}$$

? Exercise 12

$\vec{u}$ and $\vec{v}$ are two non-zero vectors of dimension n . Prove that if there is a scalar q such that $\vec{v} = q \vec{u}$, then $\vec{u}$ and $\vec{v}$ are linearly dependent.

Answer

Hint:

Since

$$\begin{aligned} \vec{v} &= q \vec{u} \\ \vec{v} - q \vec{u} &= \vec{0} \end{aligned}$$

$q \neq 0$, otherwise $\vec{v} = \vec{0}$

So the equation

$$k_1 \vec{v} + k_2 \vec{u} = \vec{0}$$

has a non trivial solution of

$$k_1 = 1, k_2 = q \neq 0.$$

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3: Binary Matrix Operations

Learning Objectives

After reading this chapter, you should be able to:

1. add, subtract, and multiply matrices, and
2. apply rules of binary operations on matrices.

How do you add two matrices?

Two matrices $[A]$ and $[B]$ can be added only if they are the same size. The addition is then shown as

$$[C] = [A] + [B]$$

where

$$c_{ij} = a_{ij} + b_{ij}$$

✓ Example 1

Add the following two matrices.

$$[A] = \begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix} \quad [B] = \begin{bmatrix} 6 & 7 & -2 \\ 3 & 5 & 19 \end{bmatrix}$$

Solution

$$\begin{aligned} [C] &= [A] + [B] \\ &= \begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix} + \begin{bmatrix} 6 & 7 & -2 \\ 3 & 5 & 19 \end{bmatrix} \\ &= \begin{bmatrix} 5+6 & 2+7 & 3-2 \\ 1+3 & 2+5 & 7+19 \end{bmatrix} \\ &= \begin{bmatrix} 11 & 9 & 1 \\ 4 & 7 & 26 \end{bmatrix} \end{aligned}$$

✓ Example 2

Blowout r'us store has two store locations A and B , and their sales of tires are given by make (in rows) and quarters (in columns) as shown below.

$$[A] = \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix}$$
$$[B] = \begin{bmatrix} 20 & 5 & 4 & 0 \\ 3 & 6 & 15 & 21 \\ 4 & 1 & 7 & 20 \end{bmatrix}$$

where the rows represent the sale of Tirestone, Michigan and Copper tires respectively and the columns represent the quarter number: 1, 2, 3 and 4. What are the total tire sales for the two locations by make and quarter?

Solution

$$\begin{aligned}
 [C] &= [A] + [B] \\
 &= \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix} + \begin{bmatrix} 20 & 5 & 4 & 0 \\ 3 & 6 & 15 & 21 \\ 4 & 1 & 7 & 20 \end{bmatrix} \\
 &= \begin{bmatrix} (25+20) & (20+5) & (3+4) & (2+0) \\ (5+3) & (10+6) & (15+15) & (25+21) \\ (6+4) & (16+1) & (7+7) & (27+20) \end{bmatrix} \\
 &= \begin{bmatrix} 45 & 25 & 7 & 2 \\ 8 & 16 & 30 & 46 \\ 10 & 17 & 14 & 47 \end{bmatrix}
 \end{aligned}$$

So, if one wants to know the total number of Copper tires sold in quarter 4 at the two locations, we would look at Row 3 – Column 4 to give $c_{34} = 47$.

How do you subtract two matrices?

Two matrices $[A]$ and $[B]$ can be subtracted only if they are the same size. The subtraction is then shown as

$$[D] = [A] - [B]$$

where

$$d_{ij} = a_{ij} - b_{ij}$$

✓ Example 3

Subtract matrix $[B]$ from matrix $[A]$.

$$\begin{aligned}
 [A] &= \begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix} \\
 [B] &= \begin{bmatrix} 6 & 7 & -2 \\ 3 & 5 & 19 \end{bmatrix}
 \end{aligned}$$

Solution

$$\begin{aligned}
 [D] &= [A] - [B] \\
 &= \begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix} - \begin{bmatrix} 6 & 7 & -2 \\ 3 & 5 & 19 \end{bmatrix} \\
 &= \begin{bmatrix} (5-6) & (2-7) & (3-(-2)) \\ (1-3) & (2-5) & (7-19) \end{bmatrix} \\
 &= \begin{bmatrix} -1 & -5 & 5 \\ -2 & -3 & -12 \end{bmatrix}
 \end{aligned}$$

✓ Example 4

Blowout r'us has two store locations A and B and their sales of tires are given by make (in rows) and quarters (in columns) as shown below.

$$\begin{aligned}
 [A] &= \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix} \\
 [B] &= \begin{bmatrix} 20 & 5 & 4 & 0 \\ 3 & 6 & 15 & 21 \\ 4 & 1 & 7 & 20 \end{bmatrix}
 \end{aligned}$$

where the rows represent the sale of Tirestone, Michigan and Copper tires respectively and the columns represent the quarter number: 1, 2, 3, and 4. How many more tires did store A sell than store B of each brand in each quarter?

Solution

$$\begin{aligned}
 [D] &= [A] - [B] \\
 &= \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix} - \begin{bmatrix} 20 & 5 & 4 & 0 \\ 3 & 6 & 15 & 21 \\ 4 & 1 & 7 & 20 \end{bmatrix} \\
 &= \begin{bmatrix} (25-20) & (20-5) & (3-4) & (2-0) \\ (5-3) & (10-6) & (15-15) & (25-21) \\ (6-4) & (16-1) & (7-7) & (27-20) \end{bmatrix} \\
 &= \begin{bmatrix} 5 & 15 & -1 & 2 \\ 2 & 4 & 0 & 4 \\ 2 & 15 & 0 & 7 \end{bmatrix}
 \end{aligned}$$

So, if you want to know how many more Copper tires were sold in quarter 4 in store A than store B , $d_{34} = 7$. Note that $d_{13} = -1$ implies that store A sold 1 less Michigan tire than store B in quarter 3.

How do I multiply two matrices?

Two matrices $[A]$ and $[B]$ can be multiplied only if the number of columns of $[A]$ is equal to the number of rows of $[B]$ to give

$$[C]_{m \times n} = [A]_{m \times p} [B]_{p \times n}$$

If $[A]$ is a $m \times p$ matrix and $[B]$ is a $p \times n$ matrix, the resulting matrix $[C]$ is a $m \times n$ matrix.

So how does one calculate the elements of $[C]$ matrix?

$$\begin{aligned}
 c_{ij} &= \sum_{k=1}^p a_{ik} b_{kj} \\
 &= a_{i1} b_{1j} + a_{i2} b_{2j} + \dots + a_{ip} b_{pj}
 \end{aligned}$$

for each $i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$.

To put it in simpler terms, the i^{th} row and j^{th} column of the $[C]$ matrix in $[C] = [A][B]$ is calculated by multiplying the i^{th} row of $[A]$ by the j^{th} column of $[B]$. That is

$$\begin{aligned}
 c_{ij} &= [a_{i1} \ a_{i2} \ \dots \ a_{ip}] \begin{bmatrix} b_{1j} \\ b_{2j} \\ \vdots \\ b_{pj} \end{bmatrix} \\
 &= a_{i1} b_{1j} + a_{i2} b_{2j} + \dots + a_{ip} b_{pj} \\
 &= \sum_{k=1}^p a_{ik} b_{kj}
 \end{aligned}$$

✓ Example 5

Given

$$\begin{aligned}
 [A] &= \begin{bmatrix} 5 & 2 & 3 \\ 1 & 2 & 7 \end{bmatrix} \\
 [B] &= \begin{bmatrix} 3 & -2 \\ 5 & -8 \\ 9 & -10 \end{bmatrix}
 \end{aligned}$$

Find

$$[C] = [A] [B]$$

Solution

c_{12} can be found by multiplying the first row of $[A]$ by the second column of $[B]$,

$$\begin{aligned} c_{12} &= [5 \quad 2 \quad 3] \begin{bmatrix} -2 \\ -8 \\ -10 \end{bmatrix} \\ &= (5)(-2) + (2)(-8) + (3)(-10) \\ &= -56 \end{aligned}$$

Similarly, one can find the other elements of $[C]$ to give

$$[C] = \begin{bmatrix} 52 & -56 \\ 76 & -88 \end{bmatrix}$$

✓ Example 6

Blowout r'us store location A and the sales of tires are given by make (in rows) and quarters (in columns) as shown below

$$[A] = \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix}$$

where the rows represent the sale of Tirestone, Michigan and Copper tires respectively and the columns represent the quarter number: 1, 2, 3, and 4. Find the per quarter sales of store A if the following are the prices of each tire:

- Tirestone = \$33.25
- Michigan = \$40.19
- Copper = \$25.03

Solution

The answer is given by multiplying the price matrix by the quantity of sales of store A . The price matrix is $[33.25 \quad 40.19 \quad 25.03]$, so the per-quarter sales of store A would be given by:

$$[C] = [33.25 \quad 40.19 \quad 25.03] \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix}$$

$$c_{ij} = \sum_{k=1}^3 a_{ik} b_{kj}$$

$$\begin{aligned} c_{11} &= \sum_{k=1}^3 a_{1k} b_{k1} \\ &= a_{11} b_{11} + a_{12} b_{21} + a_{13} b_{31} \\ &= (33.25)(25) + (40.19)(5) + (25.03)(6) \\ &= \$1182.38 \end{aligned}$$

Similarly

$$c_{12} = \$1467.38$$

$$c_{13} = \$877.81$$

$$c_{14} = \$1747.06$$

Therefore, each quarter sales of store A in dollars is given by the four columns of the row vector

$$[C] = [1182.38 \quad 1467.38 \quad 877.81 \quad 1747.06]$$

Remember, since we are multiplying a 1×3 matrix by a 3×4 matrix, the resulting matrix is a 1×4 matrix.

What is the scalar multiplication of a matrix?

If $[A]$ is a $m \times n$ matrix and k is a real number, then the multiplication $[A]$ by a scalar k is another $m \times n$ matrix $[B]$, where $b_{ij} = k a_{ij}$ for all i, j .

✓ Example 7

Let

$$[A] = \begin{bmatrix} 2.1 & 3 & 2 \\ 5 & 1 & 6 \end{bmatrix}$$

Find $2[A]$

Solution

$$\begin{aligned} 2[A] &= 2 \begin{bmatrix} 2.1 & 3 & 2 \\ 5 & 1 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 2 \times 2.1 & 2 \times 3 & 2 \times 2 \\ 2 \times 5 & 2 \times 1 & 2 \times 6 \end{bmatrix} \\ &= \begin{bmatrix} 4.2 & 6 & 4 \\ 10 & 2 & 12 \end{bmatrix} \end{aligned}$$

What is a linear combination of matrices?

If $[A_1], [A_2], \dots, [A_p]$ are matrices of the same size and $k_1, k_2, \dots, k_p$ are scalars, then

$$k_1 [A_1] + k_2 [A_2] + \dots + k_p [A_p]$$

is called a linear combination of $[A_1], [A_2], \dots, [A_p]$.

✓ Example 8

If

$$[A_1] = \begin{bmatrix} 5 & 6 & 2 \\ 3 & 2 & 1 \end{bmatrix}, [A_2] = \begin{bmatrix} 2.1 & 3 & 2 \\ 5 & 1 & 6 \end{bmatrix}, [A_3] = \begin{bmatrix} 0 & 2.2 & 2 \\ 3 & 3.5 & 6 \end{bmatrix}$$

then find

$$[A_1] + 2[A_2] - 0.5[A_3]$$

Solution

$$\begin{aligned} &[A_1] + 2[A_2] - 0.5[A_3] \\ &= \begin{bmatrix} 5 & 6 & 2 \\ 3 & 2 & 1 \end{bmatrix} + 2 \begin{bmatrix} 2.1 & 3 & 2 \\ 5 & 1 & 6 \end{bmatrix} - 0.5 \begin{bmatrix} 0 & 2.2 & 2 \\ 3 & 3.5 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 5 & 6 & 2 \\ 3 & 2 & 1 \end{bmatrix} + \begin{bmatrix} 4.2 & 6 & 4 \\ 10 & 2 & 12 \end{bmatrix} - 0.5 \begin{bmatrix} 0 & 1.1 & 1 \\ 1.5 & 1.75 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 9.2 & 10.9 & 5 \\ 11.5 & 2.25 & 10 \end{bmatrix} \end{aligned}$$

What are some of the rules of binary matrix operations?

Commutative law of addition

If $[A]$ and $[B]$ are $m \times n$ matrices, then

$$[A] + [B] = [B] + [A]$$

Associative law of addition

If $[A]$, $[B]$, and $[C]$ are all $m \times n$ matrices, then

$$[A] + ([B] + [C]) = ([A] + [B]) + [C]$$

Associative law of multiplication

If $[A]$, $[B]$, and $[C]$ are $m \times n$, $n \times p$, and $p \times r$ size matrices, respectively, then

$$[A] ([B] [C]) = ([A] [B]) [C]$$

and the resulting matrix size on both sides of the equation is $m \times p$.

Distributive law

If $[A]$ and $[B]$ are $m \times n$ size matrices, and $[C]$ and $[D]$ are $n \times p$ size matrices

$$[A] ([C] + [D]) = [A] [C] + [A] [D]$$

$$([A] + [B]) [C] = [A] [C] + [B] [C]$$

And the resulting matrix size on both sides of the equation is $m \times p$.

✓ Example 9

Illustrate the associative law of multiplication of matrices using

$$[A] = \begin{bmatrix} 1 & 2 \\ 3 & 5 \\ 0 & 2 \end{bmatrix}, \quad [B] = \begin{bmatrix} 2 & 5 \\ 9 & 6 \end{bmatrix}, \quad [C] = \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}$$

Solution

$$\begin{aligned} [B] [C] &= \begin{bmatrix} 2 & 5 \\ 9 & 6 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix} \\ &= \begin{bmatrix} 19 & 27 \\ 36 & 39 \end{bmatrix} \\ [A] ([B] [C]) &= \begin{bmatrix} 1 & 2 \\ 3 & 5 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 19 & 27 \\ 36 & 39 \end{bmatrix} \\ &= \begin{bmatrix} 91 & 105 \\ 237 & 276 \\ 72 & 78 \end{bmatrix} \\ [A] [B] &= \begin{bmatrix} 1 & 2 \\ 3 & 5 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ 9 & 6 \end{bmatrix} \\ &= \begin{bmatrix} 20 & 17 \\ 51 & 45 \\ 18 & 12 \end{bmatrix} \end{aligned}$$

$$\begin{aligned}
 ([A] [B]) [C] &= \begin{bmatrix} 20 & 17 \\ 51 & 45 \\ 18 & 12 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix} \\
 &= \begin{bmatrix} 91 & 105 \\ 237 & 276 \\ 72 & 78 \end{bmatrix}
 \end{aligned}$$

The above illustrates the associative law of multiplication of matrices.

Is $[A][B] = [B][A]$?

If $[A][B]$ exists, number of columns of $[A]$ has to be same as the number of rows of $[B]$ and if $[B][A]$ exists, number of columns of $[B]$ has to be same as the number of rows of $[A]$. Now for $[A][B] = [B][A]$, the resulting matrix from $[A][B]$ and $[B][A]$ has to be of the same size. This is only possible if $[A]$ and $[B]$ are square and are of the same size. Even then in general $[A][B] \neq [B][A]$

✓ Example 10

Determine if

$$[A][B] = [B][A]$$

For the following matrices

$$[A] = \begin{bmatrix} 6 & 3 \\ 2 & 5 \end{bmatrix}, \quad [B] = \begin{bmatrix} -3 & 2 \\ 1 & 5 \end{bmatrix}$$

Solution

$$\begin{aligned}
 [A][B] &= \begin{bmatrix} 6 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ 1 & 5 \end{bmatrix} \\
 &= \begin{bmatrix} -15 & 27 \\ -1 & 29 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 [B][A] &= \begin{bmatrix} -3 & 2 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} 6 & 3 \\ 2 & 5 \end{bmatrix} \\
 &= \begin{bmatrix} -14 & 1 \\ 16 & 28 \end{bmatrix}
 \end{aligned}$$

$$[A][B] \neq [B][A]$$

Binary Matrix Operations Quiz

? Quiz 1

If $[A] = \begin{bmatrix} 5 & 6 \\ 7 & -3 \end{bmatrix}$ and $[B] = \begin{bmatrix} 2 \\ -3 \end{bmatrix}$ then $[A][B] =$

(A) $\begin{bmatrix} -8 \\ 23 \end{bmatrix}$

(B) $\begin{bmatrix} 10 & 12 \\ 14 & 9 \end{bmatrix}$

(C) $\begin{bmatrix} -2 & 5 \end{bmatrix}$

(D) not possible

? Quiz 2

For the product $[A][B]$ to be possible

- (A) the number of rows of $[A]$ needs to be the same as the number of columns of $[B]$
- (B) the number of columns of $[A]$ needs to be the same as the number of rows of $[B]$
- (C) the number of rows of $[A]$ and $[B]$ needs to be the same
- (D) the number of columns of $[A]$ and $[B]$ needs to be the same

? Quiz 3

If $[A] = \begin{bmatrix} 50 & 60 \\ 20 & -30 \end{bmatrix}$ then $6[A]$ is equal to

- (A) $\begin{bmatrix} 50 & 360 \\ 120 & -180 \end{bmatrix}$
- (B) $\begin{bmatrix} 300 & 60 \\ 20 & -30 \end{bmatrix}$
- (C) $\begin{bmatrix} 300 & 360 \\ 120 & -180 \end{bmatrix}$
- (D) $\begin{bmatrix} 56 & 66 \\ 26 & -24 \end{bmatrix}$

? Quiz 4

$[A]$ and $[B]$ are square matrices of $n \times n$ order. Then $([A] - [B])([A] - [B])$ is equal to

- (A) $[A]^2 + [B]^2 - 2[A][B]$
- (B) $[A]^2 + [B]^2$
- (C) $[A]^2 - [B]^2$
- (D) $[A]^2 + [B]^2 - [A][B] - [B][A]$

? Quiz 5

Given $[A]$ is a rectangular matrix and $c[A] = [0]$, then choose the most appropriate answer.

- (A) $C = 0$
- (B) $C = 0$ and $[A] = [0]$
- (C) $C = 0$ or $[A] = [0]$
- (D) $C = 0$ and $[A]$ is a non-zero matrix

? Quiz 6

You sell Jupiter and Fickers Candy bars. The sales in January are 25 and 30 of Jupiter and Fickers, respectively. In February, the sales are 75 and 35 of Jupiter and Fickers, respectively. If a Jupiter bar costs \$2 and a Fickers bar costs \$7, then if

$$[A] = \begin{bmatrix} 25 & 30 \\ 75 & 35 \end{bmatrix}, \text{ and}$$

$$[B] = \begin{bmatrix} 2 \\ 7 \end{bmatrix},$$

the total sales amount in each month is given by

(A) $[B][A]$

(B) $[A][B]$

(C) $2[A]$

(D) $7[A]$

Binary Matrix Operations Exercise

? Exercise 1

For the following matrices

$$[A] = \begin{bmatrix} 3 & 0 \\ -1 & 2 \\ 1 & 1 \end{bmatrix}, [B] = \begin{bmatrix} 4 & -1 \\ 0 & 2 \end{bmatrix}, [C] = \begin{bmatrix} 5 & 2 \\ 3 & 5 \\ 6 & 7 \end{bmatrix}$$

Find where possible

A. $4[A] + 5[C]$

B. $[A][B]$

C. $[A] = 2[C]$

Answer

$$A. = \begin{bmatrix} 37 & 10 \\ 11 & 33 \\ 34 & 39 \end{bmatrix}$$

$$B. = \begin{bmatrix} 12 & -3 \\ -4 & 5 \\ 4 & 1 \end{bmatrix}$$

$$C. = \begin{bmatrix} -7 & -4 \\ -7 & -8 \\ -11 & -13 \end{bmatrix}$$

? Exercise 2

Food orders are taken from two engineering departments for a takeout. The order is tabulated below.

Food order:

	Chicken Sandwich	Fries	Drink
Mechanical	25	35	25
Civil	21	20	21

However they have a choice of buying this food from three different restaurants. Their prices for the three food items are tabulated below

Price Matrix:

	McFat	Burcholestrol	Kentucky Sodium
Chicken Sandwich	2.42	2.38	2.46
Fries	0.93	0.90	0.89
Drink	0.95	1.03	1.13

Show how much each department will pay for their order at each restaurant. Which restaurant would be more economical to order from for each department?

Answer

The cost in dollars is 116.80, 116.75, 120.90 for the Mechanical Department at three fast food joints. So BurCholestrol is the cheapest for the Mechanical Department. The cost in dollars is 89.37, 89.61, 93.19 for the Civil Department at three fast food joints. McFat is the cheapest for the Civil Department.

? Exercise 3

Given

$$[A] = \begin{bmatrix} 2 & 3 & 5 \\ 6 & 7 & 9 \\ 2 & 1 & 3 \end{bmatrix}$$

$$[B] = \begin{bmatrix} 3 & 5 \\ 2 & 9 \\ 1 & 6 \end{bmatrix}$$

$$[C] = \begin{bmatrix} 5 & 2 \\ 3 & 9 \\ 7 & 6 \end{bmatrix}$$

Illustrate the distributive law of binary matrix operations

$$[A] ([B] + [C]) = [A] [B] + [A] [C]$$

Answer

$$[B] + [C] = \begin{bmatrix} 3 & 5 \\ 2 & 9 \\ 1 & 6 \end{bmatrix} + \begin{bmatrix} 5 & 2 \\ 3 & 9 \\ 7 & 6 \end{bmatrix}$$

$$= \begin{bmatrix} 8 & 7 \\ 5 & 18 \\ 8 & 12 \end{bmatrix}$$

$$[A] ([B] + [C]) = \begin{bmatrix} 71 & 128 \\ 155 & 276 \\ 45 & 68 \end{bmatrix}$$

$$[A] [B] = \begin{bmatrix} 17 & 67 \\ 41 & 147 \\ 11 & 37 \end{bmatrix}$$

$$[A] [C] = \begin{bmatrix} 54 & 61 \\ 114 & 129 \\ 34 & 31 \end{bmatrix}$$

$$[A] [B] + [A] [C] = \begin{bmatrix} 71 & 128 \\ 155 & 276 \\ 45 & 68 \end{bmatrix}$$

? Exercise 4

Let $[I]$ be a $n \times n$ identity matrix. Show that $[A][I] = [I][A] = [A]$ for every $n \times n$ matrix $[A]$.

Let $[C]_{n \times n} = [A]_{n \times n} [I]_{n \times n}$

Answer

Hint: $c_{ij} = \sum_{p=1}^n a_{ip} i_{pj}$

$$= a_{i1} i_{1j} + \dots + a_{i,j-1} i_{j-1,j} + a_{ij} i_{jj} + a_{i(j+1)} i_{(j+1)j} + \dots + a_{in} i_{nj}$$

Since

$$i_{ij} = 0 \text{ for } i \neq j$$

$$= 1 \text{ for } i = j$$

$$c_{ij} = a_{ij}$$

$$\text{So } [A] = [A] [I]$$

Similarly do the other case

$$[I] [A] = [A]. \text{ Just do it!}$$

? Exercise 5

Consider there are only two computer companies in a country. The companies are named *Dude* and *Imac*. Each year, company *Dude* keeps $1/5^{\text{th}}$ of its customers, while the rest switch to *Imac*. Each year, *Imac* keeps $1/3^{\text{rd}}$ of its customers, while the rest switch to *Dude*. If in 2002, *Dude* has $1/6^{\text{th}}$ of the market and *Imac* has $5/6^{\text{th}}$ of the market.

- What is the distribution of the customers between the two companies in 2003? Write the answer first as multiplication of two matrices.
- What would be distribution when the market becomes stable?

Answer

A. At the end of 2002, *Dude* has $\frac{1}{5} \times \frac{1}{6} + \frac{2}{3} \times \frac{5}{6} = 0.589$.

Imac has $\frac{4}{5} \times \frac{1}{6} + \frac{1}{3} \times \frac{5}{6} = 0.411$

$$\text{In matrix form } \begin{bmatrix} \frac{1}{5} & \frac{2}{3} \\ \frac{4}{5} & \frac{1}{3} \end{bmatrix} \begin{bmatrix} \frac{1}{6} \\ \frac{5}{6} \end{bmatrix} = \begin{bmatrix} 0.589 \\ 0.411 \end{bmatrix}$$

B. Stable distribution is $[10/22 \ 12/22]$ (Try to do this part of the problem first by finding the distribution five years from now).

? Exercise 6

Given

$$[A] = \begin{bmatrix} 12.3 & -12.3 & 10.3 \\ 11.3 & -10.3 & -11.3 \\ 10.3 & -11.3 & -12.3 \end{bmatrix},$$

$$[B] = \begin{bmatrix} 2 & 4 \\ -5 & 6 \\ 11 & -20 \end{bmatrix}$$

$[A][B]$ matrix size is _____

Answer

$$3 \times 2$$

? Exercise 7

Given

$$[A] = \begin{bmatrix} 12.3 & -12.3 & 10.3 \\ 11.3 & -10.3 & -11.3 \\ 10.3 & -11.3 & -12.3 \end{bmatrix},$$

$$[B] = \begin{bmatrix} 2 & 4 \\ -5 & 6 \\ 11 & -20 \end{bmatrix}$$

if $[C] = [A] [B]$, then $c_{31} =$ _____

Answer

$$(10.3 \times 2) + ((-5) \times (-11.3)) + (11 \times (-12.3)) = -58.2$$

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4: Unary Matrix Operations

Learning Objectives

After reading this chapter, you should be able to:

1. know what unary operations are,
2. find the transpose of a square matrix and its relationship to symmetric matrices,
3. find the trace of a matrix, and
4. find the determinant of a matrix by the cofactor method.

What is the transpose of a matrix?

Let $[A]$ be a $m \times n$ matrix. Then $[B]$ is the transpose of $[A]$ if $b_{ji} = a_{ij}$ for all i and j . That is, the i^{th} row and the j^{th} column element of $[A]$ is the j^{th} row and i^{th} column element of $[B]$. Note, $[B]$ would be a $n \times m$ matrix. The transpose of $[A]$ is denoted by $[A]^T$.

✓ Example 1

Find the transpose of

$$[A] = \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix}$$

Solution

The transpose of $[A]$ is

$$[A]^T = \begin{bmatrix} 25 & 5 & 6 \\ 20 & 10 & 16 \\ 3 & 15 & 7 \\ 2 & 25 & 27 \end{bmatrix}$$

Note, the transpose of a row vector is a column vector and the transpose of a column vector is a row vector.

Also, note that the transpose of a transpose of a matrix is the matrix itself. That is,

$$([A]^T)^T = [A]$$

Also,

$$([A] + [B])^T = [A]^T + [B]^T; (cA)^T = cA^T$$

What is a symmetric matrix?

A square matrix $[A]$ with real elements were $a_{ij} = a_{ji}$ for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, n$ is called a symmetric matrix. This is the same as saying that if $[A] = [A]^T$, then $[A]^T$ is a symmetric matrix.

✓ Example 2

Give an example of a symmetric matrix

Solution

$$[A] = \begin{bmatrix} 21.2 & 3.2 & 6 \\ 3.2 & 21.5 & 8 \\ 6 & 8 & 9.3 \end{bmatrix}$$

Is a symmetric matrix as $a_{12} = a_{21} = 3.2$, $a_{13} = a_{31} = 6$ and $a_{23} = a_{32} = 8$.

What is a skew-symmetric matrix?

A $n \times n$ matrix is skew-symmetric if $a_{ij} = -a_{ji}$, for $i = 1, \dots, n$ and $j = 1, \dots, n$. This is the same as

$$[A] = -[A]^T$$

✓ Example 3

Give an example of a skew-symmetric matrix

Solution

$$\begin{bmatrix} 0 & 1 & 2 \\ -1 & 0 & -5 \\ -2 & 5 & 0 \end{bmatrix}$$

Is a skew symmetric matrix as $a_{12} = -a_{21} = 1$; $a_{13} = -a_{31} = 2$; $a_{23} = -a_{32} = -5$. Since $a_{ii} = -a_{ii}$ only if $a_{ii} = 0$, all the diagonal elements of a skew-symmetric matrix have to be zero.

What is the trace of a matrix?

The trace of a $n \times n$ matrix $[A]$ is the sum of the diagonal entries of $[A]$. That is

$$tr[A] = \sum_{i=1}^n a_{ii}$$

✓ Example 4

Find the trace of:

$$[A] = \begin{bmatrix} 15 & 6 & 7 \\ 2 & -4 & 2 \\ 3 & 2 & 6 \end{bmatrix}$$

Solution

$$\begin{aligned} tr[A] &= \sum_{i=1}^n a_{ii} \\ &= 15 + (-4) + 6 \\ &= 17 \end{aligned}$$

✓ Example 5

The sales of tires are given by make (rows) and quarters (columns) for Blowout r'us store location A , as shown below

$$[A] = \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix}$$

Where the rows represent the sales of Tirestone, Michigan, and Copper tires, and the columns represent the quarter numbers 1, 2, 3, 4.

Find the total yearly revenue of store A if the prices of tires vary by quarters as follows

$$[B] = \begin{bmatrix} 33.25 & 30.01 & 35.02 & 30.05 \\ 40.19 & 38.02 & 41.03 & 38.23 \\ 25.03 & 22.02 & 27.03 & 22.95 \end{bmatrix}$$

Where the rows represent the cost of each tire made by Tirestone, Michigan, and Copper, and the columns represent the quarter numbers.

Solution

$$[C] = [B]^T$$

✓ Example 6

Blowout r'us store location A and the sales of tires are given by make (in rows) and quarters (in columns) as shown below

$$\begin{aligned} [A] &= \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix} \\ &= \begin{bmatrix} 33.25 & 30.01 & 35.02 & 30.05 \\ 40.19 & 38.02 & 41.03 & 38.23 \\ 25.03 & 22.02 & 27.03 & 22.95 \end{bmatrix} \\ &= \begin{bmatrix} 33.25 & 40.19 & 25.03 \\ 30.01 & 38.02 & 22.02 \\ 35.02 & 41.03 & 27.03 \\ 30.05 & 38.23 & 22.95 \end{bmatrix} \end{aligned}$$

Recognize now that if we find $[A] [C]$, we get

$$\begin{aligned} [D] &= [A] [C] \\ &= \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix} \begin{bmatrix} 33.25 & 40.19 & 25.03 \\ 30.01 & 38.02 & 22.02 \\ 35.02 & 41.03 & 27.03 \\ 30.05 & 38.23 & 22.95 \end{bmatrix} \\ &= \begin{bmatrix} 1597 & 1965 & 1193 \\ 1743 & 2152 & 1325 \\ 1736 & 2169 & 1311 \end{bmatrix} \end{aligned}$$

The diagonal elements give the sales of each brand of tire for the whole year. That is

$$d_{11} = \$1597 \text{ (Tirestone sales)}$$

$$d_{22} = \$2152 \text{ (Michigan sales)}$$

$$d_{33} = \$1597 \text{ (Copper sales)}$$

The total yearly sales of all three brands of tires are

$$\begin{aligned} \sum_{i=1}^3 d_{ii} &= 1597 + 2152 + 1311 \\ &= \$5060 \end{aligned}$$

And this is the trace of matrix $[D]$.

Define the determinant of a matrix.

The determinant of a square matrix is a single unique real number corresponding to a matrix. For a matrix $[A]$, determinant is denoted by $|A|$ or $\det(A)$. So do not use $[A]$ and $|A|$ interchangeably.

For a 2×2 matrix

$$[A] = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

$$\det(A) = a_{11}a_{22} - a_{12}a_{21}$$

How does one calculate the determinant of any square matrix?

Let $[A]$ be a $n \times n$ matrix. The minor of entry a_{ij} is denoted by M_{ij} and is defined as the determinant of the $(n-1) \times (n-1)$ submatrix of $[A]$, where the submatrix is obtained by deleting the i^{th} row and j^{th} column of the matrix $[A]$. The determinant is then given by

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} M_{ij} \text{ for any } i = 1, 2, \dots, n$$

or

$$\det(A) = \sum_{i=1}^n (-1)^{i+j} a_{ij} M_{ij} \text{ for any } j = 1, 2, \dots, n$$

Couple that with $\det(A) = a_{11}$ for a 1×1 matrix $[A]$, we can always reduce the determinant of a matrix to determinants of 1×1 matrices. The number $(-1)^{i+j} M_{ij}$ is called the cofactor of a_{ij} and is denoted by c_{ij} . The formula for the determinant can then be written as

$$\det(A) = \sum_{j=1}^n a_{ij} c_{ij} \text{ for any } i = 1, 2, \dots, n$$

or

$$\det(A) = \sum_{i=1}^n a_{ij} c_{ij} \text{ for any } j = 1, 2, \dots, n$$

Determinants are not generally calculated using this method as it becomes computationally intensive for large matrices. For a $n \times n$ matrix, it requires arithmetic operations proportional to $n!$.

✓ Example 6

Find the determinant of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

Solution

Method 1:

$$\det(A) = \sum_{j=1}^3 (-1)^{i+j} a_{ij} M_{ij} \text{ for any } i = 1, 2, 3$$

Let us choose $i = 1$ in the formula

$$\begin{aligned} \det(A) &= \sum_{j=1}^3 (-1)^{1+j} a_{1j} M_{1j} \\ &= (-1)^{1+1} a_{11} M_{11} + (-1)^{1+2} a_{12} M_{12} + (-1)^{1+3} a_{13} M_{13} \\ &= a_{11} M_{11} - a_{12} M_{12} + a_{13} M_{13} \\ M_{11} &= \begin{vmatrix} 8 & 1 \\ 12 & 1 \end{vmatrix} \\ &= -4 \end{aligned}$$

$$M_{12} = \begin{vmatrix} 64 & 1 \\ 144 & 1 \end{vmatrix} \\ = -80$$

$$M_{13} = \begin{vmatrix} 64 & 8 \\ 144 & 12 \end{vmatrix} \\ = -384$$

$$\det(A) = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13} \\ = 25(-4) - 5(-80) + 1(-384) \\ = -100 + 400 - 384 \\ = -84$$

Also for $i = 1$,

$$\det(A) = \sum_{j=1}^3 a_{1j}C_{1j}$$

$$C_{11} = (-1)^{1+1}M_{11} \\ = M_{11} \\ = -4$$

$$C_{12} = (-1)^{1+2}M_{12} \\ = -M_{12} \\ = 80$$

$$C_{13} = (-1)^{1+3}M_{13} \\ = M_{13} \\ = -384$$

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \\ = (25)(-4) + (5)(80) + (1)(-384) \\ = -100 + 400 - 384 \\ = -84$$

Method 2:

$$\det(A) = \sum_{i=1}^3 (-1)^{i+j} a_{ij} M_{ij} \text{ for any } j = 1, 2, 3$$

Let us choose $j = 2$ in the formula

$$\det(A) = \sum_{i=1}^3 (-1)^{i+2} a_{i2} M_{i2} \\ = (-1)^{1+2} a_{12} M_{12} + (-1)^{2+2} a_{22} M_{22} + (-1)^{3+2} a_{32} M_{32} \\ = -a_{12} M_{12} + a_{22} M_{22} - a_{32} M_{32}$$

$$M_{12} = \begin{vmatrix} 64 & 1 \\ 144 & 1 \end{vmatrix} \\ = -80$$

$$M_{22} = \begin{vmatrix} 25 & 1 \\ 144 & 1 \end{vmatrix} \\ = -119$$

$$M_{32} = \begin{vmatrix} 25 & 1 \\ 64 & 1 \end{vmatrix} \\ = -39$$

$$\begin{aligned} \det(A) &= -a_{12}M_{12} + a_{22}M_{22} - a_{32}M_{32} \\ &= -5(-80) + 8(-119) - 12(-39) \\ &= 400 - 952 + 468 \\ &= -84 \end{aligned}$$

In terms of cofactors for $j = 2$,

$$\det(A) = \sum_{i=1}^3 a_{i2}C_{i2}$$

$$\begin{aligned} C_{12} &= (-1)^{1+2}M_{12} \\ &= -M_{12} \\ &= 80 \end{aligned}$$

$$\begin{aligned} C_{22} &= (-1)^{2+2}M_{22} \\ &= M_{22} \\ &= -119 \end{aligned}$$

$$\begin{aligned} C_{32} &= (-1)^{3+2}M_{32} \\ &= -M_{32} \\ &= 39 \end{aligned}$$

$$\begin{aligned} \det(A) &= a_{12}C_{12} + a_{22}C_{22} + a_{32}C_{32} \\ &= (5)(80) + (8)(-119) + (12)(39) \\ &= 400 - 952 + 468 \\ &= -84 \end{aligned}$$

Is there a relationship between $\det(AB)$, and $\det(A)$ and $\det(B)$?

Yes, if $[A]$ and $[B]$ are square matrices of same size, then

$$\det(AB) = \det(A)\det(B)$$

Are there some other theorems that are important in finding the determinant of a square matrix?

Theorem 1: If a row or a column in a $n \times n$ matrix $[A]$ is zero, then $\det(A) = 0$.

Theorem 2: Let $[A]$ be a $n \times n$ matrix. If a row is proportional to another row, then $\det(A) = 0$.

Theorem 3: Let $[A]$ be a $n \times n$ matrix. If a column is proportional to another column, then $\det(A) = 0$.

Theorem 4: Let $[A]$ be a $n \times n$ matrix. If a column or row is multiplied by k to result in matrix k , then $\det(B) = k\det(A)$.

Theorem 5: Let $[A]$ be a $n \times n$ upper or lower triangular matrix, then $\det(A) = \prod_{i=1}^n a_{ii}$.

✓ Example 7

What is the determinant of

$$[A] = \begin{bmatrix} 0 & 2 & 6 & 3 \\ 0 & 3 & 7 & 4 \\ 0 & 4 & 9 & 5 \\ 0 & 5 & 2 & 1 \end{bmatrix}$$

Solution

Since one of the columns (first column in the above example) of $[A]$ is a zero, $\det(A) = 0$.

✓ Example 8

What is the determinant of

$$[A] = \begin{bmatrix} 2 & 1 & 6 & 4 \\ 3 & 2 & 7 & 6 \\ 5 & 4 & 2 & 10 \\ 9 & 5 & 3 & 18 \end{bmatrix}$$

Solution

$\det(A)$ is zero because the fourth column

$$\begin{bmatrix} 4 \\ 6 \\ 10 \\ 18 \end{bmatrix}$$

is 2 times the first column

$$\begin{bmatrix} 2 \\ 3 \\ 5 \\ 9 \end{bmatrix}$$

✓ Example 9

If the determinant of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

is -84 , then what is the determinant of

$$[B] = \begin{bmatrix} 25 & 10.5 & 1 \\ 64 & 16.8 & 1 \\ 144 & 25.2 & 1 \end{bmatrix}$$

Solution

Since the second column of $[B]$ is 2.1 times the second column of $[A]$

$$\begin{aligned} \det(B) &= 2.1 \det(A) \\ &= (2.1)(-84) \\ &= -176.4 \end{aligned}$$

✓ Example 10

Given the determinant of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

is -84 , what is the determinant of

$$[B] = \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 144 & 12 & 1 \end{bmatrix}$$

Solution

Since $[B]$ is simply obtained by subtracting the second row of $[A]$ by 2.56 times the first row of $[A]$,

$$\begin{aligned} \det(B) &= \det(A) \\ &= -84 \end{aligned}$$

✓ Example 11

What is the determinant of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix}$$

Solution

Since $[A]$ is an upper triangular matrix

$$\begin{aligned} \det(A) &= \prod_{i=1}^3 a_{ii} \\ &= a_{11} \times a_{22} \times a_{33} \\ &= 25 \times (-4.8) \times 0.7 \\ &= -84 \end{aligned}$$

Unary Matrix Operations Quiz

? Quiz 1

If the determinant of a 4×4 matrix is given as 20, then the determinant of 5 is

- (A) 100
- (B) 12500
- (C) 25
- (D) 62500

? Quiz 2

If the matrix product $[A] [B] [C]$ is defined, then $([A] [B] [C])^T$ is

- (A) $[C]^T [B]^T [A]^T$
- (B) $[A]^T [B]^T [C]^T$
- (C) $[A] [B] [C]^T$
- (D) $[A]^T [B] [C]$

? Quiz 3

The trace of a matrix

$$\begin{bmatrix} 5 & 6 & -7 \\ 9 & -11 & 13 \\ -17 & 19 & 23 \end{bmatrix}$$

is

- (A) 17
- (B) 39
- (C) 40
- (D) 110

? Quiz 4

A square $n \times n$ matrix $[A]$ is symmetric if

- (A) $a_{ij} = a_{ji}$, $i = j$ for all i, j
- (B) $a_{ij} = a_{ji}$, $i \neq j$ for all i, j
- (C) $a_{ij} = -a_{ji}$, $i = j$ for all i, j
- (D) $a_{ij} = -a_{ji}$, $i \neq j$ for all i, j

? Quiz 5

The determinant of the matrix

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & 3 & 8 \\ 0 & 9 & a \end{bmatrix}$$

is 50. The value of a is then

- (A) 0.6667
- (B) 24.67
- (C) -23.33
- (D) 5.556

? Quiz 6

$[A]$ is a 5×5 matrix and a matrix $[B]$ is obtained by the row operations of replacing *Row 1* with *Row 3*, and then *Row 3* is replaced by a linear combination of $2 \times \text{Row 3} + 4 \times \text{Row 2}$. If $\det(A) = 17$, then $\det(B)$ is equal to

- (A) 12
- (B) -34
- (C) -112
- (D) 112

Unary Matrix Operations Exercise

? Exercise 1

Let

$$[A] = \begin{bmatrix} 25 & 3 & 6 \\ 7 & 9 & 2 \end{bmatrix}.$$

Find $[A]^T$

Answer

$$\begin{bmatrix} 25 & 7 \\ 3 & 9 \\ 6 & 2 \end{bmatrix}$$

? Exercise 2

If $[A]$ and $[B]$ are two $n \times n$ symmetric matrices, show that $[A] + [B]$ is also symmetric. Hint: Let $[C] = [A] + [B]$

Answer

$$c_{ij} = a_{ij} + b_{ij} \text{ for all } i, j.$$

and

$$c_{ji} = a_{ji} + b_{ji} \text{ for all } i, j.$$

$$c_{ji} = a_{ij} + b_{ij} \text{ as } [A] \text{ and } [B] \text{ are symmetric}$$

$$\text{Hence } c_{ji} = c_{ij}.$$

? Exercise 3

Give an example of a 4×4 symmetric matrix.

? Exercise 4

Give an example of a 4×4 skew-symmetric matrix.

? Exercise 5

What is the trace of

$$\text{A. } [A] = \begin{bmatrix} 7 & 2 & 3 & 4 \\ -5 & -5 & -5 & -5 \\ 6 & 6 & 7 & 9 \\ -5 & 2 & 3 & 10 \end{bmatrix}$$

B. For

$$[A] = \begin{bmatrix} 10 & -7 & 0 \\ -3 & 2.099 & 6 \\ 5 & -1 & 5 \end{bmatrix}$$

Find the determinant of $[A]$ using the cofactor method.

Answer

a) 19

b) -150.05

? Exercise 6

$\det(3[A])$ of a $n \times n$ matrix is

A. $3\det(A)$

B. $3\det(A)$

C. $3^n \det(A)$

D. $9\det(A)$

Answer

C

? Exercise 7

For a 5×5 matrix $[A]$, the first row is interchanged with the fifth row, the determinant of the resulting matrix $[B]$ is

- A. $\det(A)$
- B. $-\det(A)$
- C. $5\det(A)$
- D. $2\det(A)$

Answer

A

? Exercise 8

$$\det \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \text{ is}$$

- A. 0
- B. 1
- C. -1
- D. ∞

Answer

C

? Exercise 9

Without using the cofactor method of finding determinants, find the determinant of

$$[A] = \begin{bmatrix} 0 & 0 & 0 \\ 2 & 3 & 5 \\ 6 & 9 & 2 \end{bmatrix}$$

Answer

0: Can you answer why?

? Exercise 10

Without using the cofactor method of finding determinants, find the determinant of

$$[A] = \begin{bmatrix} 0 & 0 & 2 & 3 \\ 0 & 2 & 3 & 5 \\ 6 & 7 & 2 & 3 \\ 6.6 & 7.7 & 2.2 & 3.3 \end{bmatrix}$$

Answer

0: Can you answer why?

? Exercise 11

Without using the cofactor method of finding determinants, find the determinant of

$$[A] = \begin{bmatrix} 5 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 2 & 5 & 6 & 0 \\ 1 & 2 & 3 & 9 \end{bmatrix}$$

Answer

$5 \times 3 \times 6 \times 9 = 810$: Can you answer why?

? Exercise 12

Given the matrix

$$[A] = \begin{bmatrix} 125 & 25 & 5 & 1 \\ 512 & 64 & 8 & 1 \\ 1157 & 89 & 13 & 1 \\ 8 & 4 & 2 & 1 \end{bmatrix}$$

and

$$\det(A) = -32400$$

find the determinant of

$$\text{A. } [A] = \begin{bmatrix} 125 & 25 & 5 & 1 \\ 512 & 64 & 8 & 1 \\ 1141 & 81 & 9 & -1 \\ 8 & 4 & 2 & 1 \end{bmatrix}$$

$$\text{B. } [A] = \begin{bmatrix} 125 & 25 & 1 & 5 \\ 512 & 64 & 1 & 8 \\ 1157 & 89 & 1 & 13 \\ 8 & 4 & 1 & 2 \end{bmatrix}$$

$$\text{C. } [B] = \begin{bmatrix} 125 & 25 & 5 & 1 \\ 1157 & 89 & 13 & 1 \\ 512 & 64 & 8 & 1 \\ 8 & 4 & 2 & 1 \end{bmatrix}$$

$$\text{D. } [C] = \begin{bmatrix} 125 & 25 & 5 & 1 \\ 1157 & 89 & 13 & 1 \\ 8 & 4 & 2 & 1 \\ 512 & 64 & 8 & 1 \end{bmatrix}$$

$$\text{E. } [D] = \begin{bmatrix} 125 & 25 & 5 & 1 \\ 512 & 64 & 8 & 1 \\ 1157 & 89 & 13 & 1 \\ 16 & 8 & 4 & 2 \end{bmatrix}$$

Answer

A) -32400 B) 32400 C) 32400 D) -32400 E) -64800

? Exercise 13

What is the transpose of

$$[A] = \begin{bmatrix} 25 & 20 & 3 & 2 \\ 5 & 10 & 15 & 25 \\ 6 & 16 & 7 & 27 \end{bmatrix}$$

Answer

$$[A]^T = \begin{bmatrix} 25 & 5 & 6 \\ 20 & 10 & 16 \\ 3 & 15 & 7 \\ 2 & 25 & 27 \end{bmatrix}$$

? Exercise 14

What values of the missing numbers will make this a skew-symmetric matrix?

$$[A] = \begin{bmatrix} 0 & 3 & ? \\ ? & 0 & ? \\ 21 & ? & 0 \end{bmatrix}$$

Answer

$$\begin{bmatrix} 0 & 3 & -21 \\ -3 & 0 & 4 \\ 21 & -4 & 0 \end{bmatrix}$$

? Exercise 15

What values of the missing number will make this a symmetric matrix?

$$[A] = \begin{bmatrix} 2 & 3 & ? \\ ? & 6 & 7 \\ 21 & ? & 5 \end{bmatrix}$$

Answer

$$\begin{bmatrix} 2 & 3 & 21 \\ 3 & 6 & 7 \\ 21 & 7 & 5 \end{bmatrix}$$

? Exercise 16

Find the determinant of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 5 \end{bmatrix}$$

Answer

The determinant of $[A]$ is,

$$25 \begin{vmatrix} 8 & 1 \\ 12 & 5 \end{vmatrix} - 5 \begin{vmatrix} 64 & 1 \\ 144 & 5 \end{vmatrix} + 1 \begin{vmatrix} 64 & 8 \\ 144 & 12 \end{vmatrix}$$

$$\begin{aligned} &= 25(28) - 5(176) + 1(-384) \\ &= -564 \end{aligned}$$

? Exercise 17

What is the determinant of an upper triangular matrix $[A]$ that is of order $n \times n$?

Answer

The determinant of an upper triangular matrix is the product of its diagonal elements, $\prod_{i=1}^n a_{ii}$

? Exercise 18

Given the determinant of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & a \end{bmatrix}$$

is -564 , find a .

Answer

$$\det(A) = -120a + 36$$

$$120a + 36 = 564$$

$$a = 5$$

? Exercise 19

Why is the determinant of the following matrix zero?

$$[A] = \begin{bmatrix} 0 & 0 & 0 \\ 2 & 3 & 5 \\ 6 & 9 & 2 \end{bmatrix}$$

Answer

The first row of the matrix is zero, hence, the determinant of the matrix is zero.

? Exercise 20

Why is the determinant of the following matrix zero?

$$[A] = \begin{bmatrix} 0 & 0 & 2 & 3 \\ 0 & 2 & 3 & 5 \\ 6 & 7 & 2 & 3 \\ 6.6 & 7.7 & 2.2 & 3.3 \end{bmatrix}$$

Answer

Row 4 of the matrix is 1.1 times Row 3. Hence, its determinant is zero.

? Exercise 21

Show that if $[A][B] = [I]$, where $[A]$, $[B]$ and $[I]$ are matrices of $n \times n$ size and $[I]$ is an identity matrix, then $\det(A) \neq 0$ and $\det(B) \neq 0$.

Answer

We know that $\det(AB) = \det(A)\det(B)$.

$$[A][B] = [I]$$

$$\det(AB) = \det(I)$$

$$\det(I) = \prod_{i=1}^n a_{ii} = \prod_{i=1}^n 1 = 1$$

$$\det(A)\det(B) = 1$$

Therefore,

$$\det(A) \neq 0 \text{ and}$$

$$\det(B) \neq 0.$$

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5: System of Equations

Learning Objectives

After reading this chapter, you should be able to:

1. setup simultaneous linear equations in matrix form and vice-versa,
2. understand the concept of the inverse of a matrix,
3. know the difference between a consistent and inconsistent system of linear equations, and
4. learn that a system of linear equations can have a unique solution, no solution or infinite solutions.

Matrix algebra is used for solving systems of equations. Can you illustrate this concept?

Matrix algebra is used to solve a system of simultaneous linear equations. In fact, for many mathematical procedures such as the solution to a set of nonlinear equations, interpolation, integration, and differential equations, the solutions reduce to a set of simultaneous linear equations. Let us illustrate with an example for interpolation.

✓ Example 1

The upward velocity of a rocket is given at three different times on the following table.

Table 5.1. Velocity vs. time data for a rocket

Time, t	Velocity, v
(s)	(m/s)
5	106.8
8	177.2
12	279.2

The velocity data is approximated by a polynomial as

$$v(t) = at^2 + bt + c, 5 \leq t \leq 12$$

Set up the equations in matrix form to find the coefficients a, b, c of the velocity profile.

Solution

The polynomial is going through three data points (t_1, v_1) , (t_2, v_2) , and (t_3, v_3) where from table 5.1.

$$t_1 = 5, v_1 = 106.8$$

$$t_2 = 8, v_2 = 177.2$$

$$t_3 = 12, v_3 = 279.2$$

Requiring that $v(t) = at^2 + bt + c$ passes through the three data points gives

$$v(t_1) = v_1 = at_1^2 + bt_1 + c$$

$$v(t_2) = v_2 = at_2^2 + bt_2 + c$$

$$v(t_3) = v_3 = at_3^2 + bt_3 + c$$

Substituting the data (t_1, v_1) , (t_2, v_2) , and (t_3, v_3) gives

$$a(5^2) + b(5) + c = 106.8$$

$$a(8^2) + b(8) + c = 177.2$$

$$a(12^2) + b(12) + c = 279.2$$

or

$$25a + 5b + c = 106.8$$

$$64a + 8b + c = 177.2$$

$$144a + 12b + c = 279.2$$

This set of equations can be rewritten in the matrix form as

$$\begin{bmatrix} 25a + & 5b + & c \\ 64a + & 8b + & c \\ 144a + & 12b + & c \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

The above equation can be written as a linear combination as follows

$$a \begin{bmatrix} 25 \\ 64 \\ 144 \end{bmatrix} + b \begin{bmatrix} 5 \\ 8 \\ 12 \end{bmatrix} + c \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

and further using matrix multiplication gives

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

The above is an illustration of why matrix algebra is needed. The complete solution to the set of equations is given later in this chapter.

A general set of m linear equations and n unknowns,

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = c_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = c_2$$

.....

.....

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = c_m$$

can be rewritten in the matrix form as

$$\begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2n} \\ \vdots & & & & \vdots \\ \vdots & & & & \vdots \\ a_{m1} & a_{m2} & \cdot & \cdot & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \cdot \\ \cdot \\ x_n \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ \cdot \\ \cdot \\ c_m \end{bmatrix}$$

Denoting the matrices by $[A]$, $[X]$, and $[C]$, the system of equation is

$[A][X] = [C]$, where $[A]$ is called the coefficient matrix, $[C]$ is called the right hand side vector and $[X]$ is called the solution vector.

Sometimes $[A][X] = [C]$ systems of equations are written in the augmented form. That is

$$[A : C] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & \vdots & c_1 \\ a_{21} & a_{22} & \dots & a_{2n} & \vdots & c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} & \vdots & c_n \end{bmatrix}$$

A system of equations can be consistent or inconsistent. What does that mean?

A system of equations $[A] [X] = [C]$ is consistent if there is a solution, and it is inconsistent if there is no solution. However, a consistent system of equations does not mean a unique solution, that is, a consistent system of equations may have a unique solution or infinite solutions (Figure 1).

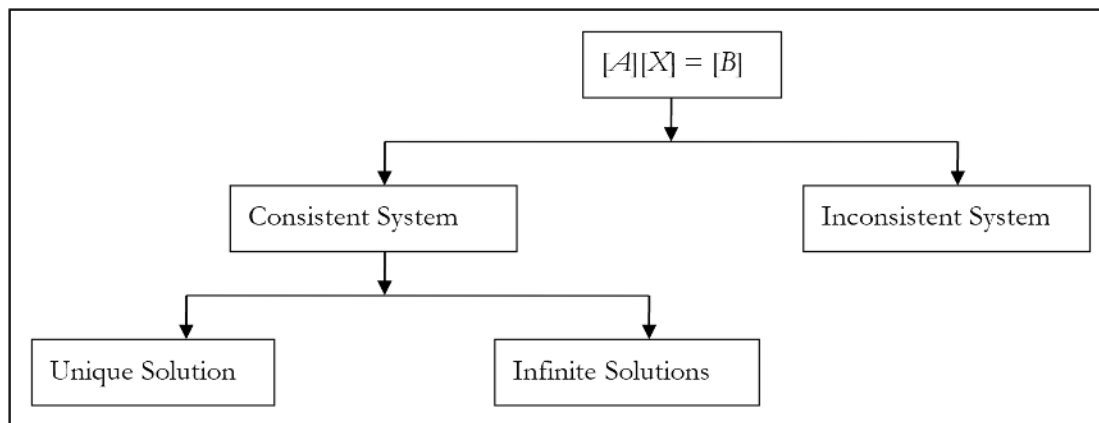


Figure 5.1: Consistent and inconsistent system of equations flow chart.

✓ Example 2

Give examples of consistent and inconsistent system of equations.

Solution

a. The system of equations

$$\begin{bmatrix} 2 & 4 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

is a consistent system of equations as it has a unique solution, that is,

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

b. The system of equations

$$\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

is also a consistent system of equations but it has infinite solutions as given as follows.

Expanding the above set of equations,

$$2x + 4y = 6$$

$$x + 2y = 3$$

you can see that they are the same equation. Hence, any combination of (x, y) that satisfies

$$2x + 4y = 6$$

is a solution. For example $(x, y) = (1, 1)$ is a solution. Other solutions include $(x, y) = (0.5, 1.25)$, $(x, y) = (0, 1.5)$, and so on.

c. The system of equations

$$\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \end{bmatrix}$$

is inconsistent as no solution exists.

How can one distinguish between a consistent and inconsistent system of equations?

A system of equations $[A] [X] = [C]$ is *consistent* if the rank of A is equal to the rank of the augmented matrix $[A : C]$.

A system of equations $[A] [X] = [C]$ is *inconsistent* if the rank of A is less than the rank of the augmented matrix $[A : C]$.

But, what do you mean by rank of a matrix?

The rank of a matrix is defined as the order of the largest square submatrix whose determinant is not zero.

✓ Example 3

What is the rank of

$$[A] = \begin{bmatrix} 3 & 1 & 2 \\ 2 & 0 & 5 \\ 1 & 2 & 3 \end{bmatrix} ?$$

Solution

The largest square submatrix possible is of order 3 and that is $[A]$ itself. Since $\det(A) = -23 \neq 0$, the rank of $[A] = 3$.

✓ Example 4

What is the rank of

$$[A] = \begin{bmatrix} 3 & 1 & 2 \\ 2 & 0 & 5 \\ 5 & 1 & 7 \end{bmatrix} ?$$

Solution

The largest square submatrix of $[A]$ is of order 3 and that is $[A]$ itself. Since $\det(A) = 0$, the rank of $[A]$ is less than 3. The next largest square submatrix would be a 2×2 matrix. One of the square submatrices of $[A]$ is

$$[B] = \begin{bmatrix} 3 & 1 \\ 2 & 0 \end{bmatrix}$$

and $\det(B) = -2 \neq 0$. Hence the rank of $[A]$ is 2. There is no need to look at other 2×2 submatrices to establish that the rank of $[A]$ is 2.

✓ Example 5

How do I now use the concept of rank to find if

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

is a consistent or inconsistent system of equations?

Solution

The coefficient matrix is

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

and the right-hand side vector is

$$[C] = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

The augmented matrix is

$$[B] = \begin{bmatrix} 25 & 5 & 1 & \vdots & 106.8 \\ 64 & 8 & 1 & \vdots & 177.2 \\ 144 & 12 & 1 & \vdots & 279.2 \end{bmatrix}$$

Since there are no square submatrices of order 4 as $[B]$ is a 3×3 matrix, the rank of $[B]$ is at most 3. So let us look at the square submatrices of $[B]$ of order 3; if any of these square submatrices have determinant not equal to zero, then the rank is 3. For example, a submatrix of the augmented matrix $[B]$ is

$$[D] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

$\text{hasdet}(D) = -84 \neq 0$.

Hence the rank of the augmented matrix $[B]$ is 3. Since $[A] = [D]$, the rank of $[A]$ is 3. Since the rank of the augmented matrix $[B]$ equals the rank of the coefficient matrix $[A]$, the system of equations is consistent.

✓ Example 6

Use the concept of rank of matrix to find if

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 89 & 13 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 284.0 \end{bmatrix}$$

is consistent or inconsistent?

Solution

The coefficient matrix is given by

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 89 & 13 & 2 \end{bmatrix}$$

and the right-hand side

$$[C] = \begin{bmatrix} 106.8 \\ 177.2 \\ 284.0 \end{bmatrix}$$

The augmented matrix is

$$[B] = \begin{bmatrix} 25 & 5 & 1 & : & 106.8 \\ 64 & 8 & 1 & : & 177.2 \\ 89 & 13 & 2 & : & 284.0 \end{bmatrix}$$

Since there are no square submatrices of order 4 as $[B]$ is a 4×3 matrix, the rank of the augmented $[B]$ is at most 3. So let us look at square submatrices of the augmented matrix $[B]$ of order 3 and see if any of these have determinants not equal to zero. For example, a square submatrix of the augmented matrix $[B]$ is

$$[D] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 89 & 13 & 2 \end{bmatrix}$$

has $\det(D) = 0$. This means, we need to explore other square submatrices of order 3 of the augmented matrix $[B]$ and find their determinants.

That is,

$$[E] = \begin{bmatrix} 5 & 1 & 106.8 \\ 8 & 1 & 177.2 \\ 13 & 2 & 284.0 \end{bmatrix}$$

$$\det(E) = 0$$

$$[F] = \begin{bmatrix} 25 & 5 & 106.8 \\ 64 & 8 & 177.2 \\ 89 & 13 & 284.0 \end{bmatrix}$$

$$\det(F) = 0$$

$$[G] = \begin{bmatrix} 25 & 1 & 106.8 \\ 64 & 1 & 177.2 \\ 89 & 2 & 284.0 \end{bmatrix}$$

$$\det(G) = 0$$

All the square submatrices of order 3×3 of the augmented matrix $[B]$ have a zero determinant. So the rank of the augmented matrix $[B]$ is less than 3. Is the rank of augmented matrix $[B]$ equal to 2?. One of the 2×2 submatrices of the augmented matrix $[B]$ is

$$[H] = \begin{bmatrix} 25 & 5 \\ 64 & 8 \end{bmatrix}$$

and

$$\det(H) = -120 \neq 0$$

So, the rank of the augmented matrix $[B]$ is 2.

Now we need to find the rank of the coefficient matrix $[A]$.

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 89 & 13 & 2 \end{bmatrix}$$

and

$$\det(A) = 0$$

So, the rank of the coefficient matrix $[A]$ is less than 3. A square submatrix of the coefficient matrix $[A]$ is

$$[J] = \begin{bmatrix} 5 & 1 \\ 8 & 1 \end{bmatrix}$$

$$\det(J) = -3 \neq 0$$

So, the rank of the coefficient matrix $[A]$ is 2.

Hence, rank of the coefficient matrix $[A]$ equals the rank of the augmented matrix $[B]$. So the system of equations $[A][X] = [C]$ is consistent.

✓ Example 7

Use the concept of rank to find if

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 89 & 13 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 280.0 \end{bmatrix}$$

is consistent or inconsistent.

Solution

The augmented matrix is

$$[B] = \begin{bmatrix} 25 & 5 & 1 & : & 106.8 \\ 64 & 8 & 1 & : & 177.2 \\ 89 & 13 & 2 & : & 280.0 \end{bmatrix}$$

Since there are no square submatrices of order 4×4 as the augmented matrix $[B]$ is a 4×3 matrix, the rank of the augmented matrix $[B]$ is at most 3. So let us look at square submatrices of the augmented matrix (B) of order 3 and see if any of the 3×3 submatrices have a determinant not equal to zero. For example, a square submatrix of order 3×3 of $[B]$

$$[D] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 89 & 13 & 2 \end{bmatrix}$$

$$\det(D) = 0$$

So, it means, we need to explore other square submatrices of the augmented matrix $[B]$

$$[E] = \begin{bmatrix} 5 & 1 & 106.8 \\ 8 & 1 & 177.2 \\ 13 & 2 & 280.0 \end{bmatrix}$$

$$\det(E) = 12.0 \neq 0.$$

So, the rank of the augmented matrix $[B]$ is 3.

The rank of the coefficient matrix $[A]$ is 2 from the previous example.

Since the rank of the coefficient matrix $[A]$ is less than the rank of the augmented matrix $[B]$, the system of equations is inconsistent. Hence, no solution exists for $[A][X] = [C]$.

If a solution exists, how do we know whether it is unique?

In a system of equations $[A][X] = [C]$ that is consistent, the rank of the coefficient matrix $[A]$ is the same as the augmented matrix $[A|C]$. If in addition, the rank of the coefficient matrix $[A]$ is same as the number of unknowns, then the solution is unique; if the rank of the coefficient matrix $[A]$ is less than the number of unknowns, then infinite solutions exist.

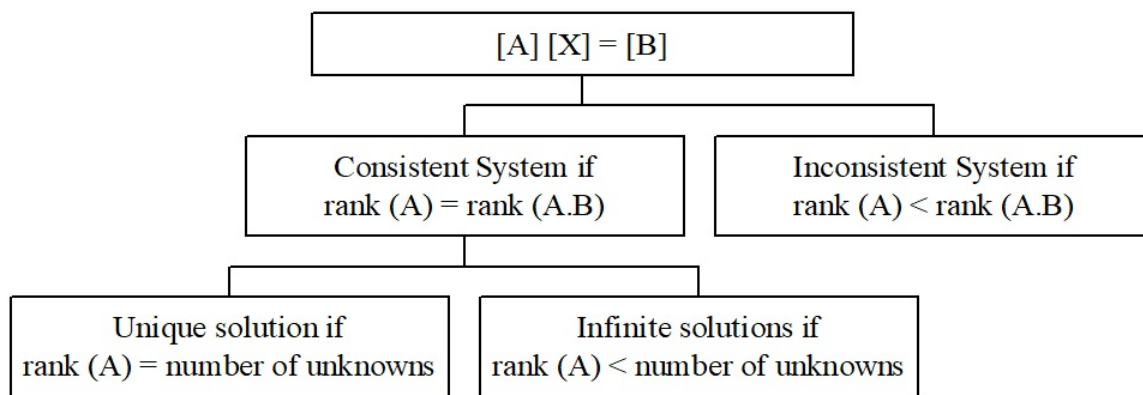


Figure 5.2: Flow chart of conditions for consistent and inconsistent system of equations.

✓ Example 8

We found that the following system of equations

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

is a consistent system of equations. Does the system of equations have a unique solution or does it have infinite solutions?

Solution

The coefficient matrix is

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

and the right-hand side is

$$[C] = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

While finding out whether the above equations were consistent in an earlier example, we found that the rank of the coefficient matrix (A) equals rank of augmented matrix $[A : C]$ equals 3.

The solution is unique as the number of unknowns = 3 = rank of (A).

✓ Example 9

We found that the following system of equations

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 89 & 13 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 284.0 \end{bmatrix}$$

is a consistent system of equations. Is the solution unique or does it have infinite solutions.

Solution

While finding out whether the above equations were consistent, we found that the rank of the coefficient matrix [A] equals the rank of augmented matrix $(A : C)$ equals 2

Since the rank of $[A] = 2 < \text{number of unknowns} = 3$, infinite solutions exist.

If we have more equations than unknowns in $[A][X] = [C]$, does it mean the system is inconsistent?

No, it depends on the rank of the augmented matrix $[A : C]$ and the rank of $[A]$.

a. For example

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \\ 89 & 13 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \\ 284.0 \end{bmatrix}$$

is consistent, since

- rank of augmented matrix = 3
- rank of coefficient matrix = 3

Now since the rank of $(A) = 3 =$ number of unknowns, the solution is not only consistent but also unique.

b. For example

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \\ 89 & 13 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \\ 280.0 \end{bmatrix}$$

is inconsistent, since

- rank of augmented matrix = 4
- rank of coefficient matrix = 3

c. For example

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 50 & 10 & 2 \\ 89 & 13 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 213.6 \\ 280.0 \end{bmatrix}$$

is consistent, since

- rank of augmented matrix = 2
- rank of coefficient matrix = 2

But since the rank of $[A] = 2 <$ the number of unknowns = 3, infinite solutions exist.

Consistent systems of equations can only have a unique solution or infinite solutions. Can a system of equations have more than one but not infinite number of solutions?

No, you can only have either a unique solution or infinite solutions. Let us suppose $[A][X] = [C]$ has two solutions $[Y]$ and $[Z]$ so that

$$[A][Y] = [C]$$

$$[A][Z] = [C]$$

If r is a constant, then from the two equations

$$r[A][Y] = r[C]$$

$$(1-r)[A][Z] = (1-r)[C]$$

Adding the above two equations gives

$$r[A][Y] + (1-r)[A][Z] = r[C] + (1-r)[C]$$

$$[A](r[Y] + (1-r)[Z]) = [C]$$

Hence

$$r[Y] + (1-r)[Z]$$

is a solution to

$$[A][X] = [C]$$

Since r is any scalar, there are infinite solutions for $[A][X] = [C]$ of the form

$$r[Y] + (1-r)[Z]$$

Can you divide two matrices?

If $[A][B] = [C]$ is defined, it might seem intuitive that $[A] = \frac{[C]}{[B]}$, but matrix division is not defined like that. However an inverse of a matrix can be defined for certain types of square matrices. The inverse of a square matrix $[A]$, if existing, is denoted by $[A]^{-1}$ such that

$$[A][A]^{-1} = [I] = [A]^{-1}[A]$$

Where $[I]$ is the identity matrix.

In other words, let $[A]$ be a square matrix. If $[B]$ is another square matrix of the same size such that $[B][A] = [I]$, then $[B]$ is the inverse of $[A]$. $[A]$ is then called to be invertible or nonsingular. If $[A]^{-1}$ does not exist, $[A]$ is called noninvertible or singular.

If $[A]$ and $[B]$ are two $n \times n$ matrices such that $[B][A] = [I]$, then these statements are also true

$[B]$ is the inverse of $[A]$

$[A]$ is the inverse of $[B]$

$[A]$ and $[B]$ are both invertible

$[A][B] = [I]$.

$[A]$ and $[B]$ are both nonsingular

all columns of $[A]$ and $[B]$ are linearly independent

all rows of $[A]$ and $[B]$ are linearly independent.

✓ Example 10

Determine if

$$[B] = \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix}$$

is the inverse of

$$[A] = \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix}$$

Solution

$$\begin{aligned} [B][A] &= \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= [I] \end{aligned}$$

Since

$$[B][A] = [I],$$

$[B]$ is the inverse of $[A]$ and $[A]$ is the inverse of $[B]$.

But, we can also show that

$$\begin{aligned}[A][B] &= \begin{bmatrix} -3 & 2 \\ 5 & -3 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 5 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= [I]\end{aligned}$$

to show that $[A]$ is the inverse of $[B]$.

Can I use the concept of the inverse of a matrix to find the solution of a set of equations $[A][X] = [C]$?

Yes, if the number of equations is the same as the number of unknowns, the coefficient matrix $[A]$ is a square matrix.

Given

$$[A][X] = [C]$$

Then, if $[A]^{-1}$ exists, multiplying both sides by $[A]^{-1}$.

$$[A]^{-1}[A][X] = [A]^{-1}[C]$$

$$[I][X] = [A]^{-1}[C]$$

$$[X] = [A]^{-1}[C]$$

This implies that if we are able to find $[A]^{-1}$, the solution vector of $[A][X] = [C]$ is simply a multiplication of $[A]^{-1}$ and the right hand side vector, $[C]$.

How do I find the inverse of a matrix?

If $[A]$ is a $n \times n$ matrix, then $[A]^{-1}$ is a $n \times n$ matrix and according to the definition of inverse of a matrix

$$[A][A]^{-1} = [I]$$

Denoting

$$\begin{aligned}[A] &= \begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n1} & a_{n2} & \cdot & \cdot & a_{nn} \end{bmatrix} \\ [A]^{-1} &= \begin{bmatrix} a'_{11} & a'_{12} & \cdot & \cdot & a'_{1n} \\ a'_{21} & a'_{22} & \cdot & \cdot & a'_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a'_{n1} & a'_{n2} & \cdot & \cdot & a'_{nn} \end{bmatrix} \\ [I] &= \begin{bmatrix} 1 & 0 & \cdot & \cdot & 0 \\ 0 & 1 & & & 0 \\ 0 & & \cdot & & \cdot \\ \cdot & & & 1 & \cdot \\ \cdot & & & \cdot & \cdot \\ 0 & \cdot & \cdot & \cdot & 1 \end{bmatrix}\end{aligned}$$

Using the definition of matrix multiplication, the first column of the $[A]^{-1}$ matrix can then be found by solving

$$\begin{bmatrix} a_{11} & a_{12} & \cdot & \cdot & a_{1n} \\ a_{21} & a_{22} & \cdot & \cdot & a_{2n} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ a_{n1} & a_{n2} & \cdot & \cdot & a_{nn} \end{bmatrix} \begin{bmatrix} a'_{11} \\ a'_{21} \\ \cdot \\ \cdot \\ a'_{n1} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \cdot \\ \cdot \\ 0 \end{bmatrix}$$

Similarly, one can find the other columns of the $[A]^{-1}$ matrix by changing the right hand side accordingly.

✓ Example 11

The upward velocity of the rocket is given by

Table 5.2. Velocity vs time data for a rocket

Time, t (s)	Velocity, v (m/s)
5	106.8
8	177.2
12	279.2

In an earlier example, we wanted to approximate the velocity profile by

$$v(t) = at^2 + bt + c, 5 \leq t \leq 12$$

We found that the coefficients a , b , and c in $v(t)$ are given by

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

First, find the inverse of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

and then use the definition of inverse to find the coefficients a , b , and c .

Solution

If

$$[A]^{-1} = \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ a'_{21} & a'_{22} & a'_{23} \\ a'_{31} & a'_{32} & a'_{33} \end{bmatrix}$$

is the inverse of $[A]$, then

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a'_{11} & a'_{12} & a'_{13} \\ a'_{21} & a'_{22} & a'_{23} \\ a'_{31} & a'_{32} & a'_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

gives three sets of equations

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a'_{11} \\ a'_{21} \\ a'_{31} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a'_{12} \\ a'_{22} \\ a'_{32} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a'_{13} \\ a'_{23} \\ a'_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Solving the above three sets of equations separately gives

$$\begin{bmatrix} a'_{11} \\ a'_{21} \\ a'_{31} \end{bmatrix} = \begin{bmatrix} 0.04762 \\ -0.9524 \\ 4.571 \end{bmatrix}$$

$$\begin{bmatrix} a'_{12} \\ a'_{22} \\ a'_{32} \end{bmatrix} = \begin{bmatrix} -0.08333 \\ 1.417 \\ -5.000 \end{bmatrix}$$

$$\begin{bmatrix} a'_{13} \\ a'_{23} \\ a'_{33} \end{bmatrix} = \begin{bmatrix} 0.03571 \\ -0.4643 \\ 1.429 \end{bmatrix}$$

Hence

$$[A]^{-1} = \begin{bmatrix} 0.04762 & -0.08333 & 0.03571 \\ -0.9524 & 1.417 & -0.4643 \\ 4.571 & -5.000 & 1.429 \end{bmatrix}$$

Now

$$[A][X] = [C]$$

where

$$[X] = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$[C] = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

Using the definition of $[A]^{-1}$,

$$[A]^{-1}[A][X] = [A]^{-1}[C]$$

$$[X] = [A]^{-1}[C]$$

$$\begin{bmatrix} 0.04762 & -0.08333 & 0.03571 \\ -0.9524 & 1.417 & -0.4643 \\ 4.571 & -5.000 & 1.429 \end{bmatrix} \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

Hence

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0.2905 \\ 19.69 \\ 1.086 \end{bmatrix}$$

So

$$v(t) = 0.2905t^2 + 19.69t + 1.086, 5 \leq t \leq 12$$

Is there another way to find the inverse of a matrix?

For finding the inverse of small matrices, the inverse of an invertible matrix can be found by

$$[A]^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

where

$$\text{adj}(A) = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \cdots & C_{nn} \end{bmatrix}^T$$

where C_{ij} are the cofactors of a_{ij} . The matrix

$$\begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1n} \\ C_{21} & C_{22} & \cdots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & \cdots & \cdots & C_{nn} \end{bmatrix}$$

itself is called the matrix of cofactors from $[A]$. Cofactors are defined in Chapter 4.

✓ Example 12

Find the inverse of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

Solution

From Example 4.6 in Chapter 04.06, we found

$$\det(A) = -84$$

Next we need to find the adjoint of $[A]$. The cofactors of A are found as follows.

The minor of entry a_{11} is

$$\begin{aligned} M_{11} &= \begin{vmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{vmatrix} \\ &= \begin{vmatrix} 8 & 1 \\ 12 & 1 \end{vmatrix} \\ &= -4 \end{aligned}$$

The cofactors of entry a_{11} is

$$\begin{aligned} C_{11} &= (-1)^{1+1} M_{11} \\ &= M_{11} \\ &= -4 \end{aligned}$$

The minor of entry a_{12} is

$$\begin{aligned}
 M_{12} &= \begin{vmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{vmatrix} \\
 &= \begin{vmatrix} 64 & 1 \\ 144 & 1 \end{vmatrix} \\
 &= -80
 \end{aligned}$$

The cofactor of entry a_{12} is

$$\begin{aligned}
 C_{12} &= (-1)^{1+2} M_{12} \\
 &= -M_{12} \\
 &= -(-80) \\
 &= 80
 \end{aligned}$$

Similarly

$$C_{13} = -384$$

$$C_{21} = 7$$

$$C_{22} = -119$$

$$C_{23} = 420$$

$$C_{31} = -3$$

$$C_{32} = 39$$

$$C_{33} = -120$$

Hence the matrix of cofactors of $[A]$ is

$$[C] = \begin{bmatrix} -4 & 80 & -384 \\ 7 & -119 & 420 \\ -3 & 39 & -120 \end{bmatrix}$$

The adjoint of matrix $[A]$ is $[C]^T$,

$$\begin{aligned}
 adj(A) &= [C]^T \\
 &= \begin{bmatrix} -4 & 7 & -3 \\ 80 & -119 & 39 \\ -384 & 420 & -120 \end{bmatrix}
 \end{aligned}$$

Hence

$$\begin{aligned}
 [A]^{-1} &= \frac{1}{\det(A)} adj(A) \\
 &= \frac{1}{-84} \begin{bmatrix} -4 & 7 & -3 \\ 80 & -119 & 39 \\ -384 & 420 & -120 \end{bmatrix} \\
 &= \begin{bmatrix} 0.04762 & -0.08333 & 0.03571 \\ -0.9524 & 1.417 & -0.4643 \\ 4.571 & -5.000 & 1.429 \end{bmatrix}
 \end{aligned}$$

If the inverse of a square matrix $[A]$ exists, is it unique?

Yes, the inverse of a square matrix is unique, if it exists. The proof is as follows. Assume that the inverse of $[A]$ is $[B]$ and if this inverse is not unique, then let another inverse of $[A]$ exist called $[C]$.

If $[B]$ is the inverse of $[A]$, then

$$[B] [A] = [I]$$

Multiply both sides by $[C]$,

$$[B] [A] [C] = [I] [C]$$

$$[B] [A] [C] = [C]$$

Since $[C]$ is inverse of $[A]$,

$$[A] [C] = [I]$$

Multiply both sides by $[B]$,

$$[B] [I] = [C]$$

$$[B] = [C]$$

This shows that $[B]$ and $[C]$ are the same. So the inverse of $[A]$ is unique.

System of Equations Quiz

? Quiz 1

A 3×4 matrix can have a rank of at most

- (A) 3
- (B) 4
- (C) 5
- (D) 12

? Quiz 2

Three kids – Jim, Corey and David receive an inheritance of \$2,253,453. The money is put in three trusts but is not divided equally to begin with. Corey gets three times what David gets because Corey made an “A” in Dr. Kaw’s class. Each trust is put in an interest generating investment. The three trusts of Jim, Corey and David pay an interest of 6%, 8%, 11%, respectively. The total interest of all the three trusts combined at the end of the first year is \$190,740.57. How much money was invested in each trust? The equations in a matrix form are

$$(A) \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \\ .06 & .08 & .11 \end{bmatrix} \begin{bmatrix} J \\ C \\ D \end{bmatrix} = \begin{bmatrix} 2253543 \\ 0 \\ 190740.57 \end{bmatrix}$$

$$(B) \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ .06 & .08 & .11 \end{bmatrix} \begin{bmatrix} J \\ C \\ D \end{bmatrix} = \begin{bmatrix} 2253543 \\ 0 \\ 190740.57 \end{bmatrix}$$

$$(C) \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ 6 & 8 & 11 \end{bmatrix} \begin{bmatrix} J \\ C \\ D \end{bmatrix} = \begin{bmatrix} 2253543 \\ 0 \\ 190740.57 \end{bmatrix}$$

$$(D) \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -3 \\ .06 & .08 & .11 \end{bmatrix} \begin{bmatrix} J \\ C \\ D \end{bmatrix} = \begin{bmatrix} 2253543 \\ 0 \\ 190740.57 \end{bmatrix}$$

? Quiz 3

Which of the following matrices does not have an inverse?

(A) $\begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix}$

(B) $\begin{bmatrix} 6 & 7 \\ 12 & 14 \end{bmatrix}$

(C) $\begin{bmatrix} 6 & 0 \\ 0 & 7 \end{bmatrix}$

(D) $\begin{bmatrix} 0 & 6 \\ 7 & 0 \end{bmatrix}$

? Quiz 4

The set of equations

$$\begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & 7 \\ 5 & 8 & 19 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 18 \\ 26 \\ 70 \end{bmatrix}$$

has

- (A) no solution
- (B) finite number of solutions
- (C) a unique solution
- (D) infinite solutions

? Quiz 5

Given a system of $[A][X] = [C]$ where $[A]$ is $n \times n$ matrix and $[X]$ and $[C]$ are $n \times 1$ matrices, a unique solution $[X]$ exists if

- (A) rank of $[A] = \text{rank of } \begin{bmatrix} A \\ C \end{bmatrix}$
- (B) rank of $[A] = \text{rank of } \begin{bmatrix} A \\ C \end{bmatrix} = n$
- (C) rank of $[A] < \text{rank of } \begin{bmatrix} A \\ C \end{bmatrix}$
- (D) rank of $[A] = \text{rank of } \begin{bmatrix} A \\ C \end{bmatrix} < n$

? Quiz 6

If $[A][X] = \begin{bmatrix} -13 \\ 76 \\ 38 \end{bmatrix}$ and $[A]^{-1} = \begin{bmatrix} 1 & 2 & -4 \\ -8 & 2 & 16 \\ 2 & 4 & 8 \end{bmatrix}$ then

(A) $[X] = \begin{bmatrix} -13.000 \\ 864.00 \\ 582.00 \end{bmatrix}$

(B) one cannot find a unique $[X]$.

$$(C) [X] = \begin{bmatrix} -1.0000 \\ 2.0000 \\ 4.0000 \end{bmatrix}$$

(D) no solutions of $[X]$ are possible

System of Equations Exercise

? Exercise 1

For a set of equations $[A] [X] = [B]$, a unique solution exists if

- A. $\text{rank}(A) = \text{rank} \begin{pmatrix} A & B \end{pmatrix}$
- B. $\text{rank}(A) = \text{rank} \begin{pmatrix} A & B \end{pmatrix}$ and $\text{rank}(A) = \text{number of unknowns}$
- C. $\text{rank}(A) = \text{rank} \begin{pmatrix} A & B \end{pmatrix}$ and $\text{rank}(A) = \text{number of rows of } (A)$.

Answer

B

? Exercise 2

The rank of matrix

$$A = \begin{bmatrix} 4 & 4 & 4 & 4 \\ 4 & 4 & 4 & 4 \\ 4 & 4 & 4 & 4 \\ 4 & 4 & 4 & 4 \end{bmatrix} \text{ is}$$

- A. 1
- B. 2
- C. 3
- D. 4

? Exercise 3

A 3×4 matrix can have a rank of at most

- A. 3
- B. 4
- C. 5
- D. 12

? Exercise 4

If $[A] [X] = [C]$ has a unique solution, where the order of $[A]$ is 3×3 , $[X]$ is 3×1 , then the rank of $[A]$ is

- A. 2
- B. 3
- C. 4
- D. 5

? Exercise 5

Show if the following system of equations is consistent or inconsistent. If they are consistent, determine if the solution would be unique or infinite ones exist.

$$\begin{bmatrix} 1 & 2 & 5 \\ 7 & 3 & 9 \\ 8 & 5 & 14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 19 \\ 27 \end{bmatrix}$$

Answer

Consistent; Infinite solutions

? Exercise 6

Show if the following system of equations is consistent or inconsistent. If they are consistent, determine if the solution would be unique or infinite ones exist.

$$\begin{bmatrix} 1 & 2 & 5 \\ 7 & 3 & 9 \\ 8 & 5 & 14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 19 \\ 28 \end{bmatrix}$$

Answer

Inconsistent

? Exercise 7

Show if the following system of equations is consistent or inconsistent. If they are consistent, determine if the solution would be unique or infinite ones exist.

$$\begin{bmatrix} 1 & 2 & 5 \\ 7 & 3 & 9 \\ 8 & 5 & 13 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 19 \\ 28 \end{bmatrix}$$

Answer

Consistent; Unique

? Exercise 8

The set of equations

$$\begin{bmatrix} 1 & 2 & 5 \\ 7 & 3 & 9 \\ 8 & 5 & 14 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 19 \\ 27 \end{bmatrix}$$

has

- A. Unique solution
- B. No solution
- C. Infinite solutions

Answer

C

? Exercise 9

For what values of a will the following equation have

$$x_1 + x_2 + x_3 = 4$$

$$x_3 = 2$$

$$(a^2 - 4)x_1 + x_3 = a - 2$$

- A. Unique solution
- B. No solution
- C. Infinite solutions

Answer

If $a \neq +2$ or -2 , then there will be a unique solution. If $a = +2$ or -2 , then there will be no solution. Possibility of infinite solutions does not exist.

? Exercise 10

Find if

$$[A] = \begin{bmatrix} 5 & -2.5 \\ -2 & 3 \end{bmatrix}$$

and

$$[B] = \begin{bmatrix} 0.3 & 0.25 \\ 0.2 & 0.5 \end{bmatrix}$$

are inverse of each other.

Answer

Yes

? Exercise 11

Find if

$$[A] = \begin{bmatrix} 5 & 2.5 \\ 2 & 3 \end{bmatrix}$$

and

$$[B] = \begin{bmatrix} 0.3 & -0.25 \\ 0.2 & 0.5 \end{bmatrix}$$

are inverse of each other.

Answer

No

? Exercise 12

Find the

- A. cofactor matrix
- B. adjoint matrix

of

$$[A] = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{bmatrix}$$

Answer

$$\begin{bmatrix} -34 & -18 & 58 \\ -19 & 7 & 29 \\ 3 & 5 & -29 \end{bmatrix} \begin{bmatrix} -34 & -19 & 3 \\ -18 & 7 & 5 \\ 58 & 29 & -29 \end{bmatrix}$$

? Exercise 13

Find $[A]^{-1}$ using any method for

$$[A] = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{bmatrix}$$

Answer

$$[A]^{-1} = \begin{bmatrix} 2.931 \times 10^{-1} & 1.638 \times 10^{-1} & -2.586 \times 10^{-2} \\ 1.552 \times 10^{-1} & -6.034 \times 10^{-2} & -4.310 \times 10^{-2} \\ -5.000 \times 10^{-1} & -2.500 \times 10^{-1} & 2.500 \times 10^{-1} \end{bmatrix}$$

? Exercise 14

Prove that if $[A]$ and $[B]$ are both invertible and are square matrices of same order, then

$$([A][B])^{-1} = [B]^{-1}[A]^{-1}$$

Answer

$$([A][B])^{-1} = [B]^{-1}[A]^{-1}$$

Let

$$[C] = [A][B]$$

$$\begin{aligned} [C][B]^{-1} &= [A][B][B]^{-1} \\ &= [A][I] \\ &= [A] \end{aligned}$$

Again

$$[C] = [A][B]$$

$$\begin{aligned} [A]^{-1}[C] &= [A]^{-1}[A][B] \\ &= [I][B] \\ &= [B] \end{aligned}$$

So

$$[C][B]^{-1} = [A]; \quad (1)$$

$$[A]^{-1}[C] = [B]; \quad (2)$$

From (1) and (2)

$$[C][B]^{-1}[A]^{-1}[C] = [A][B]$$

$$\begin{aligned}
 [A] [B] [B]^{-1} [A]^{-1} [A] [B] &= [A] [B] \\
 [A]^{-1} [A] [B] [B]^{-1} [A]^{-1} [A] [B] &= [A^{-1}] [A] [B] \\
 [B] [B]^{-1} [A^{-1}] [A] [B] &= [B] \\
 [B^{-1}] [B] [B]^{-1} [A^{-1}] [A] [B] &= [B]^{-1} [B] \\
 [B]^{-1} [A^{-1}] [A] [B] &= [I].
 \end{aligned}$$

? Exercise 15

What is the inverse of a square diagonal matrix? Does it always exist?

Answer

Hint: Inverse of a square $n \times n$ diagonal matrix $[A]$ is $[A]^{-1} =$

$$\begin{bmatrix} \frac{1}{a_{11}} & 0 & \dots & 0 \\ 0 & \frac{1}{a_{22}} & \dots & 0 \\ 0 & & & \vdots \\ \vdots & \dots & \dots & \frac{1}{a_{nn}} \end{bmatrix}$$

So inverse exists only if $a_{ii} \neq 0$ for all i .

? Exercise 16

$[A]$ and $[B]$ are square matrices. If $[A] [B] = [0]$ and $[A]$ is invertible, show $[B] = 0$.

Answer

$$\begin{aligned}
 [A] [B] &= [0] \\
 [A^{-1}] [A] [B] &= [A^{-1}] [0]
 \end{aligned}$$

? Exercise 17

If $[A] [B] [C] = [I]$, where $[A]$, $[B]$ and $[C]$ are of the same size, show that $[B]$ is invertible.

Answer

Hint: $\det(AB) = \det(A)\det(B)$

? Exercise 18

Prove if $[B]$ is invertible, $[A] [B]^{-1} = [B]^{-1} [A]$ if and only if $[A] [B] = [B] [A]$

Answer

Hint: Multiply by $[B]^{-1}$ on both sides, $[A] [B] [B]^{-1} = [B]^{-1} [A] [B]^{-1}$

? Exercise 19

For

$$[A] = \begin{bmatrix} 10 & -7 & 0 \\ -3 & 2.099 & 6 \\ 5 & -1 & 5 \end{bmatrix}$$

$$[A]^{-1} = \begin{bmatrix} -0.1099 & -0.2333 & 0.2799 \\ -0.2999 & -0.3332 & 0.3999 \\ 0.04995 & 0.1666 & 6.664 \times 10^{-5} \end{bmatrix}$$

Show

$$\det(A) = \frac{1}{\det(A^{-1})}.$$

? Exercise 20

For what values of a does the linear system have

$$\begin{aligned} x + y &= 2 \\ 6x + 6y &= a \end{aligned}$$

- A. infinite solutions
- B. unique solution

Answer

- A. 12
- B. not possible

? Exercise 21

Three kids - Jim, Corey and David receive an inheritance of \$2, \$253, \$453 The money is put in three trusts but is not divided equally to begin with. Corey gets three times more than David because Corey made an "A" in Dr. Kaw's class. Each trust is put in an interest generating investment. The three trusts of Jim, Corey and David pays an interest of 6%, 8%, 11%, respectively. The total interest of all the three trusts combined at the end of the first year is \$190, \$740.57 How much money was invested in each trust? Set the following as equations in a matrix form. Identify the unknowns. Do not solve for the unknowns.

Answer

$$J + C + D = \$2, \$253, \$453$$

$$C = 3D$$

$$0.06J + 0.08C + 0.11D = \$190, 740.57$$

In matrix form

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & -3 \\ 0.06 & 0.08 & 0.11 \end{bmatrix} \begin{bmatrix} J \\ C \\ D \end{bmatrix} = \begin{bmatrix} 2, 253, 453 \\ 0 \\ 190, 740.57 \end{bmatrix}$$

? Exercise 22

What is the rank of

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 6 & 7 \\ 6 & 10 & 13 \end{bmatrix} ?$$

Justify your answer.

Answer

In the above matrix, $2(\text{Row } 1) + \text{Row } 2 = \text{Row } 3$. Hence, rank is less than 3. Row 1 and Row 2 are linearly independent. Hence, the rank of the matrix is 2.

? Exercise 23

What is the rank of

$$\begin{bmatrix} 1 & 2 & 3 & 6 \\ 4 & 6 & 7 & 17 \\ 6 & 10 & 13 & 29 \end{bmatrix} ?$$

Justify your answer.

Answer

The determinant of all the 3×3 sub-matrices is zero. Hence, the rank is less than 3. Determinant of

$$\begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix} = -4 \neq 0.$$

? Exercise 24

What is the rank of

$$\begin{bmatrix} 1 & 2 & 3 & 6 \\ 4 & 6 & 7 & 18 \\ 6 & 10 & 13 & 30 \end{bmatrix} ?$$

Justify your answer.

Answer

In the above matrix, $2(\text{Row } 1) + \text{Row } 2 = \text{Row } 3$. Hence, rank is less than 3 as the 3 rows are linearly dependent. Determinant of

$$\begin{bmatrix} 2 & 3 \\ 6 & 7 \end{bmatrix} = -4 \neq 0.$$

Hence, the rank is 2.

? Exercise 25

How many solutions does the following system of equations have

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 6 & 7 \\ 6 & 10 & 13 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \\ 29 \end{bmatrix} ?$$

Justify your answer.

Answer

Rank of $A = 2$ \ Rank of $A|C = 2$ \ Number of unknowns = 3. \ There are infinite solutions since rank of A is less than the number of unknowns.

? Exercise 26

How many solutions does the following system of equations have

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 6 & 7 \\ 6 & 10 & 13 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 6 \\ 18 \\ 30 \end{bmatrix} ?$$

Justify your answer.

Answer

Rank of $A = 2$ \ Rank of $A|C = 2$ \ Number of unknowns = 3. \ There are infinite solutions since rank of A is less than the number of unknowns.

? Exercise 27

By any scientific method, find the second column of the inverse of

$$\begin{bmatrix} 1 & 2 & 0 \\ 4 & 5 & 0 \\ 0 & 0 & 13 \end{bmatrix} .$$

Answer

$$\begin{bmatrix} 1 & 2 & 0 \\ 4 & 5 & 0 \\ 0 & 0 & 13 \end{bmatrix} \begin{bmatrix} X & a'_{12} & X \\ X & a'_{22} & X \\ X & a'_{32} & X \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$a'_{12} + 2a'_{22} = 0$$

$$4a'_{12} + 5a'_{22} = 1$$

$$13a'_{32} = 0$$

Simplifying,

$$\begin{bmatrix} a'_{12} \\ a'_{22} \\ a'_{32} \end{bmatrix} = \begin{bmatrix} 0.667 \\ -0.333 \\ 0 \end{bmatrix}$$

? Exercise 28

Just write out the inverse of (no need to show any work)

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

Answer

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{4} & 0 \\ 0 & 0 & 0 & \frac{1}{5} \end{bmatrix}$$

? Exercise 29

Solve $[A][X] = [B]$ for $[X]$ if

$$[A]^{-1} = \begin{bmatrix} 10 & -7 & 0 \\ 2 & 2 & 5 \\ 2 & 0 & 6 \end{bmatrix}$$

and

$$[B] = \begin{bmatrix} 7 \\ 2.5 \\ 6.012 \end{bmatrix}$$

Answer

$$\begin{aligned} [X] &= [A]^{-1}[B] = \begin{bmatrix} 10 & -7 & 0 \\ 2 & 2 & 5 \\ 2 & 0 & 6 \end{bmatrix} \begin{bmatrix} 7 \\ 2.5 \\ 6.012 \end{bmatrix} \\ &= \begin{bmatrix} 52.5 \\ 49.06 \\ 50.072 \end{bmatrix} \end{aligned}$$

? Exercise 30

Let $[A]$ be a 3×3 matrix. Suppose

$$[X] = \begin{bmatrix} 7 \\ 2.5 \\ 6.012 \end{bmatrix}$$

is a solution to the homogeneous set of equations $[A][X] = [0]$ (the right hand side is a zero vector of order 3×1). Does $[A]$ have an inverse? Justify your answer.

Answer

Given

$$[A][X] = [0]$$

If $[A]^{-1}$ exists, then

$$[A]^{-1}[A][X] = [A]^{-1}[0]$$

$$[I][X] = [0]$$

$$[X] = [0]$$

This contradicts the given value of $[X]$. Hence, $[A]^{-1}$ does not exist.

? Exercise 31

Is the set of vectors

$$\vec{A} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \vec{B} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}, \vec{C} = \begin{bmatrix} 1 \\ 4 \\ 25 \end{bmatrix}$$

linearly independent? Justify your answer.

Answer

The set of vectors are linearly independent.

? Exercise 32

What is the rank of the set of vectors

$$\vec{A} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \vec{B} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}, \vec{C} = \begin{bmatrix} 1 \\ 3 \\ 6 \end{bmatrix}?$$

Justify your answer.

Answer

Since, the 3 vectors are linearly independent as proved above, the rank of the 3 vectors is 3.

? Exercise 33

What is the rank of

$$\vec{A} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \vec{B} = \begin{bmatrix} 2 \\ 2 \\ 4 \end{bmatrix}, \vec{C} = \begin{bmatrix} 3 \\ 3 \\ 5 \end{bmatrix}?$$

Justify your answer.

Answer

By inspection, $\vec{C} = \vec{A} + \vec{B}$. Hence, the 3 vectors are linearly dependent, and the rank is less than 3. Linear combination of $\vec{A}$ and $\vec{B}$, that is, $K_1 \vec{A} + K_2 \vec{B} = 0$ has only one solution $K_1 = K_2 = 0$. Therefore, the rank is 2.

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6: Gaussian Elimination Method for Solving Simultaneous Linear Equations

Learning Objectives

After successful completion of this lesson, you should be able to

1. write the algorithm to solve a set of simultaneous linear equations using Naïve Gauss elimination method
2. solve a set of simultaneous linear equations using Naïve Gauss elimination.
3. use the forward elimination steps of Gauss elimination method to find determinant of a square matrix,
4. enumerate theorems related to determinant of matrices,
5. relate the zero and non-zero value of the determinant of a square matrix to the existence or non-existence of the matrix inverse.
6. enumerate the pitfalls of the Naïve Gauss elimination method
7. show the pitfalls of Naïve Gauss elimination method through examples
8. write the algorithm to solve a set of simultaneous linear equations using Gaussian elimination with Partial Pivoting.
9. solve a set of simultaneous linear equations using Gauss elimination method with partial pivoting

How is a set of equations solved numerically by Gaussian elimination method?

One of the most popular techniques for solving simultaneous linear equations is the Gaussian elimination method. The approach is designed to solve a general set of n equations and n unknowns

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n &= b_2 \\ \vdots & \\ a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \dots + a_{nn}x_n &= b_n \end{aligned}$$

Gaussian elimination consists of two steps

1. Forward Elimination of Unknowns: In this step, the unknown is eliminated in each equation starting with the first equation. This way, the equations are *reduced* to one equation and one unknown in each equation.
2. Back Substitution: In this step, starting from the last equation, each of the unknowns is found.

Forward Elimination of Unknowns:

In the first step of forward elimination, the first unknown, x_1 is eliminated from all rows below the first row. The first equation is selected as the pivot equation to eliminate x_1 . So, to eliminate x_1 in the second equation, one divides the first equation by a_{11} (hence called the pivot element) and then multiplies it by a_{21} . This is the same as multiplying the first equation by a_{21}/a_{11} to give

$$a_{21}x_1 + \frac{a_{21}}{a_{11}}a_{12}x_2 + \dots + \frac{a_{21}}{a_{11}}a_{1n}x_n = \frac{a_{21}}{a_{11}}b_1$$

Now, this equation can be subtracted from the second equation to give

$$\left(a_{22} - \frac{a_{21}}{a_{11}}a_{12}\right)x_2 + \dots + \left(a_{2n} - \frac{a_{21}}{a_{11}}a_{1n}\right)x_n = b_2 - \frac{a_{21}}{a_{11}}b_1$$

or

$$a'_{22}x_2 + \dots + a'_{2n}x_n = b'_2$$

where

$$\begin{aligned} a'_{22} &= a_{22} - \frac{a_{21}}{a_{11}}a_{12} \\ &\vdots \\ a'_{2n} &= a_{2n} - \frac{a_{21}}{a_{11}}a_{1n} \end{aligned}$$

This procedure of eliminating x_1 , is now repeated for the third equation to the n^{th} equation to reduce the set of equations as

$$\begin{aligned}
 a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n &= b_1 \\
 a'_{22}x_2 + a'_{23}x_3 + \dots + a'_{2n}x_n &= b'_2 \\
 a'_{32}x_2 + a'_{33}x_3 + \dots + a'_{3n}x_n &= b'_3 \\
 \vdots & \\
 a'_{n2}x_2 + a'_{n3}x_3 + \dots + a'_{nn}x_n &= b'_n
 \end{aligned}$$

This is the end of the first step of forward elimination. Now for the second step of forward elimination, we start with the second equation as the pivot equation and a'_{22} as the pivot element. So, to eliminate x_2 in the third equation, one divides the second equation by a'_{22} (the pivot element) and then multiply it by a'_{32} . This is the same as multiplying the second equation by a'_{32}/a'_{22} and subtracting it from the third equation. This makes the coefficient of x_2 zero in the third equation. The same procedure is now repeated for the fourth equation till the n^{th} equation to give

$$\begin{aligned}
 a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n &= b_1 \\
 a'_{22}x_2 + a'_{23}x_3 + \dots + a'_{2n}x_n &= b'_2 \\
 a''_{33}x_3 + \dots + a''_{3n}x_n &= b''_3 \\
 \vdots & \\
 a''_{n3}x_3 + \dots + a''_{nn}x_n &= b''_n
 \end{aligned}$$

The next steps of forward elimination are conducted by using the third equation as a pivot equation and so on. That is, there will be a total of $n - 1$ steps of forward elimination. At the end of $n - 1$ steps of forward elimination, we get a set of equations that look like

$$\begin{aligned}
 a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n &= b_1 \\
 a'_{22}x_2 + a'_{23}x_3 + \dots + a'_{2n}x_n &= b'_2 \\
 a''_{33}x_3 + \dots + a''_{3n}x_n &= b''_3 \\
 \vdots & \\
 a^{(n-1)}_{nn}x_n &= b^{(n-1)}_n
 \end{aligned}$$

Back Substitution:

Now the equations are solved starting from the last equation as it has only one unknown.

$$x_n = \frac{b^{(n-1)}_n}{a^{(n-1)}_{nn}}$$

Then the second last equation, that is the $(n - 1)^{\text{th}}$ equation, has two unknowns: x_n and x_{n-1} , but x_n is already known. This reduces the $(n - 1)^{\text{th}}$ equation also to one unknown. Back substitution hence can be represented for all equations by the formula

$$x_i = \frac{b^{(i-1)}_i - \sum_{j=i+1}^n a^{(i-1)}_{ij} x_j}{a^{(i-1)}_{ii}} \text{ for } i = n - 1, n - 2, \dots, 1$$

and

$$x_n = \frac{b^{(n-1)}_n}{a^{(n-1)}_{nn}}$$

✓ Example 1

The upward velocity of a rocket is given at three different times in Table 1.

Table 1: Velocity vs. time data.

Time, t (s)	Velocity, v (m/s)
5	106.8

Time, t (s)	Velocity, v (m/s)
8	177.2
12	279.2

The velocity data is approximated by a polynomial as

$$v(t) = a_1 t^2 + a_2 t + a_3, 5 \leq t \leq 12$$

The coefficients a_1 , a_2 , and a_3 for the above expression are given by

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

Find the values of a_1 , a_2 , and a_3 using the Naïve Gauss elimination method. Find the velocity at $t = 6, 7.5, 9, 11$ seconds.

Solution

The augmented matrix is

$$\left[\begin{array}{ccc|c} 25 & 5 & 1 & 106.8 \\ 64 & 8 & 1 & 177.2 \\ 144 & 12 & 1 & 279.2 \end{array} \right]$$

Forward Elimination of Unknowns

Since there are three equations, there will be two steps of forward elimination of unknowns.

First step

Divide Row 1 by 25 and then multiply it by 64, that is, multiply Row 1 by $64/25 = 2.56$

$$\begin{aligned} &([25 \quad 5 \quad 1 \quad | \quad 106.8]) \times 2.56 \text{ gives Row 1 as} \\ &[64 \quad 12.8 \quad 2.56 \quad | \quad 273.408] \end{aligned}$$

Subtract the result from Row 2

$$\begin{array}{r} [64 \quad 8 \quad 1 \quad | \quad 177.2] \\ - [64 \quad 12.8 \quad 2.56 \quad | \quad 273.408] \\ \hline 0 \quad -4.8 \quad -1.56 \quad -96.208 \end{array}$$

to get the resulting equations as

$$\left[\begin{array}{ccc|c} 25 & 5 & 1 & 106.8 \\ 0 & -4.8 & -1.56 & -96.208 \\ 144 & 12 & 1 & 279.2 \end{array} \right]$$

Divide Row 1 by 25 and then multiply it by 144, that is, multiply Row 1 by $144/25 = 5.76$

$$\begin{aligned} &([25 \quad 5 \quad 1 \quad | \quad 106.8]) \times 5.76 \text{ gives Row 1 as} \\ &[144 \quad 28.8 \quad 5.76 \quad | \quad 615.168] \end{aligned}$$

Subtract the result from Row 3

$$\begin{array}{r} [144 \quad 12 \quad 1 \quad | \quad 279.2] \\ - [144 \quad 28.8 \quad 5.76 \quad | \quad 615.168] \\ \hline 0 \quad -16.8 \quad -4.76 \quad -335.968 \end{array}$$

to get the resulting equations as

$$\left[\begin{array}{ccc|c} 25 & 5 & 1 & 106.8 \\ 0 & -4.8 & -1.56 & -96.208 \\ 0 & -16.8 & -4.76 & -335.968 \end{array} \right]$$

Second step

We now divide Row 2 by -4.8 and then multiply by -16.8 , that is, multiply Row 2 by $-16.8 / -4.8 = 3.5$.

$$([0 \ -4.8 \ -1.56 \ | \ -96.208]) \times 3.5 \text{ gives Row 2 as}$$

$$[0 \ -16.8 \ -5.46 \ | \ -336.728]$$

Subtract the result from Row 3

$$\begin{array}{r} [0 \ -16.8 \ -4.76 \ | \ -335.968] \\ - [0 \ -16.8 \ -5.46 \ | \ -336.728] \\ \hline 0 \quad 0 \quad 0.7 \quad 0.76 \end{array}$$

to get the resulting equations as

$$\left[\begin{array}{ccc|c} 25 & 5 & 1 & 106.8 \\ 0 & -4.8 & -1.56 & -96.208 \\ 0 & 0 & 0.7 & 0.76 \end{array} \right]$$

$$\left[\begin{array}{ccc} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{array} \right] \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ -96.208 \\ 0.76 \end{bmatrix}$$

Back substitution

From the third equation

$$0.7a_3 = 0.76$$

$$a_3 = \frac{0.76}{0.7}$$

$$= 1.08571$$

Substituting the value of a_3 in the second equation,

$$-4.8a_2 - 1.56a_3 = -96.208$$

$$a_2 = \frac{-96.208 + 1.56a_3}{-4.8}$$

$$= \frac{-96.208 + 1.56 \times 1.08571}{-4.8}$$

$$= 19.6905$$

Substituting the value of a_2 and a_3 in the first equation,

$$25a_1 + 5a_2 + a_3 = 106.8$$

$$a_1 = \frac{106.8 - 5a_2 - a_3}{25}$$

$$= \frac{106.8 - 5 \times 19.6905 - 1.08571}{25}$$

$$= 0.290472$$

Hence the solution vector is

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0.290472 \\ 19.6905 \\ 1.08571 \end{bmatrix}$$

The polynomial that passes through the three data points is then

$$\begin{aligned} v(t) &= a_1 t^2 + a_2 t + a_3 \\ &= 0.290472 t^2 + 19.6905 t + 1.08571, \quad 5 \leq t \leq 12 \end{aligned}$$

Since we want to find the velocity at $t = 6, 7.5, 9$ and 11 seconds, we could simply substitute each value of t in $v(t) = 0.290472 t^2 + 19.6905 t + 1.08571$ and find the corresponding velocity. For example, at $t = 6$

$$\begin{aligned} v(6) &= 0.290472(6)^2 + 19.6905(6) + 1.08571 \\ &= 129.686 \text{ m/s} \end{aligned}$$

However, we could also find all the needed values of velocity at $t = 6, 7.5, 9, 11$ seconds using matrix multiplication.

$$v(t) = [0.290472 \quad 19.6905 \quad 1.08571] \begin{bmatrix} t^2 \\ t \\ 1 \end{bmatrix}$$

So, if we want to find $v(6), v(7.5), v(9), v(11)$, it is given by

$$\begin{aligned} [v(6) \ v(7.5) \ v(9) \ v(11)] &= [0.290472 \quad 19.6905 \quad 1.08571] \begin{bmatrix} 6^2 & 7.5^2 & 9^2 & 11^2 \\ 6 & 7.5 & 9 & 11 \\ 1 & 1 & 1 & 1 \end{bmatrix} \\ &= [0.290472 \quad 19.6905 \quad 1.08571] \begin{bmatrix} 36 & 56.25 & 81 & 121 \\ 6 & 7.5 & 9 & 11 \\ 1 & 1 & 1 & 1 \end{bmatrix} \\ &= [129.686 \quad 165.104 \quad 201.828 \quad 252.828] \\ v(6) &= 129.686 \text{ m/s} \\ v(7.5) &= 165.104 \text{ m/s} \\ v(9) &= 201.828 \text{ m/s} \\ v(11) &= 252.828 \text{ m/s} \end{aligned}$$

Can we use Naive Gauss elimination methods to find the determinant of a square matrix?

One of the more efficient ways to find the determinant of a square matrix is by taking advantage of the following two theorems on a determinant of matrices coupled with Naive Gauss elimination.

Theorem 6.1

Let $[A]$ be a $n \times n$ matrix. Then, if $[B]$ is a $n \times n$ matrix that results from adding or subtracting a multiple of one row to another row, then $\det(A) = \det(B)$ (The same is true for column operations also).

Theorem 6.2

Let $[A]$ be a $n \times n$ matrix that is upper triangular, lower triangular or diagonal, then

$$\begin{aligned} \det(A) &= a_{11} \times a_{22} \times \dots \times a_{ii} \times \dots \times a_{nn} \\ &= \prod_{i=1}^n a_{ii} \end{aligned}$$

This implies that if we apply the forward elimination steps of the Naive Gauss elimination method, the determinant of the matrix stays the same according to Theorem 6.1. Then since at the end of the forward elimination steps, the resulting matrix is upper triangular, the determinant will be given by Theorem 6.2.

✓ Example 2

Find the determinant of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

Solution

Remember in Example 1, we conducted the steps of forward elimination of unknowns using the Naive Gauss elimination method on $[A]$ to give

$$[B] = \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix}$$

According to Theorem 2

$$\begin{aligned} \det(A) &= \det(B) \\ &= 25 \times (-4.8) \times 0.7 \\ &= -84.00 \end{aligned}$$

What if I cannot find the determinant of the matrix using the Naive Gauss elimination method, for example, if I get division by zero problems during the Naive Gauss elimination method?

Well, you can apply Gaussian elimination with partial pivoting. However, the determinant of the resulting upper triangular matrix may differ by a sign. The following theorem applies in addition to the previous two to find the determinant of a square matrix.

Theorem 6.3

Let $[A]$ be a $n \times n$ matrix. Then, if $[B]$ is a matrix that results from switching one row with another row, then $\det(B) = -\det(A)$.

✓ Example 3

Find the determinant of

$$[A] = \begin{bmatrix} 10 & -7 & 0 \\ -3 & 2.099 & 6 \\ 5 & -1 & 5 \end{bmatrix}$$

Solution

The end of the forward elimination steps of Gaussian elimination with partial pivoting, we would obtain

$$[B] = \begin{bmatrix} 10 & -7 & 0 \\ 0 & 2.5 & 5 \\ 0 & 0 & 6.002 \end{bmatrix}$$

$$\begin{aligned} \det(B) &= 10 \times 2.5 \times 6.002 \\ &= 150.05 \end{aligned}$$

Since rows were switched once during the forward elimination steps of Gaussian elimination with partial pivoting,

$$\begin{aligned} \det(A) &= -\det(B) \\ &= -150.05 \end{aligned}$$

✓ Example 4

Prove

$$\det(A) = \frac{1}{\det(A^{-1})}$$

Solution

$$[A][A]^{-1} = [I]$$

$$\det(AA^{-1}) = \det(I)$$

$$\det(A) \det(A^{-1}) = 1$$

$$\det(A) = \frac{1}{\det(A^{-1})}$$

If $[A]$ is a $n \times n$ matrix and $\det(A) \neq 0$, what other statements are equivalent to it?

1. $[A]$ is invertible.
2. $[A]^{-1}$ exists.
3. $[A][X] = [C]$ has a unique solution.
4. $[A][X] = [0]$ solution is $[X] = [0]$.
5. $[A][A]^{-1} = [I] = [A]^{-1}[A]$.

Are there any pitfalls of the Naïve Gauss elimination method?

Yes, there are two pitfalls of the Naïve Gauss elimination method.

Division by zero: It is possible for division by zero to occur during the beginning of the $n - 1$ steps of forward elimination.

For example

$$5x_2 + 6x_3 = 11$$

$$4x_1 + 5x_2 + 7x_3 = 16$$

$$9x_1 + 2x_2 + 3x_3 = 15$$

will result in division by zero in the first step of forward elimination as the coefficient of x_1 in the first equation is zero as is evident when we write the equations in matrix form.

$$\begin{bmatrix} 0 & 5 & 6 \\ 4 & 5 & 7 \\ 9 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 11 \\ 16 \\ 15 \end{bmatrix}$$

But what about the equations below: Is division by zero a problem?

$$5x_1 + 6x_2 + 7x_3 = 18$$

$$10x_1 + 12x_2 + 3x_3 = 25$$

$$20x_1 + 17x_2 + 19x_3 = 56$$

Written in matrix form,

$$\begin{bmatrix} 5 & 6 & 7 \\ 10 & 12 & 3 \\ 20 & 17 & 19 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 18 \\ 25 \\ 56 \end{bmatrix}$$

there is no issue of division by zero in the first step of forward elimination. The pivot element is the coefficient of x_1 in the first equation, 5, and that is a non-zero number. However, at the end of the first step of forward elimination, we get the following equations in matrix form

$$\begin{bmatrix} 5 & 6 & 7 \\ 0 & 0 & -11 \\ 0 & -7 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 18 \\ -11 \\ -16 \end{bmatrix}$$

Now at the beginning of the 2nd step of forward elimination, the coefficient of x_2 in Equation 2 would be used as the pivot element. That element is zero and hence would create the division by zero problem.

So it is important to consider that the possibility of division by zero can occur at the beginning of any step of forward elimination.

Round-off error

The Naïve Gauss elimination method is prone to round-off errors. This is true when there are large numbers of equations as errors propagate. Also, if there is subtraction of numbers from each other, it may create large errors. See the example below.

✓ Example 5

Remember [Example 2](#) where we used Naïve Gauss elimination to solve

$$20x_1 + 15x_2 + 10x_3 = 45$$

$$-3x_1 - 2.249x_2 + 7x_3 = 1.751$$

$$5x_1 + x_2 + 3x_3 = 9$$

using six significant digits with chopping in your calculations? Repeat the problem, but now use five significant digits with chopping in your calculations.

Solution

Writing in the matrix form

$$\begin{bmatrix} 20 & 15 & 10 \\ -3 & -2.249 & 7 \\ 5 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 45 \\ 1.751 \\ 9 \end{bmatrix}$$

Forward Elimination of Unknowns

First step

Divide Row 1 by 20 and then multiply it by -3 , that is, multiply Row 1 by $-3/20 = -0.15$.

$$([20 \ 15 \ 10] \ [45]) \times -0.15 \text{ gives Row 1 as}$$

$$[-3 \ -2.25 \ -1.5] \quad [-6.75]$$

Subtract the result from Row 2

$$\begin{array}{r} [-3 \quad -2.249 \quad 7 \quad | \quad 1.751] \\ -[-3 \quad -2.25 \quad -1.5 \quad | \quad -6.75] \\ \hline 0 \quad 0.001 \quad 8.5 \quad 8.501 \end{array}$$

to get the resulting equations as

$$\begin{bmatrix} 20 & 15 & 10 \\ 0 & 0.001 & 8.5 \\ 5 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 45 \\ 8.501 \\ 9 \end{bmatrix}$$

Divide Row 1 by 20 and then multiply it by 5, that is, multiply Row 1 by $5/20 = 0.25$.

$$([20 \ 15 \ 10] \ [45]) \times 0.25 \text{ gives Row 1 as}$$

$$[5 \ 3.75 \ 2.5] \quad [11.25]$$

Subtract the result from Row 3

$$\begin{array}{ccc|c} 5 & 1 & 3 & 9 \\ -[5 & 3.75 & 2.5 & 11.25] \\ \hline 0 & -2.75 & 0.5 & -2.25 \end{array}$$

to get the resulting equations as

$$\begin{bmatrix} 20 & 15 & 10 \\ 0 & 0.001 & 8.5 \\ 0 & -2.75 & 0.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 45 \\ 8.501 \\ -2.25 \end{bmatrix}$$

Second step

Now for the second step of forward elimination, we will use Row 2 as the pivot equation and eliminate Row 3: Column 2.

Divide Row 2 by 0.001 and then multiply it by -2.75 , that is, multiply Row 2 by $-2.75/0.001 = -2750$

$$([0 \quad 0.001 \quad 8.5] \quad [8.501]) \times -2750 \text{ gives Row 2 as}$$

$$[0 \quad -2.75 \quad -23375] \quad [-23377.75]$$

Rewriting within 5 significant digits with chopping

$$[0 \quad -2.75 \quad -23375] \quad [-23377]$$

Subtract the result from Row 3

$$\begin{array}{ccc|c} 0 & -2.75 & 0.5 & -2.25 \\ -[0 & -2.75 & -23375 & -23377] \\ \hline 0 & 0 & 23375 & 23374 \end{array}$$

Rewriting within 6 significant digits with chopping

$$[0 \quad 0 \quad 23375] \quad [-23374]$$

to get the resulting equations as

$$\begin{bmatrix} 20 & 15 & 10 \\ 0 & 0.001 & 8.5 \\ 0 & 0 & 23375 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 45 \\ 8.501 \\ 23374 \end{bmatrix}$$

This is the end of the forward elimination steps.

Back substitution

We can now solve the above equations by back substitution. From the third equation,

$$\begin{aligned} 23375x_3 &= 23374 \\ x_3 &= \frac{23374}{23375} \\ &= 0.99995 \end{aligned}$$

Substituting the value of x_3 in the second equation

$$0.001x_2 + 8.5x_3 = 8.501$$

$$\begin{aligned}
 x_2 &= \frac{8.501 - 8.5x_3}{0.001} \\
 &= \frac{8.501 - 8.5 \times 0.99995}{0.001} \\
 &= \frac{8.501 - 8.499575}{0.001} \\
 &= \frac{0.001425}{0.001} \\
 &= 1.425 \\
 &= 1.5
 \end{aligned}$$

Substituting the value of x_3 and x_2 in the first equation,

$$\begin{aligned}
 20x_1 + 15x_2 + 10x_3 &= 45 \\
 x_1 &= \frac{45 - 15x_2 - 10x_3}{20} \\
 &= \frac{45 - 15 \times 1.5 - 10 \times 0.99995}{20} \\
 &= \frac{45 - 22.5 - 9.9995}{20} \\
 &= \frac{22.5 - 9.9995}{20} \\
 &= \frac{12.5005}{20} \\
 &= \frac{12.500}{20} \\
 &= 0.625
 \end{aligned}$$

Hence the solution is

$$\begin{aligned}
 [X] &= \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\
 &= \begin{bmatrix} 0.625 \\ 1.5 \\ 0.99995 \end{bmatrix}
 \end{aligned}$$

Compare this with the exact solution of

$$[X] = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

What are some techniques for improving the Naïve Gauss elimination method?

As seen in [Example 3](#), round off errors were large when five significant digits were used as opposed to six significant digits. One method of decreasing the round-off error would be to use more significant digits, that is, use double or quad precision for representing the numbers. However, this would not avoid possible division by zero errors in the Naïve Gauss elimination method. To avoid division by zero as well as reduce (not eliminate) round-off error, Gaussian elimination with partial pivoting is the method of choice.

How does Gaussian elimination with partial pivoting differ from Naïve Gauss elimination?

The two methods are the same, except in the beginning of each step of forward elimination, a row switching is done based on the following criterion. If there are n equations, then there are $n - 1$ forward elimination steps. At the beginning of the k^{th} step of forward elimination, one finds the maximum of

$$|a_{kk}|, |a_{k+1,k}|, \dots, |a_{nk}|$$

Then if the maximum of these values is $|a_{pk}|$ in the p^{th} row, $k \leq p \leq n$, then switch rows p and k .

The other steps of forward elimination are the same as the Naïve Gauss elimination method. The back substitution steps stay exactly the same as the Naïve Gauss elimination method.

✓ Example 6

In the previous two examples, we used Naïve Gauss elimination to solve

$$\begin{aligned} 20x_1 + 15x_2 + 10x_3 &= 45 \\ -3x_1 - 2.249x_2 + 7x_3 &= 1.751 \\ 5x_1 + x_2 + 3x_3 &= 9 \end{aligned}$$

using five and six significant digits with chopping in the calculations. Using five significant digits with chopping, the solution found was

$$\begin{aligned} [X] &= \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ &= \begin{bmatrix} 0.625 \\ 1.5 \\ 0.99995 \end{bmatrix} \end{aligned}$$

This is different from the exact solution of

$$\begin{aligned} [X] &= \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \end{aligned}$$

Find the solution using Gaussian elimination with partial pivoting using five significant digits with chopping in your calculations.

Solution

$$\begin{bmatrix} 20 & 15 & 10 \\ -3 & -2.249 & 7 \\ 5 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 45 \\ 1.751 \\ 9 \end{bmatrix}$$

Forward Elimination of Unknowns

Now for the first step of forward elimination, the absolute value of the first column elements below Row 1 is

$$|20|, |-3|, |5|$$

or

$$20, 3, 5$$

So, the largest absolute value is in the Row 1. So as per Gaussian elimination with partial pivoting, the switch is between Row 1 and Row 1 to give

$$\begin{bmatrix} 20 & 15 & 10 \\ -3 & -2.249 & 7 \\ 5 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 45 \\ 1.751 \\ 9 \end{bmatrix}$$

Divide Row 1 by 20 and then multiply it by -3 , that is, multiply Row 1 by $-3/20 = -0.15$.

$$([20 \ 15 \ 10] \ [45]) \times -0.15 \text{ gives Row 1 as}$$

$$[-3 \ -2.25 \ -1.5] \quad [-6.75]$$

Subtract the result from Row 2

$$\begin{array}{r} [-3 \quad -2.249 \quad 7 \quad | \quad 1.751] \\ -[-3 \quad -2.25 \quad -1.5 \quad | \quad -6.75] \\ \hline 0 \quad 0.001 \quad 8.5 \quad 8.501 \end{array}$$

to get the resulting equations as

$$\begin{bmatrix} 20 & 15 & 10 \\ 0 & 0.001 & 8.5 \\ 5 & 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 45 \\ 8.501 \\ 9 \end{bmatrix}$$

Divide Row 1 by 20 and then multiply it by 5, that is, multiply Row 1 by $5/20 = 0.25$.

$$([20 \ 15 \ 10] \ [45]) \times 0.25 \text{ gives Row 1 as}$$

$$[5 \ 3.75 \ 2.5] \quad [11.25]$$

Subtract the result from Row 3

$$\begin{array}{r} [5 \quad 1 \quad 3 \quad | \quad 9] \\ -[5 \quad 3.75 \quad 2.5 \quad | \quad 11.25] \\ \hline 0 \quad -2.75 \quad 0.5 \quad -2.25 \end{array}$$

to get the resulting equations as

$$\begin{bmatrix} 20 & 15 & 10 \\ 0 & 0.001 & 8.5 \\ 0 & -2.75 & 0.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 45 \\ 8.501 \\ -2.25 \end{bmatrix}$$

This is the end of the first step of forward elimination.

Now for the second step of forward elimination, the absolute value of the second column elements below Row 1 is

$$|0.001|, |-2.75|$$

or

$$0.001, 2.75$$

So, the largest absolute value is in Row 3. So, Row 2 is switched with Row 3 to give

$$\begin{bmatrix} 20 & 15 & 10 \\ 0 & -2.75 & 0.5 \\ 0 & 0.001 & 8.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 45 \\ -2.25 \\ 8.501 \end{bmatrix}$$

Divide Row 2 by -2.75 and then multiply it by 0.001, that is, multiply Row 2 by $0.001/-2.75 = -0.00036363$

$$([0 \ -2.75 \ 0.5] \ [-2.25]) \times -0.00036363 \text{ gives Row 2 as}$$

$$[0 \ 0.00099998 \ -0.00018182] \quad [0.00081816]$$

Subtract the result from Row 3

$$\begin{array}{r} [0 \quad 0.001 \quad 8.5 \quad | \quad 8.501] \\ -[0 \quad 0.00099998 \quad -0.00018182 \quad | \quad 0.00081816] \\ \hline 0 \quad 0 \quad 8.50018182 \quad 8.50018184 \end{array}$$

Rewriting within 5 significant digits with chopping

$$\begin{bmatrix} 0 & 0 & 8.5001 \end{bmatrix} \quad [8.5001]$$

to get the resulting equations as

$$\begin{bmatrix} 20 & 15 & 10 \\ 0 & -2.75 & 0.5 \\ 0 & 0 & 8.5001 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 45 \\ -2.25 \\ 8.5001 \end{bmatrix}$$

Back substitution

$$8.5001x_3 = 8.5001$$

$$x_3 = \frac{8.5001}{8.5001} = 1$$

Substituting the value of x_3 in Row 2

$$-2.75x_2 + 0.5x_3 = -2.25$$

$$\begin{aligned} x_2 &= \frac{-2.25 - 0.5x_3}{-2.75} \\ &= \frac{-2.25 - 0.5 \times 1}{-2.75} \\ &= \frac{-2.25 - 0.5}{-2.75} \\ &= \frac{-2.75}{-2.75} \\ &= 1 \end{aligned}$$

Substituting the value of x_3 and x_2 in Row 1

$$20x_1 + 15x_2 + 10x_3 = 45$$

$$\begin{aligned} x_1 &= \frac{45 - 15x_2 - 10x_3}{20} \\ &= \frac{45 - 15 \times 1 - 10 \times 1}{20} \\ &= \frac{45 - 15 - 10}{20} \\ &= \frac{20}{20} \\ &= 1 \end{aligned}$$

Gaussian Elimination Method for Solving Simultaneous Linear Equations Quiz

? Quiz 1

The goal of forward elimination steps in the Naïve Gauss elimination method is to reduce the coefficient matrix to a (an) _____ matrix.

- (A) diagonal
- (B) identity
- (C) lower triangular
- (D) upper triangular

? Quiz 2

Division by zero during forward elimination steps in Naïve Gaussian elimination of the set of equations $[A][X] = [C]$ implies the coefficient matrix $[A]$

- (A) is invertible
- (B) is nonsingular
- (C) may be singular or nonsingular
- (D) is singular

? Quiz 3

Using a computer with four significant digits with chopping, the Naïve Gauss elimination solution to

$$0.0030x_1 + 55.23x_2 = 58.12$$

$$6.239x_1 - 7.123x_2 = 47.23$$

is

- (A) $x_1 = 26.66$; $x_2 = 1.051$
- (B) $x_1 = 8.769$; $x_2 = 1.051$
- (C) $x_1 = 8.800$; $x_2 = 1.000$
- (D) $x_1 = 8.771$; $x_2 = 1.052$

? Quiz 4

Using a computer with four significant digits with chopping, the Gaussian elimination with partial pivoting solution to

$$0.0030x_1 + 55.23x_2 = 58.12$$

$$6.239x_1 - 7.123x_2 = 47.23$$

is

- (A) $x_1 = 26.66$; $x_2 = 1.051$
- (B) $x_1 = 8.769$; $x_2 = 1.051$
- (C) $x_1 = 8.800$; $x_2 = 1.000$
- (D) $x_1 = 8.771$; $x_2 = 1.052$

? Quiz 5

At the end of the forward elimination steps of the Naïve Gauss elimination method on the following equations

$$\begin{bmatrix} 4.2857 \times 10^7 & -9.2307 \times 10^5 & 0 & 0 \\ 4.2857 \times 10^7 & -5.4619 \times 10^5 & -4.2857 \times 10^7 & 5.4619 \times 10^5 \\ -6.5 & -0.15384 & 6.5 & 0.15384 \\ 0 & 0 & 4.2857 \times 10^7 & -3.6057 \times 10^5 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} -7.887 \times 10^3 \\ 0 \\ 0.007 \\ 0 \end{bmatrix}$$

the resulting equations in matrix form are given by

$$\begin{bmatrix} 4.2857 \times 10^7 & -9.2307 \times 10^5 & 0 & 0 \\ 0 & 3.7688 \times 10^5 & -4.2857 \times 10^7 & 5.4619 \times 10^5 \\ 0 & 0 & -26.9140 & 0.579684 \\ 0 & 0 & 0 & 5.62500 \times 10^5 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \\ c_4 \end{bmatrix} = \begin{bmatrix} -7.887 \times 10^3 \\ 7.887 \times 10^3 \\ 1.19530 \times 10^{-2} \\ 1.90336 \times 10^4 \end{bmatrix}$$

The determinant of the original coefficient matrix is

- (A) 0.00
- (B) 4.2857×10^7
- (C) 5.486×10^{19}
- (D) -2.445×10^{20}

? Quiz 6

The following data is given for the velocity of the rocket as a function of time. To find the velocity at $t = 21s$, you are asked to use a quadratic polynomial, $v(t) = at^2 + bt + c$ to approximate the velocity profile.

t	(s)	0	14	15	20	30	35
$v(t)$	(m/s)	0	227.04	362.78	517.35	602.97	901.67

The correct set of equations that will find a , b and c are

- (A) $\begin{bmatrix} 176 & 14 & 1 \\ 225 & 15 & 1 \\ 400 & 20 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 227.04 \\ 362.78 \\ 517.35 \end{bmatrix}$
- (B) $\begin{bmatrix} 225 & 15 & 1 \\ 400 & 20 & 1 \\ 900 & 30 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 362.78 \\ 517.35 \\ 602.97 \end{bmatrix}$
- (C) $\begin{bmatrix} 0 & 0 & 1 \\ 225 & 15 & 1 \\ 400 & 20 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 362.78 \\ 517.35 \end{bmatrix}$
- (D) $\begin{bmatrix} 400 & 20 & 1 \\ 900 & 30 & 1 \\ 1225 & 35 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 517.35 \\ 602.97 \\ 901.67 \end{bmatrix}$

Gaussian Elimination Method for Solving Simultaneous Linear Equations Exercise

? Exercise 1

Use Naïve Gauss elimination to solve

$$4x_1 + x_2 - x_3 = -2$$

$$5x_1 + x_2 + 2x_3 = 4$$

$$6x_1 + x_2 + x_3 = 6$$

? Exercise 2

Assume that you are using a computer with four significant digits with chopping. Use Naïve Gauss elimination method to solve

$$4x_1 + x_2 - x_3 = -2$$

$$5x_1 + x_2 + 2x_3 = 4$$

$$6x_1 + x_2 + x_3 = 6$$

? Exercise 3

For

$$[A] = \begin{bmatrix} 10 & -7 & 0 \\ -3 & 2.099 & 6 \\ 5 & -1 & 5 \end{bmatrix}$$

Find the determinant of $[A]$ using forward elimination step of naïve Gauss elimination method.

? Exercise 4

At the end of forward elimination steps using naïve Gauss elimination method on the coefficient matrix

$$[A] = \begin{bmatrix} 25 & c & 1 \\ 64 & a & 1 \\ 144 & b & 1 \end{bmatrix}$$

$[A]$ reduces to

$$[B] = \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix}$$

What is the determinant of $[A]$?

? Exercise 5

Using Gaussian elimination with partial pivoting to solve

$$4x_1 + x_2 - x_3 = -2$$

$$5x_1 + x_2 + 2x_3 = 4$$

$$6x_1 + x_2 + x_3 = 6$$

? Exercise 6

Assume that you are using a computer with four significant digits with chopping, use Gaussian elimination with partial pivoting to solve

$$4x_1 + x_2 - x_3 = -2$$

$$5x_1 + x_2 + 2x_3 = 4$$

$$6x_1 + x_2 + x_3 = 6$$

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7: LU Decomposition Method for Solving Simultaneous Linear Equations

Learning Objectives

After successful completion of this section, you should be able to

1. solve a set of simultaneous linear equations using LU decomposition method
2. decompose a nonsingular matrix into LU form.
3. solve a set of simultaneous linear equations using LU decomposition method
4. decompose a nonsingular matrix into LU form.
5. find the inverse of a matrix using LU decomposition method.
6. justify why using LU decomposition method is more efficient than Gaussian elimination in some cases.

I hear about LU decomposition used as a method to solve a set of simultaneous linear equations. What is it?

We already studied two numerical methods of finding the solution to simultaneous linear equations – Naïve Gauss elimination and Gaussian elimination with partial pivoting. Then, why do we need to learn another method? To appreciate why LU decomposition could be a better choice than the Gauss elimination techniques in some cases, let us discuss first what LU decomposition is about.

For a nonsingular matrix $[A]$ on which one can successfully conduct the Naïve Gauss elimination forward elimination steps, one can always write it as

$$[A] = [L] [U]$$

where

$[L]$ = Lower triangular matrix

$[U]$ = Upper triangular matrix

Then if one is solving a set of equations

$$[A] [X] = [C],$$

then

$$[L] [U] [X] = [C] \text{ as } ([A] = [L] [U])$$

Multiplying both sides by $[L]^{-1}$,

$$[L]^{-1} [L] [U] [X] = [L]^{-1} [C]$$

$$[I] [U] [X] = [L]^{-1} [C] \text{ as } ([L]^{-1} [L] = [I])$$

$$[U] [X] = [L]^{-1} [C] \text{ as } ([I] [U] = [U])$$

Let

$$[L]^{-1} [C] = [Z]$$

then

$$[L] [Z] = [C] \quad (1)$$

and

$$[U] [X] = [Z] \quad (2)$$

So we can solve Equation (1) first for $[Z]$ by using forward substitution and then use Equation (2) to calculate the solution vector $[X]$ by back substitution.

How do I decompose a non-singular matrix $[A]$, that is, how do I find $[A] = [L][U]$?

If forward elimination steps of the Naïve Gauss elimination methods can be applied on a nonsingular matrix, then $[A]$ can be decomposed into LU as

$$[A] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \dots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \dots & 0 \\ l_{21} & 1 & \dots & 0 \\ \vdots & \vdots & \dots & \vdots \\ l_{n1} & l_{n2} & \dots & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & \dots & u_{1n} \\ 0 & u_{22} & \dots & u_{2n} \\ \vdots & \vdots & \dots & \vdots \\ 0 & 0 & \dots & u_{nn} \end{bmatrix}$$

The elements of the $[U]$ matrix are exactly the same as the coefficient matrix one obtains at the end of the forward elimination steps in Naïve Gauss elimination.

The lower triangular matrix $[L]$ has 1 in its diagonal entries. The non-zero elements on the non-diagonal elements in $[L]$ are multipliers that made the corresponding entries zero in the upper triangular matrix $[U]$ during forward elimination.

Let us look at this using the same example as used in Naïve Gaussian elimination.

✓ Example 7.1

Find the LU decomposition of the matrix

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

Solution

$$[A] = [L][U] = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

The $[U]$ matrix is the same as found at the end of the forward elimination of Naïve Gauss elimination method, that is

$$[U] = \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix}$$

To find l_{21} and l_{31} , find the multiplier that was used to make the a_{21} and a_{31} elements zero in the first step of forward elimination of the Naïve Gauss elimination method. It was

$$\begin{aligned} l_{21} &= \frac{64}{25} \\ &= 2.56 \\ l_{31} &= \frac{144}{25} \\ &= 5.76 \end{aligned}$$

To find l_{32} , what multiplier was used to make a_{32} element zero? Remember a_{32} element was made zero in the second step of forward elimination. The $[A]$ matrix at the beginning of the second step of forward elimination was

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & -16.8 & -4.76 \end{bmatrix}$$

So

$$l_{32} = \frac{-16.8}{-4.8} \\ = 3.5$$

Hence

$$[L] = \begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix}$$

Confirm $[L][U] = [A]$.

$$\begin{aligned} [L][U] &= \begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} \\ &= \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \end{aligned}$$

✓ Example 7.2

Use the LU decomposition method to solve the following simultaneous linear equations.

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

Solution

Recall that

$$[A][X] = [C]$$

and if

$$[A] = [L][U]$$

then first solving

$$[L][Z] = [C]$$

and then

$$[U][X] = [Z]$$

gives the solution vector $[X]$.

Now in the previous example, we showed

$$\begin{aligned} [A] &= [L][U] \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} \end{aligned}$$

First solve

$$[L] [Z] = [C]$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

to give

$$z_1 = 106.8$$

$$2.56z_1 + z_2 = 177.2$$

$$5.76z_1 + 3.5z_2 + z_3 = 279.2$$

Forward substitution starting from the first equation gives

$$z_1 = 106.8$$

$$\begin{aligned} z_2 &= 177.2 - 2.56z_1 \\ &= 177.2 - 2.56 \times 106.8 \\ &= -96.208 \end{aligned}$$

$$\begin{aligned} z_3 &= 279.2 - 5.76z_1 - 3.5z_2 \\ &= 279.2 - 5.76 \times 106.8 - 3.5 \times (-96.208) \\ &= 0.76 \end{aligned}$$

Hence

$$\begin{aligned} [Z] &= \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} \\ &= \begin{bmatrix} 106.8 \\ -96.208 \\ 0.76 \end{bmatrix} \end{aligned}$$

This matrix is same as the right-hand side obtained at the end of the forward elimination steps of Naïve Gauss elimination method. Is this a coincidence?

Now solve

$$[U] [X] = [Z]$$

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ -96.208 \\ 0.76 \end{bmatrix}$$

$$25a_1 + 5a_2 + a_3 = 106.8$$

$$-4.8a_2 - 1.56a_3 = -96.208$$

$$0.7a_3 = 0.76$$

From the third equation

$$0.7a_3 = 0.76$$

$$\begin{aligned} a_3 &= \frac{0.76}{0.7} \\ &= 1.0857 \end{aligned}$$

Substituting the value of a_3 in the second equation,

$$-4.8a_2 - 1.56a_3 = -96.208$$

$$\begin{aligned} a_2 &= \frac{-96.208 + 1.56a_3}{-4.8} \\ &= \frac{-96.208 + 1.56 \times 1.0857}{-4.8} \\ &= 19.691 \end{aligned}$$

Substituting the value of a_2 and a_3 in the first equation,

$$\begin{aligned} 25a_1 + 5a_2 + a_3 &= 106.8 \\ a_1 &= \frac{106.8 - 5a_2 - a_3}{25} \\ &= \frac{106.8 - 5 \times 19.691 - 1.0857}{25} \\ &= 0.29048 \end{aligned}$$

Hence the solution vector is

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0.29048 \\ 19.691 \\ 1.0857 \end{bmatrix}$$

How do I find the inverse of a square matrix using LU decomposition?

A matrix $[B]$ is the inverse of $[A]$ if

$$[A][B] = [I] = [B][A]$$

How can we use LU decomposition to find the inverse of the matrix? Assume the first column of $[B]$ (the inverse of $[A]$) is

$$[b_{11} \ b_{12} \ \dots \ \dots b_{n1}]^T$$

Then from the above definition of an inverse and the definition of matrix multiplication

$$[A] \begin{bmatrix} b_{11} \\ b_{21} \\ \vdots \\ b_{n1} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Similarly, the second column of $[B]$ is given by

$$[A] \begin{bmatrix} b_{12} \\ b_{22} \\ \vdots \\ b_{n2} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}$$

Similarly, all columns of $[B]$ can be found by solving n different sets of equations with the column of the right-hand side being the n columns of the identity matrix.

✓ Example 7.3

Use LU decomposition to find the inverse of

$$[A] = \begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

Solution

Knowing that

$$\begin{aligned}
 [A] &= [L] [U] \\
 &= \begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix}
 \end{aligned}$$

We can solve for the first column of $[B] = [A]^{-1}$ by solving for

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

First solve

$$[L] [Z] = [C],$$

that is

$$\begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

to give

$$z_1 = 1$$

$$2.56z_1 + z_2 = 0$$

$$5.76z_1 + 3.5z_2 + z_3 = 0$$

Forward substitution starting from the first equation gives

$$z_1 = 1$$

$$\begin{aligned}
 z_2 &= 0 - 2.56z_1 \\
 &= 0 - 2.56(1) \\
 &= -2.56
 \end{aligned}$$

$$\begin{aligned}
 z_3 &= 0 - 5.76z_1 - 3.5z_2 \\
 &= 0 - 5.76(1) - 3.5(-2.56) \\
 &= 3.2
 \end{aligned}$$

Hence

$$\begin{aligned}
 [Z] &= \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} \\
 &= \begin{bmatrix} 1 \\ -2.56 \\ 3.2 \end{bmatrix}
 \end{aligned}$$

Now solve

$$[U] [X] = [Z]$$

that is

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} \begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} = \begin{bmatrix} 1 \\ -2.56 \\ 3.2 \end{bmatrix}$$

$$25b_{11} + 5b_{21} + b_{31} = 1$$

$$-4.8b_{21} - 1.56b_{31} = -2.56$$

$$0.7b_{31} = 3.2$$

Backward substitution starting from the third equation gives

$$b_{31} = \frac{3.2}{0.7} \\ = 4.571$$

$$b_{21} = \frac{-2.56 + 1.56b_{31}}{-4.8} \\ = \frac{-2.56 + 1.56(4.571)}{-4.8} \\ = -0.9524$$

$$b_{11} = \frac{1 - 5b_{21} - b_{31}}{25} \\ = \frac{1 - 5(-0.9524) - 4.571}{25} \\ = 0.04762$$

Hence the first column of the inverse of $[A]$ is

$$\begin{bmatrix} b_{11} \\ b_{21} \\ b_{31} \end{bmatrix} = \begin{bmatrix} 0.04762 \\ -0.9524 \\ 4.571 \end{bmatrix}$$

Similarly, solving

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} b_{12} \\ b_{22} \\ b_{32} \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \text{ gives } \begin{bmatrix} b_{12} \\ b_{22} \\ b_{32} \end{bmatrix} = \begin{bmatrix} -0.08333 \\ 1.417 \\ -5.000 \end{bmatrix}$$

and solving

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} b_{13} \\ b_{23} \\ b_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ gives } \begin{bmatrix} b_{13} \\ b_{23} \\ b_{33} \end{bmatrix} = \begin{bmatrix} 0.03571 \\ -0.4643 \\ 1.429 \end{bmatrix}$$

Hence

$$[A]^{-1} = \begin{bmatrix} 0.04762 & -0.08333 & 0.03571 \\ -0.9524 & 1.417 & -0.4643 \\ 4.571 & -5.000 & 1.429 \end{bmatrix}$$

Can you confirm the following for the above example?

$$[A] [A]^{-1} = [I] = [A]^{-1} [A]$$

LU decomposition looks more complicated than Gaussian elimination. Do we use LU decomposition because it is computationally more efficient than Gaussian elimination to solve a set of n equations given by $[A][X] = [C]$?

For a square matrix $[A]$ of $n \times n$ size, the computational time $CT|_{DE}$ to decompose the $[A]$ matrix to $[L][U]$ form is given by

$$CT|_{DE} = T \left(\frac{8n^3}{3} + 4n^2 - \frac{20n}{3} \right),$$

where

$$T = \text{clock cycle time}^2$$

The computational time $CT|_{FS}$ to solve by forward substitution $[L][Z] = [C]$ is given by

$$CT|_{FS} = T(4n^2 - 4n)$$

The computational time $CT|_{BS}$ to solve by back substitution $[U][X] = [Z]$ is given by

$$CT|_{BS} = T(4n^2 + 12n)$$

So, the total computational time to solve a set of equations by LU decomposition is

$$\begin{aligned} CT|_{LU} &= CT|_{DE} + CT|_{FS} + CT|_{BS} \\ &= T\left(\frac{8n^3}{3} + 4n^2 - \frac{20n}{3}\right) + T(4n^2 - 4n) + T(4n^2 + 12n) \\ &= T\left(\frac{8n^3}{3} + 12n^2 + \frac{4n}{3}\right) \end{aligned}$$

Now let us look at the computational time taken by Gaussian elimination. The computational time $CT|_{FE}$ for the forward elimination part,

$$CT|_{FE} = T\left(\frac{8n^3}{3} + 8n^2 - \frac{32n}{3}\right),$$

and the computational time $CT|_{BS}$ for the back substitution part is

$$CT|_{BS} = T(4n^2 + 12n)$$

So, the total computational time $CT|_{GE}$ to solve a set of equations by Gaussian Elimination is

$$\begin{aligned} CT|_{GE} &= CT|_{FE} + CT|_{BS} \\ &= T\left(\frac{8n^3}{3} + 8n^2 - \frac{32n}{3}\right) + T(4n^2 + 12n) \\ &= T\left(\frac{8n^3}{3} + 12n^2 + \frac{4n}{3}\right) \end{aligned}$$

The computational time for Gaussian elimination and LU decomposition is identical.

This has confused me further! Why should I learn LU decomposition method when it takes the same computational time as Gaussian elimination, and that too when the two methods are closely related? Please convince me that LU decomposition has its place in solving linear equations!

We now have the knowledge to convince you that LU decomposition method has its place in the solution of simultaneous linear equations. Let us look at an example where the LU decomposition method is computationally more efficient than Gaussian elimination. Remember in trying to find the inverse of the matrix $[A]$ in Chapter 04.01, the problem reduces to solving n sets of equations with the n columns of the identity matrix as the RHS vector. For calculations of each column of the inverse of the $[A]$ matrix, the coefficient matrix $[A]$ matrix in the set of equation $[A][X] = [C]$ does not change. So, if we use the LU decomposition method, the $[A] = [L][U]$ decomposition needs to be done only once, the forward substitution (Equation 1) n times, and the back substitution (Equation 2) n times.

Therefore, the total computational time $CT|_{inverse LU}$ required to find the inverse of a matrix using LU decomposition is

$$\begin{aligned} CT|_{inverse LU} &= 1 \times CT|_{DE} + n \times CT|_{FS} + n \times CT|_{BS} \\ &= 1 \times T\left(\frac{8n^3}{3} + 4n^2 - \frac{20n}{3}\right) + n \times T(4n^2 - 4n) + n \times T(4n^2 + 12n) \\ &= T\left(\frac{32n^3}{3} + 12n^2 - \frac{20n}{3}\right) \end{aligned}$$

In comparison, if Gaussian elimination method were used to find the inverse of a matrix, the forward elimination as well as the back substitution will have to be done n times. The total computational time $CT|_{inverse GE}$ required to find the inverse of a matrix

by using Gaussian elimination then is

$$\begin{aligned}
 CT|_{inverse\ GE} &= n \times CT|_{FE} + n \times CT|_{BS} \\
 &= n \times T \left(\frac{8n^3}{3} + 8n^2 - \frac{32n}{3} \right) + n \times T (4n^2 + 12n) \\
 &= T \left(\frac{8n^4}{3} + 12n^3 + \frac{4n^2}{3} \right)
 \end{aligned}$$

Clearly for large n , $CT|_{inverse\ GE} \gg CT|_{inverse\ LU}$ as $CT|_{inverse\ GE}$ has the dominating terms of n^4 and $CT|_{inverse\ LU}$ has the dominating terms of n^3 . For large values of n , Gaussian elimination method would take more computational time (approximately $n/4$ times – prove it) than the LU decomposition method. Typical values of the ratio of the computational time for different values of n are given in Table 1.

Table 1. Comparing computational times of finding inverse of a matrix using LU decomposition and Gaussian elimination.

n	10	100	1000	10000
$CT _{inverse\ GE} / CT _{inverse\ LU}$	3.28	25.83	250.8	2501

Are you convinced now that LU decomposition has its place in solving systems of equations when we have to solve for multiple right-hand side vectors while the coefficient matrix stays unchanged?

The time is calculated by first separately calculating the number of additions, subtractions, multiplications, and divisions in a procedure such as back substitution, etc. We then assume 4 clock cycles each for an add, subtract, or multiply operation, and 16 clock cycles for a divide operation as is the case for a typical AMD®-K7 chip.

http://www.isi.edu/~draper/papers/mwscas07_kwon.pdf

LU Decomposition Method for Solving Simultaneous Linear Equations Quiz

? Quiz 1

The $[L] [U]$ decomposition method is computationally more efficient than Naïve Gauss elimination for solving

- (A). a single set of simultaneous linear equations.
- (B). multiple sets of simultaneous linear equations with different coefficient matrices and the same right-hand side vectors.
- (C). multiple sets of simultaneous linear equations with the same coefficient matrix and different right-hand side vectors.
- (D). less than ten simultaneous linear equations.

? Quiz 2

The lower triangular matrix $[L]$ in the $[L] [U]$ decomposition of the matrix given below

$$\begin{bmatrix} 25 & 5 & 4 \\ 10 & 8 & 16 \\ 8 & 12 & 22 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

is

- (A) $\begin{bmatrix} 1 & 0 & 0 \\ 0.40000 & 1 & 0 \\ 0.32000 & 1.7333 & 1 \end{bmatrix}$
- (B) $\begin{bmatrix} 25 & 5 & 4 \\ 0 & 6 & 14.400 \\ 0 & 0 & -4.2400 \end{bmatrix}$

$$(C) \begin{bmatrix} 1 & 0 & 0 \\ 10 & 1 & 0 \\ 8 & 12 & 0 \end{bmatrix}$$

$$(D) \begin{bmatrix} 1 & 0 & 0 \\ 0.40000 & 1 & 0 \\ 0.32000 & 1.5000 & 1 \end{bmatrix}$$

? Quiz 3

The upper triangular matrix $[U]$ in the $[L][U]$ decomposition of the matrix given below

$$\begin{bmatrix} 25 & 5 & 4 \\ 0 & 8 & 16 \\ 0 & 12 & 22 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

is

$$(A) \begin{bmatrix} 1 & 0 & 0 \\ 0.40000 & 1 & 0 \\ 0.32000 & 1.7333 & 1 \end{bmatrix}$$

$$(B) \begin{bmatrix} 25 & 5 & 4 \\ 0 & 6 & 14.400 \\ 0 & 0 & -4.2400 \end{bmatrix}$$

$$(C) \begin{bmatrix} 25 & 5 & 4 \\ 0 & 8 & 16 \\ 0 & 0 & -2 \end{bmatrix}$$

$$(D) \begin{bmatrix} 1 & 0.2000 & 0.16000 \\ 0 & 1 & 2.4000 \\ 0 & 0 & -4.240 \end{bmatrix}$$

$$[>$$

? Quiz 4

For a given 2000×2000 matrix $[A]$, assume that it takes about 15 seconds to find the inverse of $[A]$ by the use of the $[L][U]$ decomposition method, that is, finding the $[L][U]$ once, and then doing forward substitution and back substitution 2000 times using the 2000 columns of the identity matrix as the right hand side vector. The approximate time, in seconds, that it will take to find the inverse if found by repeated use of the Naive Gauss elimination method, that is, doing forward elimination and back substitution 2000 times by using the 2000 columns of the identity matrix as the right hand side vector is most nearly

- (A) 300
- (B) 1500
- (C) 7500
- (D) 30000

? Quiz 5

The algorithm for solving a set of n equations $[A][X] = [C]$, where $[A] = [L][U]$ involves solving $[L][Z] = [C]$ by forward substitution. The algorithm to solve $[L][Z] = [C]$ is given by

- (A) $z_1 = c_1/l_{11}$
- for i from 2 to n do
- sum = 0


```

for  $j$  from 1 to  $i$  do
sum = sum +  $l_{ij} * z_j$ 
end do
 $z_i = (c_i - \text{sum}) / l_{ii}$ 
end do
(B)  $z_1 = c_1 / l_{11}$ 
for  $i$  from 2 to  $n$  do
sum = 0
for  $j$  from 1 to  $(i - 1)$  do
sum = sum +  $l_{ij} * z_j$ 
end do
 $z_i = (c_i - \text{sum}) / l_{ii}$ 
end do
(C)  $z_1 = c_1 / l_{11}$ 
for  $i$  from 2 to  $n$  do
for  $j$  from 1 to  $(i - 1)$  do
sum = sum +  $l_{ij} * z_j$ 
end do
 $z_i = (c_i - \text{sum}) / l_{ii}$ 
end do
(D) for  $i$  from 2 to  $n$  do
sum = 0
for  $j$  from 1 to  $(i - 1)$  do
sum = sum +  $l_{ij} * z_j$ 
end do
 $z_i = (c_i - \text{sum}) / l_{ii}$ 
end do

```

? Quiz 6

To solve boundary value problems, a numerical method based on finite difference method is used. This results in simultaneous linear equations with tridiagonal coefficient matrices. These are solved using a specialized $[L][U]$ decomposition method.

Choose the set of equations that approximately solves the boundary value problem

$$\frac{d^2 y}{dx^2} = 6x - 0.5x^2, \quad y(0) = 0, \quad y(12) = 0, \quad 0 \leq x \leq 12$$

The second derivative in the above equation is approximated by the second-order accurate central divided difference approximation as learned in the differentiation module (Chapter 02.02). A step size of $h = 4$ is used, and hence the value of y can be found approximately at equidistantly placed 4 nodes between $x = 0$ and $x = 12$.

$$(A) \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.0625 & 0.125 & 0.0625 & 0 \\ 0 & 0.0625 & 0.125 & 0.0625 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 16.0 \\ 16.0 \\ 0 \end{bmatrix}$$

$$(B) \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0.0625 & -0.125 & 0.0625 & 0 \\ 0 & 0.0625 & -0.125 & 0.0625 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 16.0 \\ 16.0 \\ 0 \end{bmatrix}$$

$$(C) \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0.0625 & -0.125 & 0.0625 & 0 \\ 0 & 0.0625 & -0.125 & 0.0625 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 16.0 \\ 16.0 \\ 0 \end{bmatrix}$$

$$(D) \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0.0625 & 0.125 & 0.0625 & 0 \\ 0 & 0.0625 & 0.125 & 0.0625 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 16.0 \\ 16.0 \end{bmatrix}$$

LU Decomposition Method for Solving Simultaneous Linear Equations Exercise

? Exercise 1

Show that LU decomposition is computationally a more efficient way of finding the inverse of a square matrix than using Gaussian elimination.

? Exercise 2

Use LU decomposition to find [L] and [U]

$$4x_1 + x_2 - x_3 = -2$$

$$5x_1 + x_2 + 2x_3 = 4$$

$$6x_1 + x_2 + x_3 = 6$$

? Exercise 3

Find the inverse

$$[A] = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 5 \end{bmatrix}$$

using LU decomposition.

? Exercise 4

Fill in the blanks for the unknowns in the LU decomposition of the matrix given below

$$\begin{bmatrix} 25 & 5 & 4 \\ 75 & 7 & 16 \\ 12.5 & 12 & 22 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 25 & 5 & 4 \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

? Exercise 5

Show that the nonsingular matrix

$$[A] = \begin{bmatrix} 0 & 2 \\ 2 & 0 \end{bmatrix}$$

cannot be decomposed into LU form.

? Exercise 6

The LU decomposition of

$$[A] = \begin{bmatrix} 4 & 1 & -1 \\ 5 & 1 & 2 \\ 6 & 1 & 1 \end{bmatrix}$$

is given by

$$\begin{bmatrix} 4 & 1 & -1 \\ 5 & 1 & 2 \\ 6 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1.25 & 1 & 0 \\ 1.5 & 2 & 1 \end{bmatrix} \begin{bmatrix} ?? & ?? & ?? \\ 0 & ?? & ?? \\ 0 & 0 & ?? \end{bmatrix}$$

Find the upper triangular matrix in the above decomposition?

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8: Gauss-Seidel Method

Learning Objectives

After reading this chapter, you should be able to:

1. solve a set of equations using the Gauss-Seidel method,
2. recognize the advantages and pitfalls of the Gauss-Seidel method, and
3. determine under what conditions the Gauss-Seidel method always converges.

Why do we need another method to solve a set of simultaneous linear equations?

In certain cases, such as when a system of equations is large, iterative methods of solving equations are more advantageous. Elimination methods, such as Gaussian elimination, are prone to large round-off errors for a large set of equations. Iterative methods, such as the Gauss-Seidel method, give the user control of the round-off error. Also, if the physics of the problem are well known, initial guesses needed in iterative methods can be made more judiciously leading to faster convergence.

What is the algorithm for the Gauss-Seidel method? Given a general set of n equations and n unknowns, we have

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \dots + a_{1n}x_n = c_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \dots + a_{2n}x_n = c_2$$

$$\vdots \quad \quad \quad \vdots$$

$$a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \dots + a_{nn}x_n = c_n$$

If the diagonal elements are non-zero, each equation is rewritten for the corresponding unknown, that is, the first equation is rewritten with x_1 on the left hand side, the second equation is rewritten with x_2 on the left hand side and so on as follows

$$x_1 = \frac{c_1 - a_{12}x_2 - a_{13}x_3 - \dots - a_{1n}x_n}{a_{11}}$$

$$x_2 = \frac{c_2 - a_{21}x_1 - a_{23}x_3 - \dots - a_{2n}x_n}{a_{22}}$$

$$\vdots$$

$$\vdots$$

$$x_{n-1} = \frac{c_{n-1} - a_{n-1,1}x_1 - a_{n-1,2}x_2 - \dots - a_{n-1,n-2}x_{n-2} - a_{n-1,n}x_n}{a_{n-1,n-1}}$$

$$x_n = \frac{c_n - a_{n1}x_1 - a_{n2}x_2 - \dots - a_{n,n-1}x_{n-1}}{a_{nn}}$$

These equations can be rewritten in a summation form as

$$x_1 = \frac{c_1 - \sum_{j=2}^n a_{1j}x_j}{a_{11}}$$

$$x_2 = \frac{c_2 - \sum_{j=1}^n a_{2j}x_j}{a_{22}}$$

$$\vdots$$

$$x_{n-1} = \frac{c_{n-1} - \sum_{j=1}^n a_{n-1,j}x_j}{a_{n-1,n-1}}$$

$$x_n = \frac{c_n - \sum_{j=1, j \neq n}^n a_{nj}x_j}{a_{nn}}$$

Hence for any row i ,

$$x_i = \frac{c_i - \sum_{j=1, j \neq i}^n a_{ij}x_j}{a_{ii}}, i = 1, 2, \dots, n.$$

Now to find x_i 's, one assumes an initial guess for the x_i 's and then uses the rewritten equations to calculate the new estimates. Remember, one always uses the most recent estimates to calculate the next estimates, x_i . At the end of each iteration, one calculates the absolute relative approximate error for each x_i as

$$|\epsilon_a|_i = \left| \frac{x_i^{\text{new}} - x_i^{\text{old}}}{x_i^{\text{new}}} \right| \times 100$$

where x_i^{new} is the recently obtained value of x_i , and x_i^{old} is the previous value of x_i .

When the absolute relative approximate error for each x_i is less than the pre-specified tolerance, the iterations are stopped.

✓ Example 1

The upward velocity of a rocket is given at three different times in the following table

Table 1: Velocity vs. time data.

Time, $t(s)$	Velocity, $v(m/s)$
5	106.8
8	177.2
12	279.2

The velocity data is approximated by a polynomial as

$$v(t) = a_1t^2 + a_2t + a_3, 5 \leq t \leq 12$$

Find the values of a_1 , a_2 , and a_3 using the Gauss-Seidel method. Assume an initial guess of the solution as

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

and conduct two iterations.

Solution

The polynomial is going through three data points (t_1, v_1) , (t_2, v_2) , and (t_3, v_3) where from the above table

$$t_1 = 5, v_1 = 106.8$$

$$t_2 = 8, v_2 = 177.2$$

$$t_3 = 12, v_3 = 279.2$$

Requiring that $v(t) = a_1t^2 + a_2t + a_3$ passes through the three data points gives

$$v(t_1) = v_1 = a_1t_1^2 + a_2t_1 + a_3$$

$$v(t_2) = v_2 = a_1t_2^2 + a_2t_2 + a_3$$

$$v(t_3) = v_3 = a_1t_3^2 + a_2t_3 + a_3$$

Substituting the data (t_1, v_1) , (t_2, v_2) , and (t_3, v_3) gives

$$a_1 (5^2) + a_2 (5) + a_3 = 106.8$$

$$a_1 (8^2) + a_2 (8) + a_3 = 177.2$$

$$a_1 (12^2) + a_2 (12) + a_3 = 279.2$$

or

$$25a_1 + 5a_2 + a_3 = 106.8$$

$$64a_1 + 8a_2 + a_3 = 177.2$$

$$144a_1 + 12a_2 + a_3 = 279.2$$

The coefficients a_1 , a_2 , and a_3 for the above expression are given by

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

Rewriting the equations gives

$$a_1 = \frac{106.8 - 5a_2 - a_3}{25}$$

$$a_2 = \frac{177.2 - 64a_1 - a_3}{8}$$

$$a_3 = \frac{279.2 - 144a_1 - 12a_2}{1}$$

Iteration #1

Given the initial guess of the solution vector as

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

we get

$$\begin{aligned} a_1 &= \frac{106.8 - 5(2) - (5)}{25} \\ &= 3.6720 \end{aligned}$$

$$\begin{aligned} a_2 &= \frac{177.2 - 64(3.6720) - (5)}{8} \\ &= -7.8150 \end{aligned}$$

$$\begin{aligned} a_3 &= \frac{279.2 - 144(3.6720) - 12(-7.8150)}{1} \\ &= -155.36 \end{aligned}$$

The absolute relative approximate error for each x_i then is

$$\begin{aligned} |\epsilon_a|_1 &= \left| \frac{3.6720 - 1}{3.6720} \right| \times 100 \\ &= 72.76\% \end{aligned}$$

$$\begin{aligned} |\epsilon_a|_2 &= \left| \frac{-7.8150 - 2}{-7.8150} \right| \times 100 \\ &= 125.47\% \end{aligned}$$

$$|\epsilon_a|_3 = \left| \frac{-155.36 - 5}{-155.36} \right| \times 100 \\ = 103.22\%$$

At the end of the first iteration, the estimate of the solution vector is

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 3.6720 \\ -7.8510 \\ -155.36 \end{bmatrix}$$

and the maximum absolute relative approximate error is 125.47.

Iteration #2

The estimate of the solution vector at the end of Iteration #1 is

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 3.6720 \\ -7.8510 \\ -155.36 \end{bmatrix}$$

Now we get

$$a_1 = \frac{106.8 - 5(-7.8510) - (-155.36)}{25} \\ = 12.056 \\ a_2 = \frac{177.2 - 64(12.056) - (-155.36)}{8} \\ = -54.882 \\ a_3 = \frac{279.2 - 144(12.056) - 12(-54.882)}{1} \\ = -798.34$$

The absolute relative approximate error for each x_i then is

$$|\epsilon_a|_1 = \left| \frac{12.056 - 3.6720}{12.056} \right| \times 100 \\ = 69.543\% \\ |\epsilon_a|_2 = \left| \frac{-54.882 - (-7.8510)}{-54.882} \right| \times 100 \\ = 85.695\% \\ |\epsilon_a|_3 = \left| \frac{-798.34 - (-155.36)}{-798.34} \right| \times 100 \\ = 80.540\%$$

At the end of the second iteration the estimate of the solution vector is

$$\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 12.056 \\ -54.882 \\ -798.54 \end{bmatrix}$$

and the maximum absolute relative approximate error is 85.695%

Conducting more iterations gives the following values for the solution vector and the corresponding absolute relative approximate errors.

Iteration	a_1	$ \epsilon_a _1\%$	a_2	$ \epsilon_a _2\%$	a_3	$ \epsilon_a _3\%$
1	3.672	72.767	-7.8510	125.47	-155.36	103.22

Iteration	42.056	69.543	454.882	85.695	4798.34	80.54
3	37182	71.467	-235510	78521	-34489	76.332
4	193.33	75.595	-1093.4	76.632	-14440	76.116
2	12.056	69.543	-54.882	85.695	-798.34	80.54
5	800.53	75.85	-4577.2	76.112	-60072	75.963
3	47.182	74.447	-255.51	78.521	-3448.9	76.852
6	3322.6	75.906	-19049	75.972	-249580	75.931
4	193.33	75.595	-1093.4	76.632	-14440	76.116
5	800.53	75.85	-4577.2	76.112	-60072	75.963
As seen in the above table, the solution estimates are not converging to the true solution of	3322.6	75.906	-19049	75.972	-249580	75.931

$$a_1 = 0.29048$$

$$a_2 = 19.690$$

$$a_3 = 1.0857$$

The above system of equations does not seem to converge. Why?

Well, a pitfall of most iterative methods is that they may or may not converge. However, the solution to a certain class of system of simultaneous equations does always converge using the Gauss-Seidel method. This class of system of equations is where the coefficient matrix $[A]$ in $[A][X] = [C]$ is diagonally dominant, that is

$$|a_{ii}| \geq \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}| \text{ for all } i$$

$$|a_{ii}| > \sum_{\substack{j=1 \\ j \neq i}}^n |a_{ij}| \text{ for at least one } i$$

If a system of equations has a coefficient matrix that is not diagonally dominant, it may or may not converge. Fortunately, many physical systems that result in simultaneous linear equations have a diagonally dominant coefficient matrix, which then assures convergence for iterative methods such as the Gauss-Seidel method of solving simultaneous linear equations.

✓ Example 2

Find the solution to the following system of equations using the Gauss-Seidel method.

$$12x_1 + 3x_2 - 5x_3 = 1$$

$$x_1 + 5x_2 + 3x_3 = 28$$

$$3x_1 + 7x_2 + 13x_3 = 76$$

Use

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

as the initial guess and conduct two iterations.

Solution1

The coefficient matrix

$$[A] = \begin{bmatrix} 12 & 3 & -5 \\ 1 & 5 & 3 \\ 3 & 7 & 13 \end{bmatrix}$$

is diagonally dominant as

$$|a_{11}| = |12| = 12 \geq |a_{12}| + |a_{13}| = |3| + |-5| = 8$$

$$|a_{22}| = |5| = 5 \geq |a_{21}| + |a_{23}| = |1| + |3| = 4$$

$$|a_{33}| = |13| = 13 \geq |a_{31}| + |a_{32}| = |3| + |7| = 10$$

and the inequality is strictly greater than for at least one row. Hence, the solution should converge using the Gauss-Seidel method.

Rewriting the equations, we get

$$x_1 = \frac{1 - 3x_2 + 5x_3}{12}$$

$$x_2 = \frac{28 - x_1 - 3x_3}{5}$$

$$x_3 = \frac{76 - 3x_1 - 7x_2}{13}$$

Assuming an initial guess of

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Iteration #1

$$x_1 = \frac{1 - 3(0) + 5(1)}{12}$$

$$= 0.50000$$

$$x_2 = \frac{28 - (0.50000) - 3(1)}{5}$$

$$= 4.9000$$

$$x_3 = \frac{76 - 3(0.50000) - 7(4.9000)}{13}$$

$$= 3.0923$$

The absolute relative approximate error at the end of the first iteration is

$$|\epsilon_a|_1 = \left| \frac{0.50000 - 1}{0.50000} \right| \times 100$$

$$= 100.00\%$$

$$|\epsilon_a|_2 = \left| \frac{4.9000 - 0}{4.9000} \right| \times 100$$

$$= 100.00\%$$

$$|\epsilon_a|_3 = \left| \frac{3.0923 - 1}{3.0923} \right| \times 100$$

$$= 67.662\%$$

The maximum absolute relative approximate error is 100.00%

Iteration #2

$$x_1 = \frac{1 - 3(4.9000) + 5(3.0923)}{12}$$

$$= 0.14679$$

$$x_2 = \frac{28 - (0.14679) - 3(3.0923)}{5}$$

$$= 3.7153$$

$$x_3 = \frac{76 - 3(0.14679) - 7(3.7153)}{13}$$

$$= 3.8118$$

At the end of second iteration, the absolute relative approximate error is

$$|\epsilon_a|_1 = \left| \frac{0.14679 - 0.50000}{0.14679} \right| \times 100$$

$$= 240.61\%$$

$$|\epsilon_a|_2 = \left| \frac{3.7153 - 4.9000}{3.7153} \right| \times 100$$

$$= 31.889\%$$

$$|\epsilon_a|_3 = \left| \frac{3.8118 - 3.0923}{3.8118} \right| \times 100$$

$$= 18.874\%$$

The maximum absolute relative approximate error is 240.61%. This is greater than the value of 100.00% we obtained in the first iteration. Is the solution diverging? No, as you conduct more iterations, the solution converges as follows.

Iteration	a_1	$ \epsilon_a _1\%$	a_2	$ \epsilon_a _2\%$	a_3	$ \epsilon_a _3\%$
1	0.5	100	4.9	100	3.0923	67.662
2	0.14679	240.61	3.7153	31.889	3.8118	18.874
3	0.74275	80.236	3.1644	17.408	3.9708	4.0064
4	0.94675	21.546	3.0281	4.4996	3.9971	0.65772
5	0.99177	4.5391	3.0034	0.82499	4.0001	0.074383
6	0.99919	0.74307	3.0001	0.10856	4.0001	0.00101

This is close to the exact solution vector of

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$$

✓ Example 3

Given the system of equations

$$3x_1 + 7x_2 + 13x_3 = 76$$

$$x_1 + 5x_2 + 3x_3 = 28$$

$$12x_1 + 3x_2 - 5x_3 = 1$$

find the solution using the Gauss-Seidel method. Use

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

as the initial guess.

Solution1

Rewriting the equations, we get

$$\begin{aligned}x_1 &= \frac{76 - 7x_2 - 13x_3}{3} \\x_2 &= \frac{28 - x_1 - 3x_3}{5} \\x_3 &= \frac{1 - 12x_1 - 3x_2}{-5}\end{aligned}$$

Assuming an initial guess of

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

the next six iterative values are given in the table below.

Iteration	a_1	$ \epsilon_a _1\%$	a_2	$ \epsilon_a _2\%$	a_3	$ \epsilon_a _3\%$
1	21	95.238	0.8	100	50.68	98.027
2	-196.15	110.71	14.421	94.453	-462.30	110.96
3	1995	109.83	-116.02	112.43	4718.1	109.8
4	-20149	109.9	1204.6	109.63	-47636	109.9
5	2.0364×10^5	109.89	-12140	109.92	4.8144×10^5	109.89
6	-2.0579×10^6	109.89	1.2272×10^5	109.89	-4.8653×10^6	109.89

You can see that this solution is not converging and the coefficient matrix is not diagonally dominant. The coefficient matrix

$$[A] = \begin{bmatrix} 3 & 7 & 13 \\ 1 & 5 & 3 \\ 12 & 3 & -5 \end{bmatrix}$$

is not diagonally dominant as

$$|a_{11}| = |3| = 3 \leq |a_{12}| + |a_{13}| = |7| + |13| = 20$$

Hence, the Gauss-Seidel method may or may not converge.

However, it is the same set of equations as the previous example and that converged. The only difference is that we exchanged first and the third equation with each other and that made the coefficient matrix not diagonally dominant.

Therefore, it is possible that a system of equations can be made diagonally dominant if one exchanges the equations with each other. However, it is not possible for all cases. For example, the following set of equations

$$\begin{aligned}x_1 + x_2 + x_3 &= 3 \\2x_1 + 3x_2 + 4x_3 &= 9 \\x_1 + 7x_2 + x_3 &= 9\end{aligned}$$

cannot be rewritten to make the coefficient matrix diagonally dominant.

Gauss-Seidel Method Quiz

? Quiz 1

A square matrix $[A]_{n \times n}$ is diagonally dominant if

(A) $|a_{ii}| \geq \sum_{\substack{j=1 \\ i \neq j}}^n |a_{ij}|, i = 1, 2, \dots, n$

(B) $|a_{ii}| \geq \sum_{\substack{j=1 \\ i \neq j}}^n |a_{ij}|, i = 1, 2, \dots, n$ and $|a_{ii}| > \sum_{\substack{j=1 \\ i \neq j}}^n |a_{ij}|$, for any $i = 1, 2, \dots, n$

(C) $|a_{ii}| \geq \sum_{j=1}^n |a_{ij}|, i = 1, 2, \dots, n$ and $|a_{ii}| > \sum_{j=1}^n |a_{ij}|$, for any $i = 1, 2, \dots, n$

(D) $|a_{ii}| \geq \sum_{j=1}^n |a_{ij}|, i = 1, 2, \dots, n$

? Quiz 2

Using $[x_1, x_2, x_3] = [1, 3, 5]$ as the initial guess, the values of $[x_1, x_2, x_3]$ after three iterations in the Gauss-Seidel method for

$$\begin{bmatrix} 12 & 7 & 3 \\ 1 & 5 & 1 \\ 2 & 7 & -11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -5 \\ 6 \end{bmatrix}$$

are

(A) $[-2.8333 - 1.4333 - 1.9727]$

(B) $[1.4959 - 0.90464 - 0.84914]$

(C) $[0.90666 - 1.0115 - 1.0243]$

(D) $[1.2148 - 0.72060 - 0.82451]$

? Quiz 3

To ensure that the following system of equations,

$$2x_1 + 7x_2 - 11x_3 = 6$$

$$x_1 + 2x_2 + x_3 = -5$$

$$7x_1 + 5x_2 + 2x_3 = 17$$

converges using the Gauss-Seidel method, one can rewrite the above equations as follows:

(A) $\begin{bmatrix} 2 & 7 & -11 \\ 1 & 2 & 1 \\ 7 & 5 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -5 \\ 17 \end{bmatrix}$

(B) $\begin{bmatrix} 7 & 5 & 2 \\ 1 & 2 & 1 \\ 2 & 7 & -11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 17 \\ -5 \\ 6 \end{bmatrix}$

(C) $\begin{bmatrix} 7 & 5 & 2 \\ 1 & 2 & 1 \\ 2 & 7 & -11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -5 \\ 17 \end{bmatrix}$

(D) The equations cannot be rewritten in a form to ensure convergence.

? Quiz 4

For $\begin{bmatrix} 12 & 7 & 3 \\ 1 & 5 & 1 \\ 2 & 7 & -11 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 22 \\ 7 \\ -2 \end{bmatrix}$ and using $[x_1 \ x_2 \ x_3] = [1 \ 2 \ 1]$ as the initial guess, the values of $[x_1 \ x_2 \ x_3]$ are found at the end of each iteration as

Iteration #	x_1	x_2	x_3
1	0.41667	1.1167	0.96818
2	0.93990	1.0184	1.0008
3	0.98908	1.0020	0.99931
4	0.99899	1.0003	1.0000

At what first iteration number would you trust at least 1 significant digit in your solution?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

? Quiz 5

The algorithm for the Gauss-Seidel method to solve $[A][X] = [C]$ is given as follows when using n max iterations. The initial value of $[X]$ is stored in $[X]$.

```
(A) Sub
Seidel( $n, a, x, rhs, nmax$ )
For  $k = 1$  To  $nmax$ 
For  $i = 1$  To  $n$ 
For  $j = 1$  To  $n$ 
If ( $i <> j$ ) Then
Sum = Sum +  $a(i, j) * x(j)$ 
endif
Next  $j$ 
 $x(i) = (rhs(i) - Sum) / a(i, i)$ 
Next  $i$ 
Next  $j$ 
End Sub
```

```
(B) Sub
Seidel( $n, a, x, rhs, nmax$ )
For  $k = 1$  To  $nmax$ 
For  $i = 1$  To  $n$ 
Sum = 0
For  $j = 1$  To  $n$ 
If ( $i <> j$ ) Then
Sum = Sum +  $a(i, j) * x(j)$ 
endif
Next  $j$ 
 $x(i) = (rhs(i) - Sum) / a(i, i)$ 
Next  $i$ 
Next  $k$ 
End Sub
```

```
(C) Sub
Seidel(n, a, x, rhs, nmax)
For k = 1 To nmax
For i = 1 To n
Sum = 0
For j = 1 To n
Sum = Sum + a(i, j) * x(j)
Next j
x(i) = (rhs(i) - Sum) / a(i, i)
Next i
Next k
End Sub
```

```
(D) Sub
Seidel(n, a, x, rhs, nmax)
For k = 1 To {nmax}
For i = 1 To n
Sum = 0
For j = 1 To n
If (i <> j) Then
Sum = Sum + a(i, j) * x(j)
endif
Next j
x(i) = (rhs(i) - Sum) / a(i, i)
Next i
Next k
End Sub
```

? Quiz 6

Thermistors measure temperature, have a nonlinear output and are valued for a limited range. So when a thermistor is manufactured, the manufacturer supplies a resistance vs. temperature curve. An accurate representation of the curve is generally given by

$$\frac{1}{T} = a_0 + a_1 \ln(R) + a_2 \{\ln(R)\}^2 + a_3 \{\ln(R)\}^3$$

where T is temperature in Kelvin, R is resistance in ohms, and a_0, a_1, a_2, a_3 are constants of the calibration curve. Given the following for a thermistor

R	T
ohm	$^{\circ}C$
1101.0	25.113
911.3	30.131
636.0	40.120
451.1	50.128

the value of temperature in $^{\circ}C$ for a measured resistance of 900 ohms most nearly is

- (A) 30.002
- (B) 30.473
- (C) 31.272
- (D) 31.445

Gauss-Seidel Method Exercise

? Exercise 1

In a system of equation $[A][X] = [C]$, if $[A]$ is diagonally dominant, then GaussSeidal-Seidel method

- A. always converges
- B. may or may not converge
- C. always diverges

Answer

A

? Exercise 2

In a system of equations $[A][X] = [C]$, if $[A]$ is not diagonally dominant, then Gauss-Seidel method

- A. Always converges
- B. May or may not converge
- C. Always diverges.

Answer

B

? Exercise 3

In a system of equations $[A][X] = [C]$, if $[A]$ is not diagonally dominant, the system of equations can always be rewritten to make it diagonally dominant.

- A. True
- B. False

Answer

B

? Exercise 4

Solve the following system of equations using Gauss-Seidel method

$$\begin{aligned} 12x_1 + 7x_2 + 3x_3 &= 2 \\ x_1 + 5x_2 + x_3 &= -5 \\ 2x_1 + 7x_2 - 11x_3 &= 6 \end{aligned}$$

Conduct 3 iterations, calculate the maximum absolute relative approximate error at the end of each iteration and choose $[x_1 \ x_2 \ x_3] = [1 \ 3 \ 5]$ as your initial guess.

Answer

$$\begin{aligned} [x_1 \ x_2 \ x_3] &= [0.90666 \ -1.0115 \ -1.0243] \quad [|\epsilon_a|_1 \ |\epsilon_a|_2 \ |\epsilon_a|_3] = [65.001\% \ 10.564\% \ 17.099\%] \\ [|\epsilon_a|_1 \ |\epsilon_a|_2 \ |\epsilon_a|_3] &= [65.001\% \ 10.564\% \ 17.099\%] \end{aligned}$$

? Exercise 5

Solve the following system of equations using Gauss-Seidel method

$$\begin{aligned} 12x_1 + 7x_2 + 3x_3 &= 2 \\ x_1 + 5x_2 + x_3 &= -5 \\ 2x_1 + 7x_2 - 11x_3 &= 6 \end{aligned}$$

Conduct 3 iterations, calculate the maximum absolute relative approximate error at the end of each iteration, and choose $[x_1 \ x_2 \ x_3] = [1 \ 3 \ 5]$ as your initial guess.

Answer

$$[x_1 \ x_2 \ x_3] = [0.90666 \ -1.0115 \ -1.0243]$$

? Exercise 6

Solve the following system of equations using Gauss-Seidel Method

$$x_1 + 5x_2 + x_3 = 5$$

$$12x_1 + 7x_2 + 3x_3 = 2$$

$$2x_1 + 7x_2 - 11x_3 = 6$$

Conduct 3 iterations, calculate the maximum absolute relative approximate error at the end of each iteration, and choose $[x_1 \ x_2 \ x_3] = [1 \ 3 \ 5]$ as your initial guess.

Answer

$$[x_1 \ x_2 \ x_3] = [-1163.7 \ 1947.6 \ 1027.2]$$

$$[|\epsilon_a|_1 \ |\epsilon_a|_2 \ |\epsilon_a|_3] = [89.156\% \ 89.139\% \ 89.183\%]$$

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9: Adequacy of Solutions

Learning Objectives

After reading this chapter, you should be able to:

1. know the difference between ill-conditioned and well-conditioned systems of equations,
2. define and find the norm of a matrix
3. define and evaluate the condition number of an invertible square matrix
4. relate the condition number of a coefficient matrix to the ill or well conditioning of the system of simultaneous linear equations, that is, how much can you trust the solution of the simultaneous linear equations.

What do you mean by ill-conditioned and well-conditioned system of equations?

A system of equations is considered to be **well-conditioned** if a small change in the coefficient matrix or a small change in the right hand side results in a small change in the solution vector.

A system of equations is considered to be **ill-conditioned** if a small change in the coefficient matrix or a small change in the right hand side results in a large change in the solution vector.

✓ Example 1

Is this system of equations well-conditioned?

$$\begin{bmatrix} 1 & 2 \\ 2 & 3.999 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7.999 \end{bmatrix}$$

Solution

The solution to the above set of equations is

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Make a small change in the right-hand side vector of the equations

$$\begin{bmatrix} 1 & 2 \\ 2 & 3.999 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4.001 \\ 7.998 \end{bmatrix}$$

gives

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -3.999 \\ 4.000 \end{bmatrix}$$

Make a small change in the coefficient matrix of the equations

$$\begin{bmatrix} 1.001 & 2.001 \\ 2.001 & 3.998 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7.999 \end{bmatrix}$$

gives

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3.994 \\ 0.001388 \end{bmatrix}$$

This last systems of equation “looks” ill-conditioned because a small change in the coefficient matrix or the right hand side resulted in a large change in the solution vector.

✓ Example 2

Is this system of equations well-conditioned?

$$\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

Solution

The solution to the above equations is

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Make a small change in the right-hand side vector of the equations.

$$\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4.001 \\ 7.001 \end{bmatrix}$$

gives

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1.999 \\ 1.001 \end{bmatrix}$$

Make a small change in the coefficient matrix of the equations.

$$\begin{bmatrix} 1.001 & 2.001 \\ 2.001 & 3.001 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

gives

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2.003 \\ 0.997 \end{bmatrix}$$

This system of equation “looks” well-conditioned because small changes in the coefficient matrix or the right-hand side resulted in small changes in the solution vector.

So, what if the system of equations is ill conditioned or well-conditioned?

Well, if a system of equations is ill-conditioned, we cannot trust the solution as much. Revisit the velocity problem, Example 5.1 in Chapter 5. The values in the coefficient matrix $[A]$ are squares of time, etc. For example, if instead of $a_{11} = 25$, you used $a_{11} = 24.99$, would you want this small change to make a huge difference in the solution vector. If it did, would you trust the solution?

Later we will see how much (quantifiable terms) we can trust the solution in a system of equations. Every invertible square matrix has a **condition number** and coupled with the **machine epsilon**, we can quantify how many significant digits one can trust in the solution.

To calculate the condition number of an invertible square matrix, I need to know what the norm of a matrix means. How is the norm of a matrix defined?

Just like the determinant, the norm of a matrix is a simple unique scalar number. However, the norm is always positive and is defined for all matrices – square or rectangular, and invertible or noninvertible square matrices.

One of the popular definitions of a norm is the row sum norm (also called the uniform-matrix norm). For a $m \times n$ matrix $[A]$, the row sum norm of $[A]$ is defined as

$$\|A\|_{\infty} = \max_{1 \leq i \leq m} \sum_{j=1}^n |a_{ij}|$$

that is, find the sum of the absolute value of the elements of each row of the matrix $[A]$. The maximum out of the m such values is the row sum norm of the matrix $[A]$.

✓ Example 3

Find the row sum norm of the following matrix $[A]$.

$$A = \begin{bmatrix} 10 & -7 & 0 \\ -3 & 2.099 & 6 \\ 5 & -1 & 5 \end{bmatrix}$$

Solution

$$\begin{aligned} \|A\|_{\infty} &= \max_{1 \leq i \leq 3} \sum_{j=1}^3 |a_{ij}| \\ &= \max [(|10| + |-7| + |0|), (|-3| + |2.099| + |6|), (|5| + |-1| + |5|)] \\ &= \max [(10 + 7 + 0), (3 + 2.099 + 6), (5 + 1 + 5)] \\ &= \max [17, 11.099, 11] \\ &= 17 \end{aligned}$$

How is the norm related to the conditioning of the matrix?

Let us start answering this question using an example. Go back to the *ill-conditioned* system of equations,

$$\begin{bmatrix} 1 & 2 \\ 2 & 3.999 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7.999 \end{bmatrix}$$

that gives the solution as

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Denoting the above set of equations as

$$[A] [X] = [C]$$

$$\|X\|_{\infty} = 2$$

$$\|C\|_{\infty} = 7.999$$

Making a small change in the right-hand side,

$$\begin{bmatrix} 1 & 2 \\ 2 & 3.999 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4.001 \\ 7.998 \end{bmatrix}$$

gives

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -3.999 \\ 4.000 \end{bmatrix}$$

Denoting the above set of equations by

$$[A] [X'] = [C']$$

right hand side vector is found by

$$[\Delta C] = [C'] - [C]$$

and the change in the solution vector is found by

$$[\Delta X] = [X'] - [X]$$

then

$$\begin{aligned} [\Delta C] &= \begin{bmatrix} 4.001 \\ 7.998 \end{bmatrix} - \begin{bmatrix} 4 \\ 7.999 \end{bmatrix} \\ &= \begin{bmatrix} 0.001 \\ -0.001 \end{bmatrix} \end{aligned}$$

and

$$\begin{aligned} [\Delta X] &= \begin{bmatrix} -3.999 \\ 4.000 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} -5.999 \\ 3.000 \end{bmatrix} \end{aligned}$$

then

$$\|\Delta C\|_{\infty} = 0.001$$

$$\|\Delta X\|_{\infty} = 5.999$$

The relative change in the norm of the solution vector is

$$\begin{aligned} \frac{\|\Delta X\|_{\infty}}{\|X\|_{\infty}} &= \frac{5.999}{2} \\ &= 2.9995 \end{aligned}$$

The relative change in the norm of the right-hand side vector is

$$\begin{aligned} \frac{\|\Delta C\|_{\infty}}{\|C\|_{\infty}} &= \frac{0.001}{7.999} \\ &= 1.250 \times 10^{-4} \end{aligned}$$

See the small relative change of 1.250×10^{-4} in the right-hand side vector norm results in a large relative change in the solution vector norm of 2.9995.

In fact, the ratio between the relative change in the norm of the solution vector and the relative change in the norm of the right-hand side vector is

$$\begin{aligned} \frac{\|\Delta X\|_{\infty}/\|X\|_{\infty}}{\|\Delta C\|_{\infty}/\|C\|_{\infty}} &= \frac{2.9995}{1.250 \times 10^{-4}} \\ &= 23993 \end{aligned}$$

Let us now go back to the *well-conditioned* system of equations.

$$\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

gives

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Denoting the system of equations by

$$[A] [X] = [C]$$

$$\|X\|_{\infty} = 2$$

$$\|C\|_{\infty} = 7$$

Making a small change in the right-hand side vector

$$\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4.001 \\ 7.001 \end{bmatrix}$$

gives

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1.999 \\ 1.001 \end{bmatrix}$$

Denoting the above set of equations by

$$[A] [X'] = [C']$$

the change in the right-hand side vector is then found by

$$[\Delta C] = [C'] - [C]$$

and the change in the solution vector is

$$[\Delta X] = [X'] - [X]$$

then

$$\begin{aligned} [\Delta C] &= \begin{bmatrix} 4.001 \\ 7.001 \end{bmatrix} - \begin{bmatrix} 4 \\ 7 \end{bmatrix} \\ &= \begin{bmatrix} 0.001 \\ 0.001 \end{bmatrix} \end{aligned}$$

and

$$\begin{aligned} [\Delta X] &= \begin{bmatrix} 1.999 \\ 1.001 \end{bmatrix} - \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} -0.001 \\ 0.001 \end{bmatrix} \end{aligned}$$

then

$$\|\Delta C\|_{\infty} = 0.001$$

$$\|\Delta X\|_{\infty} = 0.001$$

The relative change in the norm of solution vector is

$$\begin{aligned} \frac{\|\Delta X\|_{\infty}}{\|X\|_{\infty}} &= \frac{0.001}{2} \\ &= 5 \times 10^{-4} \end{aligned}$$

The relative change in the norm of the right-hand side vector is

$$\begin{aligned} \frac{\|\Delta C\|_{\infty}}{\|C\|_{\infty}} &= \frac{0.001}{7} \\ &= 1.429 \times 10^{-4} \end{aligned}$$

See the small relative change in the right-hand side vector norm of 1.429×10^{-4} results in the small relative change in the solution vector norm of 5×10^{-4} .

In fact, the ratio between the relative change in the norm of the solution vector and the relative change in the norm of the right-hand side vector is

$$\begin{aligned} \frac{\|\Delta X\|_{\infty}/\|X\|_{\infty}}{\|\Delta C\|_{\infty}/\|C\|_{\infty}} &= \frac{5 \times 10^{-4}}{1.429 \times 10^{-4}} \\ &= 3.5 \end{aligned}$$

What are some of the properties of norms?

1. For a matrix $[A]$, $\|A\| \geq 0$
2. For a matrix $[A]$ and a scalar k , $\|kA\| = |k| \|A\|$
3. For two matrices $[A]$ and $[B]$ of same order, $\|A + B\| \leq \|A\| + \|B\|$
4. For two matrices $[A]$ and $[B]$ that can be multiplied as $[A][B]$, $\|AB\| \leq \|A\| \|B\|$

Is there a general relationship that exists between $\|\Delta X\| / \|X\|$ and $\|\Delta C\| / \|C\|$ or between $\|\Delta X\| / \|X\|$ and $\|\Delta A\| / \|A\|$? If so, it could help us identify well-conditioned and ill conditioned system of equations.

If there is such a relationship, will it help us quantify the conditioning of the matrix? That is, will it tell us how many significant digits we could trust in the solution of a system of simultaneous linear equations?

There is a relationship that exists between

$$\frac{\|\Delta X\|}{\|X\|} \text{ and } \frac{\|\Delta C\|}{\|C\|}$$

and between

$$\frac{\|\Delta X\|}{\|X\|} \text{ and } \frac{\|\Delta A\|}{\|A\|}$$

These relationships are

$$\frac{\|\Delta X\|}{\|X\|} \leq \|A\| \|A^{-1}\| \frac{\|\Delta C\|}{\|C\|}$$

and

$$\frac{\|\Delta X\|}{\|X + \Delta X\|} \leq \|A\| \|A^{-1}\| \frac{\|\Delta A\|}{\|A\|}$$

The above two inequalities show that the relative change in the norm of the right-hand side vector or the coefficient matrix can be amplified by as much as $\|A\| \|A^{-1}\|$.

This number $\|A\| \|A^{-1}\|$ is called the **condition number** of the matrix and coupled with the machine epsilon, we can quantify the accuracy of the solution of $[A][X] = [C]$.

Prove for

$$[A][X] = [C]$$

that

$$\frac{\|\Delta X\|}{\|X + \Delta X\|} \leq \|A\| \|A^{-1}\| \frac{\|\Delta A\|}{\|A\|}$$

Proof

Let

$$[A][X] = [C] \quad (1)$$

Then if $[A]$ is changed to $[A']$, the $[X]$ will change to $[X']$, such

that

$$[A'][X'] = [C] \quad (2)$$

From Equations (1) and (2),

$$[A][X] = [A'][X']$$

Denoting change in $[A]$ and $[X]$ matrices as $[\Delta A]$ and $[\Delta X]$, respectively

$$[\Delta A] = [A'] - [A]$$

$$[\Delta X] = [X'] - [X]$$

then

$$[A] [X] = ([A] + [\Delta A]) ([X] + [\Delta X])$$

Expanding the above expression

$$[A] [X] = [A] [X] + [A] [\Delta X] + [\Delta A] [X] + [\Delta A] [\Delta X]$$

$$[0] = [A] [\Delta X] + [\Delta A] ([X] + [\Delta X])$$

$$-[A] [\Delta X] = [\Delta A] ([X] + [\Delta X])$$

$$[\Delta X] = -[A]^{-1} [\Delta A] ([X] + [\Delta X])$$

Applying the theorem of norms, that the norm of multiplied matrices is less than the multiplication of the individual norms of the matrices,

$$\|\Delta X\| \leq \|A^{-1}\| \|\Delta A\| \|X + \Delta X\|$$

Multiplying both sides by $\|A\|$

$$\|A\| \|\Delta X\| \leq \|A\| \|A^{-1}\| \|\Delta A\| \|X + \Delta X\|$$

$$\frac{\|\Delta X\|}{\|X + \Delta X\|} \leq \|A\| \|A^{-1}\| \frac{\|\Delta A\|}{\|A\|}$$

How do I use the above theorems to find how many significant digits are correct in my solution vector?

The relative error in a solution vector norm is $\leq \text{Cond}(A) \times \text{relative error in right hand side vector norm}$.

The possible relative error in the solution vector norm is $\leq \text{Cond}(A) \times \epsilon_{mach}$

Hence $\text{Cond}(A) \times \epsilon_{mach}$ should give us the number of significant digits, m that are at least correct in our solution by finding out the largest value of m for which $\text{Cond}(A) \times \epsilon_{mach}$ is less than 0.5×10^{-m} .

✓ Example 4

How many significant digits can I trust in the solution of the following system of equations?

$$\begin{bmatrix} 1 & 2 \\ 2 & 3.999 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

Solution

For

$$[A] = \begin{bmatrix} 1 & 2 \\ 2 & 3.999 \end{bmatrix}$$

it can be shown

$$[A]^{-1} = \begin{bmatrix} -3999 & 2000 \\ 2000 & -1000 \end{bmatrix}$$

$$\|A\|_{\infty} = 5.999$$

$$\|A^{-1}\|_{\infty} = 5999$$

$$\begin{aligned} \text{Cond}(A) &= \|A\| \|A^{-1}\| \\ &= 5.999 \times 5999.4 \\ &= 35990 \end{aligned}$$

Assuming single precision with 23 bits used in the mantissa for real numbers, the machine epsilon is

$$\begin{aligned}\epsilon_{mach} &= 2^{-23} \\ &= 0.119209 \times 10^{-6}\end{aligned}$$

$$\begin{aligned}Cond(A) \times \epsilon_{mach} &= 35990 \times 0.119209 \times 10^{-6} \\ &= 0.4290 \times 10^{-2}\end{aligned}$$

For what maximum positive value of m , would $Cond(A) \times \epsilon_{mach}$ be less than or equal to 0.5×10^{-m}

$$\begin{aligned}0.4290 \times 10^{-2} &\leq 0.5 \times 10^{-m} \\ 0.8580 \times 10^{-2} &\leq 10^{-m} \\ \log(0.8580 \times 10^{-2}) &\leq \log(10^{-m}) \\ -2.067 &\leq -m \\ m &\leq 2.067 \\ m &\leq 2\end{aligned}$$

So, two significant digits are at least correct in the solution vector.

✓ Example 5

How many significant digits can I trust in the solution of the following system of equations?

$$\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

Solution

For

$$[A] = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$$

it can be shown

$$[A]^{-1} = \begin{bmatrix} -3 & 2 \\ 2 & -1 \end{bmatrix}$$

Then

$$\begin{aligned}\|A\|_{\infty} &= 5, \\ \|A^{-1}\|_{\infty} &= 5. \\ Cond(A) &= \|A\| \|A^{-1}\| \\ &= 5 \times 5 \\ &= 25\end{aligned}$$

Assuming single precision with 23 bits used in the mantissa for real numbers, the machine epsilon

$$\begin{aligned}\epsilon_{mach} &= 2^{-23} \\ &= 0.119209 \times 10^{-6} \\ Cond(A) \times \epsilon_{mach} &= 25 \times 0.119209 \times 10^{-6} \\ &= 0.2980 \times 10^{-5}\end{aligned}$$

For what maximum positive value of m , would $Cond(A) \times \epsilon_{mach}$ be less than or equal to 0.5×10^{-m}

$$0.2980 \times 10^{-5} \leq 0.5 \times 10^{-m}$$

$$m \leq 5$$

So, five significant digits are at least correct in the solution vector.

Adequacy of Solutions Quiz

? Quiz 1

The row sum norm of the matrix

$$[A] = \begin{bmatrix} 6 & -7 & 3 & 13 \\ 19 & -21 & 23 & -29 \\ 41 & 47 & -51 & 61 \end{bmatrix}$$

is

- (A) 29
- (B) 61
- (C) 98
- (D) 200

? Quiz 2

The adequacy of the solution of simultaneous linear equations

$$[A] [X] = [C]$$

depends on

- (A) the condition number of coefficient matrix $[A]$
- (B) the machine epsilon
- (C) the condition number for matrix $[A]$ and the machine epsilon
- (D) norm of the coefficient matrix $[A]$

? Quiz 3

Given a set of equations in matrix form

$$\begin{aligned} [A] [X] &= [C], \\ \|A\| &= 250, \|A^{-1}\| = 40 \text{ and} \\ \varepsilon_{mach} &= 0.119 \times 10^{-6}, \end{aligned}$$

then the number of significant digits you can at least trust in the solution are

- (A) 1
- (B) 2
- (C) 3
- (D) 4

? Quiz 4

The solution to a set of simultaneous linear equations

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 44 \\ 94 \\ 138 \end{bmatrix}$$

is given as

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 7 \end{bmatrix}$$

The solution to another set of simultaneous linear equations is given by (note the coefficient matrix is the same as above)

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 43.99 \\ 93.98 \\ 138.03 \end{bmatrix}$$

is given as

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 214.01 \\ -208.01 \\ 60 \end{bmatrix}$$

Based on the row sum norm, the condition number of the coefficient matrix is greater than (choose the largest possible value)

- (A) 1
- (B) 138
- (C) 4500
- (D) 139320

? Quiz 5

The condition number of the $n \times n$ identity matrix based on the row sum norm is

- (A) 0
- (B) 1
- (C) n
- (D) n^2

? Quiz 6

Let $[A] = \begin{bmatrix} 1 & 2+\delta \\ 2-\delta & 1 \end{bmatrix}$. Based on the row sum norm and given that $\delta \rightarrow 0, \delta > 0$, the condition number of the matrix is

- (A) $\frac{3-\delta}{3+\delta}$
- (B) $\frac{9-\delta^2}{3-\delta^2}$
- (C) $\frac{(3+\delta)^2}{3-\delta^2}$
- (D) $\frac{3-2\delta-\delta^2}{3-\delta^2}$

Adequacy of Solutions Exercise

? Exercise 1

The adequacy of the solution of simultaneous linear equations depends on

- A. Condition number
- B. Machine epsilon
- C. Product of condition number and machine epsilon
- D. Norm of the matrix.

Answer

C

? Exercise 2

If a system of equations $[A] [X] = [C]$ is ill conditioned, then

- A. $\det(A) = 0$
- B. $Cond(A) = 1$
- C. $Cond(A)$ is large.
- D. $\|A\|$ is large.

Answer

C

? Exercise 3

If $Cond(A) = 10^4$ and $\epsilon_{mach} = 0.119 \times 10^{-6}$, then in $[A] [X] = [C]$, at least these many significant digits are correct in your solution,

- A. 0
- B. 1
- C. 2
- D. 3

Answer

C

? Exercise 4

Make a small change in the coefficient matrix of $\begin{bmatrix} 1 & 2 \\ 2 & 3.999 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7.999 \end{bmatrix}$ and find

$$\frac{\|\Delta X\|_{\infty} / \|X\|_{\infty}}{\|\Delta A\|_{\infty} / \|A\|_{\infty}}$$

Is it a large or small number? How is this number related to the condition number of the coefficient matrix?

Answer

Changing $[A]$ to

$$\begin{bmatrix} 1.001 & 2.001 \\ 2.001 & 4.000 \end{bmatrix}$$

Results in solution of

$$\begin{bmatrix} 5999 \\ -2999 \end{bmatrix}$$

$$\frac{\frac{\|\Delta X\|_\infty}{\|X\|_\infty}}{\frac{\|\Delta A\|_\infty}{\|A\|_\infty}} = \frac{\frac{5999.7}{2}}{\frac{0.002}{5.999}}$$

$$4 = 8.994 \times 10^6 \$$$

? Exercise 5

Make a small change in the coefficient matrix of

$$\begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 7 \end{bmatrix}$$

and find

$$\frac{\|\Delta X\|_\infty / \|X\|_\infty}{\|\Delta A\|_\infty / \|A\|_\infty}.$$

Is it a large or a small number? Compare your results with the previous problem. How is this number related to the condition number of the coefficient matrix?

Answer

Changing $[A]$ to

$$\begin{bmatrix} 1.001 & 2.001 \\ 2.001 & 3.000 \end{bmatrix}$$

Results in solution of

$$\begin{bmatrix} 2.003 \\ 0.9970 \end{bmatrix}$$

$$\frac{\frac{\|\Delta X\|_\infty}{\|X\|_\infty}}{\frac{\|\Delta A\|_\infty}{\|A\|_\infty}} = \frac{\frac{0.003}{2.000}}{\frac{0.002}{5}}$$

$$= 3.75$$

? Exercise 6

Prove

$$\frac{\|\Delta X\|}{\|X\|} \|A\| \|A^{-1}\| \frac{\|\Delta C\|}{\|C\|}$$

Answer

Use theorem that if $[A][B] = [C]$ then $\|A\| \|B\| \geq \|C\|$

? Exercise 7

For

$$[A] = \begin{bmatrix} 10 & -7 & 0 \\ -3 & 2.099 & 6 \\ 5 & -1 & 5 \end{bmatrix}$$

gives

$$[A]^{-1} = \begin{bmatrix} -0.1099 & -0.2333 & 0.2799 \\ -0.2999 & -0.3332 & 0.3999 \\ 0.04995 & 0.1666 & 6.664 \times 10^{-5} \end{bmatrix}$$

- What is the condition number of $[A]$?
- How many significant digits can we at least trust in the solution of $[A] [X] = [C]$ if $\epsilon_{mach} = 0.1192 \times 10^{-6}$?
- Without calculating the inverse of the matrix $[A]$, can you estimate the condition number of $[A]$ using the theorem in problem#6?

Answer

$$\begin{aligned} \text{A. } \|A\| &= 17 \\ \|A^{-1}\| &= 1.033 \\ \text{Cond}(A) &= 17.56 \end{aligned}$$

B. 5

C. Try different values of right hand side of $C = [\pm 1 \ \pm 1 \ \pm 1]^T$ with signs chosen randomly. Then $\|A^{-1}\| \leq \|X\|$ obtained from solving equation set $[A] [X] = [C]$ as $\|C\| = 1$.

? Exercise 8

Prove that the $\text{Cond}(A) \geq 1$.

Answer

We know that

$$\|A B\| \leq \|A\| \|B\|$$

then if

$$\begin{aligned} [B] &= [A]^{-1}, \\ \|A A^{-1}\| &\leq \|A\| \|A^{-1}\| \\ \|I\| &\leq \|A\| \|A^{-1}\| \\ 1 &\leq \|A\| \|A^{-1}\| \\ \|A\| \|A^{-1}\| &\geq 1 \\ \text{Cond}(A) &\geq 1 \end{aligned}$$

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10: Eigenvalues and Eigenvectors

Learning Objectives

After reading this chapter, you should be able to:

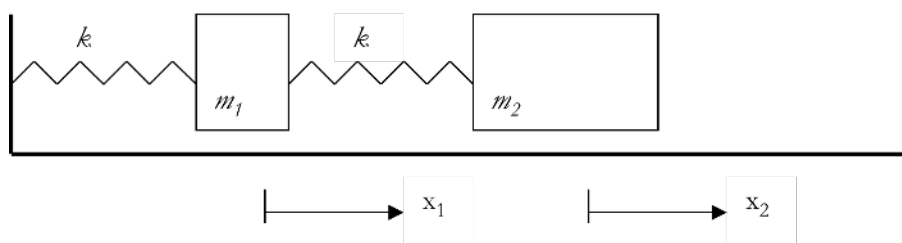
1. define eigenvalues and eigenvectors of a square matrix,
2. find eigenvalues and eigenvectors of a square matrix,
3. relate eigenvalues to the singularity of a square matrix, and
4. use the power method to numerically find the largest eigenvalue in magnitude of a square matrix and the corresponding eigenvector.

What does eigenvalue mean?

The word eigenvalue comes from the German word *Eigenwert* where Eigen means *characteristic* and Wert means *value*. However, what the word means is not on your mind! You want to know why I need to learn about eigenvalues and eigenvectors. Once I give you an example of an application of eigenvalues and eigenvectors, you will want to know how to find these eigenvalues and eigenvectors.

Can you give me a physical example application of eigenvalues and eigenvectors?

Look at the spring-mass system as shown in the picture below.



Assume each of the two mass-displacements to be denoted by x_1 and x_2 , and let us assume each spring has the same spring constant k . Then by applying Newton's 2nd and 3rd law of motion to develop a force-balance for each mass we have

$$m_1 \frac{d^2 x_1}{dt^2} = -kx_1 + k(x_2 - x_1)$$

$$m_2 \frac{d^2 x_2}{dt^2} = -k(x_2 - x_1)$$

Rewriting the equations, we have

$$m_1 \frac{d^2 x_1}{dt^2} - k(-2x_1 + x_2) = 0$$

$$m_2 \frac{d^2 x_2}{dt^2} - k(x_1 - x_2) = 0$$

Let $m_1 = 10$, $m_2 = 20$, $k = 15$

$$10 \frac{d^2 x_1}{dt^2} - 15(-2x_1 + x_2) = 0$$

$$20 \frac{d^2 x_2}{dt^2} - 15(x_1 - x_2) = 0$$

From vibration theory, the solutions can be of the form

$$x_i = A_i \sin(\omega t - \Theta)$$

where

- A_i = amplitude of the vibration of mass i ,
- ω = frequency of vibration,
- Θ = phase shift.

then

$$\frac{d^2 x_i}{dt^2} = -A_i \omega^2 \sin(\omega t - \Theta)$$

Substituting x_i and $\frac{d^2 x_i}{dt^2}$ in equations,

$$\begin{aligned} -10A_1\omega^2 - 15(-2A_1 + A_2) &= 0 \\ -20A_2\omega^2 - 15(A_1 - A_2) &= 0 \end{aligned}$$

gives

$$\begin{aligned} (-10\omega^2 + 30)A_1 - 15A_2 &= 0 \\ -15A_1 + (-20\omega^2 + 15)A_2 &= 0 \end{aligned}$$

or

$$\begin{aligned} (-\omega^2 + 3)A_1 - 1.5A_2 &= 0 \\ -0.75A_1 + (-\omega^2 + 0.75)A_2 &= 0 \end{aligned}$$

In matrix form, these equations can be rewritten as

$$\begin{bmatrix} -\omega^2 + 3 & -1.5 \\ -0.75 & -\omega^2 + 0.75 \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & -1.5 \\ -0.75 & 0.75 \end{bmatrix} \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} - \omega^2 \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Let $\omega^2 = \lambda$

$$\begin{aligned} [A] &= \begin{bmatrix} 3 & -1.5 \\ -0.75 & 0.75 \end{bmatrix} \\ [X] &= \begin{bmatrix} A_1 \\ A_2 \end{bmatrix} \\ [A] [X] - \lambda [X] &= 0 \\ [A] [X] &= \lambda [X] \end{aligned}$$

In the above equation, λ is the eigenvalue and $[X]$ is the eigenvector corresponding to λ . As you can see, if we know λ for the above example we can calculate the natural frequency of the vibration

$$\omega = \sqrt{\lambda}$$

Why are the natural frequencies of vibration important? Because you do not want to have a forcing force on the spring-mass system close to this frequency as it would make the amplitude A_i very large and make the system unstable.

What is the general definition of eigenvalues and eigenvectors of a square matrix?

If $[A]$ is a $n \times n$ matrix, then $[X] \neq \vec{0}$ is an eigenvector of $[A]$ if

$$[A] [X] = \lambda [X]$$

where λ is a scalar and $[X] \neq 0$. The scalar λ is called the eigenvalue of $[A]$ and $[X]$ is called the eigenvector corresponding to the eigenvalue λ .

How does find eigenvalues of a square matrix?

To find the eigenvalues of a $n \times n$ matrix $[A]$, we have

$$\begin{aligned}[A][X] &= \lambda[X] \\ [A][X] - \lambda[X] &= 0 \\ [A][X] - \lambda[I][X] &= 0 \\ ([A] - \lambda[I])[X] &= 0\end{aligned}$$

Now for the above set of equations to have a nonzero solution,

$$\det([A] - \lambda[I]) = 0$$

This left-hand side can be expanded to give a polynomial in λ and solving the above equation would give us values of the eigenvalues. The above equation is called the characteristic equation of $[A]$.

For a $[A]$ $n \times n$ matrix, the characteristic polynomial of A is of degree n as follows

$$\det([A] - \lambda[I]) = 0$$

giving

$$\lambda^n + c_1\lambda^{n-1} + c_2\lambda^{n-2} + \dots + c_n = 0$$

Hence, this polynomial has n roots.

✓ Example 1

Find the eigenvalues of the [physical problem discussed in the beginning of this chapter](#), that is, find the eigenvalues of the matrix

$$[A] = \begin{bmatrix} 3 & -1.5 \\ -0.75 & 0.75 \end{bmatrix}$$

Solution

$$\begin{aligned}[A] - \lambda[I] &= \begin{bmatrix} 3 - \lambda & -1.5 \\ -0.75 & 0.75 - \lambda \end{bmatrix} \\ \det([A] - \lambda[I]) &= (3 - \lambda)(0.75 - \lambda) - (-0.75)(-1.5) = 0 \\ 2.25 - 0.75\lambda - 3\lambda + \lambda^2 - 1.125 &= 0 \\ \lambda^2 - 3.75\lambda + 1.125 &= 0 \\ \lambda &= \frac{-(-3.75) \pm \sqrt{(-3.75)^2 - 4(1)(1.125)}}{2(1)} \\ &= \frac{3.75 \pm 3.092}{2} \\ &= 3.421, 0.3288\end{aligned}$$

So the eigenvalues are 3.421 and 0.3288.

✓ Example 2

Find the eigenvectors of

$$A = \begin{bmatrix} 3 & -1.5 \\ -0.75 & 0.75 \end{bmatrix}$$

Solution

The eigenvalues have already been found in Example 1 as

$$\lambda_1 = 3.421, \lambda_2 = 0.3288$$

Let

$$[X] = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

be the eigenvector corresponding to

$$\lambda_1 = 3.421$$

Hence

$$\begin{aligned} ([A] - \lambda_1[I])[X] &= 0 \\ \left\{ \begin{bmatrix} 3 & -1.5 \\ -0.75 & 0.75 \end{bmatrix} - 3.421 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right\} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= 0 \\ \begin{bmatrix} -0.421 & -1.5 \\ -0.75 & -2.671 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \end{aligned}$$

If

$$x_1 = s$$

then

$$\begin{aligned} -0.421s - 1.5x_2 &= 0 \\ x_2 &= -0.2808s \end{aligned}$$

The eigenvector corresponding to $\lambda_1 = 3.421$ then is

$$\begin{aligned} [X] &= \begin{bmatrix} s \\ -0.2808s \end{bmatrix} \\ &= s \begin{bmatrix} 1 \\ -0.2808 \end{bmatrix} \end{aligned}$$

The eigenvector corresponding to

$$\lambda_1 = 3.421$$

is

$$\begin{bmatrix} 1 \\ -0.2808 \end{bmatrix}$$

Similarly, the eigenvector corresponding to

$$\lambda_2 = 0.3288$$

is

$$\begin{bmatrix} 1 \\ 1.781 \end{bmatrix}$$

✓ Example 3

Find the eigenvalues and eigenvectors of

$$[A] = \begin{bmatrix} 1.5 & 0 & 1 \\ -0.5 & 0.5 & -0.5 \\ -0.5 & 0 & 0 \end{bmatrix}$$

Solution

The characteristic equation is given by

$$\det([A] - \lambda[I]) = 0$$

$$\det \begin{bmatrix} 1.5 - \lambda & 0 & 1 \\ -0.5 & 0.5 - \lambda & -0.5 \\ -0.5 & 0 & -\lambda \end{bmatrix} = 0$$

$$(1.5 - \lambda)[(0.5 - \lambda)(-\lambda) - (-0.5)(0)] + (1)[(-0.5)(0) - (-0.5)(0.5 - \lambda)] = 0$$

$$-\lambda^3 + 2\lambda^2 - 1.25\lambda + 0.25 = 0$$

To find the roots of the characteristic polynomial equation

$$-\lambda^3 + 2\lambda^2 - 1.25\lambda + 0.25 = 0$$

we find that the first root by observation is

$$\lambda = 1$$

as substitution of $\lambda = 1$ gives

$$(-1)^3 + 2(1)^2 - 1.25(1) + 0.25 = 0$$

$$0 = 0$$

So

$$(\lambda - 1)$$

is a factor of

$$-\lambda^3 + 2\lambda^2 - 1.25\lambda + 0.25$$

To find the other factors of the characteristic polynomial, we first conduct long division

$$\begin{array}{r} -\lambda^2 + \lambda + 0.25 \\ (\lambda - 1) \overline{) -\lambda^3 + 2\lambda^2 - 1.25\lambda + 0.25} \\ \underline{-\lambda^3 + \lambda^2} \\ \lambda^2 - 1.25\lambda + 0.25 \\ \underline{\lambda^2 - \lambda} \\ -0.25\lambda + 0.25 \\ \underline{-0.25\lambda + 0.25} \\ 0 \end{array}$$

Hence

$$-\lambda^3 + 2\lambda^2 - 1.25\lambda + 0.25 = (\lambda - 1)(-\lambda^2 + \lambda + 0.25)$$

To find zeroes of $-\lambda^2 + \lambda + 0.25$, we solve the quadratic equation,

$$-\lambda^2 + \lambda + 0.25 = 0$$

to give

$$\begin{aligned}\lambda &= \frac{-(1) \pm \sqrt{(1)^2 - (4)(-1)(0.25)}}{2(-1)} \\ &= \frac{-1 \pm \sqrt{0}}{-2} \\ &= 0.5, 0.5\end{aligned}$$

So $\lambda = 0.5$ and $\lambda = 0.5$ are the zeroes of

$$-\lambda^2 + \lambda + 0.5$$

giving

$$-\lambda^2 + \lambda + 0.25 = -(\lambda - 0.5)(\lambda - 0.5)$$

Hence

$$-\lambda^3 + 2\lambda^2 - 1.25\lambda + 0.25 = 0$$

can be rewritten as

$$-(\lambda - 1)(\lambda - 0.5)(\lambda - 0.5) = 0$$

to give the roots as

$$\lambda = 1, 0.5, 0.5$$

These are the three roots of the characteristic polynomial equation and hence the eigenvalues of matrix $[A]$.

Note that there are eigenvalues that are repeated. Since there are only two distinct eigenvalues, there are only two eigenspaces. But, corresponding to $\lambda = 0.5$ there should be two eigenvectors that form a basis for the eigenspace corresponding to $\lambda = 0.5$.

Given

$$[(A - \lambda I)] [X] = 0$$

then

$$\begin{bmatrix} 1.5 - \lambda & 0 & 1 \\ -0.5 & 0.5 - \lambda & -0.5 \\ -0.5 & 0 & -\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

For $\lambda = 0.5$,

$$\begin{bmatrix} 1 & 0 & 1 \\ -0.5 & 0 & -0.5 \\ -0.5 & 0 & -0.5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Solving this system gives

$$x_1 = -a, x_2 = b, x_3 = a$$

So

$$\begin{aligned}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} &= \begin{bmatrix} -a \\ b \\ a \end{bmatrix} \\ &= \begin{bmatrix} -a \\ 0 \\ a \end{bmatrix} + \begin{bmatrix} 0 \\ b \\ 0 \end{bmatrix} \\ &= a \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}\end{aligned}$$

So, the vectors $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ form a basis for the eigenspace for the eigenvalue $\lambda = 0.5$ and are the two eigenvectors corresponding to $\lambda = 0.5$.

For $\lambda = 1$,

$$\begin{bmatrix} 0.5 & 0 & 1 \\ -0.5 & -0.5 & -0.5 \\ -0.5 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Solving this system gives

$$x_1 = a, x_2 = -0.5a, x_3 = -0.5a$$

The eigenvector corresponding to $\lambda = 1$ is

$$\begin{bmatrix} a \\ -0.5a \\ -0.5a \end{bmatrix} = a \begin{bmatrix} 1 \\ -0.5 \\ -0.5 \end{bmatrix}$$

Hence the vector

$$\begin{bmatrix} 1 \\ -0.5 \\ -0.5 \end{bmatrix}$$

is a basis for the eigenspace for the eigenvalue of $\lambda = 1$, and is the eigenvector corresponding to $\lambda = 1$.

What are some of the theorems of eigenvalues and eigenvectors?

Theorem 1: If $[A]$ is a $n \times n$ triangular matrix – upper triangular, lower triangular or diagonal, the eigenvalues of $[A]$ are the diagonal entries of $[A]$.

Theorem 2: $\lambda = 0$ is an eigenvalue of $[A]$ if $[A]$ is a singular (noninvertible) matrix.

Theorem 3: $[A]$ and $[A]^T$ have the same eigenvalues.

Theorem 4: Eigenvalues of a symmetric matrix are real.

Theorem 5: Eigenvectors of a symmetric matrix are orthogonal, but only for distinct eigenvalues.

Theorem 6: $|\det(A)|$ is the product of the absolute values of the eigenvalues of $[A]$.

✓ Example 4

What are the eigenvalues of

$$[A] = \begin{bmatrix} 6 & 0 & 0 & 0 \\ 7 & 3 & 0 & 0 \\ 9 & 5 & 7.5 & 0 \\ 2 & 6 & 0 & -7.2 \end{bmatrix}$$

Solution

Since the matrix $[A]$ is a lower triangular matrix, the eigenvalues of $[A]$ are the diagonal elements of $[A]$. The eigenvalues are

$$\lambda_1 = 6, \lambda_2 = 3, \lambda_3 = 7.5, \lambda_4 = -7.2$$

✓ Example 5

One of the eigenvalues of

$$[A] = \begin{bmatrix} 5 & 6 & 2 \\ 3 & 5 & 9 \\ 2 & 1 & -7 \end{bmatrix}$$

is zero. Is $[A]$ invertible?

Solution

$\lambda = 0$ is an eigenvalue of $[A]$, that implies $[A]$ is singular and is not invertible.

✓ Example 6

Given the eigenvalues of

$$[A] = \begin{bmatrix} 2 & -3.5 & 6 \\ 3.5 & 5 & 2 \\ 8 & 1 & 8.5 \end{bmatrix}$$

are

$$\lambda_1 = -1.547, \lambda_2 = 12.33, \lambda_3 = 4.711$$

What are the eigenvalues of $[B]$ if

$$[B] = \begin{bmatrix} 2 & 3.5 & 8 \\ -3.5 & 5 & 1 \\ 6 & 2 & 8.5 \end{bmatrix}$$

Solution

Since $[B] = [A]^T$, the eigenvalues of $[A]$ and $[B]$ are the same. Hence eigenvalues of $[B]$ also are

$$\lambda_1 = -1.547, \lambda_2 = 12.33, \lambda_3 = 4.711$$

✓ Example 7

Given the eigenvalues of

$$[A] = \begin{bmatrix} 2 & -3.5 & 6 \\ 3.5 & 5 & 2 \\ 8 & 1 & 8.5 \end{bmatrix}$$

are

$$\lambda_1 = -1.547, \lambda_2 = 12.33, \lambda_3 = 4.711$$

Calculate the magnitude of the determinant of the matrix.

Solution

Since

$$\begin{aligned} |\det[A]| &= |\lambda_1| |\lambda_2| |\lambda_3| \\ &= |-1.547| |12.33| |4.711| \\ &= 89.88 \end{aligned}$$

How does one find eigenvalues and eigenvectors numerically?

One of the most common methods used for finding eigenvalues and eigenvectors is the power method. It is used to find the largest eigenvalue in an absolute sense. Note that if this largest eigenvalue is repeated, this method will not work. Also this eigenvalue needs to be distinct. The method is as follows:

- (1). Assume a guess $[X^{(0)}]$ for the eigenvector in

$$[A] [X] = \lambda [X]$$

equation. One of the entries of $[X^{(0)}]$ needs to be unity.

- (2). Find

$$[Y^{(1)}] = [A] [X^{(0)}]$$

- (3). Scale $[Y^{(1)}]$ so that the chosen unity component remains unity.

$$[Y^{(1)}] = \lambda^{(1)} [X^{(1)}]$$

- (4). Repeat steps (2) and (3) with

$$[X] = [X^{(1)}] \text{ to get } [X^{(2)}].$$

- (5). Repeat the steps 2 and 3 until the value of the eigenvalue converges.

If E_s is the pre-specified percentage relative error tolerance to which you would like the answer to converge to, keep iterating until

$$\left| \frac{\lambda^{(i+1)} - \lambda^{(i)}}{\lambda^{(i+1)}} \right| \times 100 \leq E_s$$

where the left-hand side of the above inequality is the definition of absolute percentage relative approximate error, denoted generally by E_s . A pre-specified percentage relative tolerance of $0.5 \times 10^{2-m}$ implies at least m significant digits are current in your answer. When the system converges, the value of λ is the largest (in absolute value) eigenvalue of $[A]$.

✓ Example 8

Using the power method, find the largest eigenvalue and the corresponding eigenvector of

$$[A] = \begin{bmatrix} 1.5 & 0 & 1 \\ -0.5 & 0.5 & -0.5 \\ -0.5 & 0 & 0 \end{bmatrix}$$

Solution

Assume

$$[X^{(0)}] = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{aligned} [A] [X^{(0)}] &= \begin{bmatrix} 1.5 & 0 & 1 \\ -0.5 & 0.5 & -0.5 \\ -0.5 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 2.5 \\ -0.5 \\ -0.5 \end{bmatrix} \end{aligned}$$

$$Y^{(1)} = 2.5 \begin{bmatrix} 1 \\ -0.2 \\ -0.2 \end{bmatrix}$$

$$\lambda^{(1)} = 2.5$$

We will choose the first element of $[X^{(0)}]$ to be unity.

$$\begin{aligned}
 [X^{(1)}] &= \begin{bmatrix} 1 \\ -0.2 \\ -0.2 \end{bmatrix} \\
 [A] [X^{(1)}] &= \begin{bmatrix} 1.5 & 0 & 1 \\ -0.5 & 0.5 & -0.5 \\ -0.5 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -0.2 \\ -0.2 \end{bmatrix} \\
 &= \begin{bmatrix} 1.3 \\ -0.5 \\ -0.5 \end{bmatrix} \\
 [X^{(2)}] &= 1.3 \begin{bmatrix} 1 \\ -0.3846 \\ -0.3846 \end{bmatrix} \\
 \lambda^{(2)} &= 1.3 \\
 [X^{(2)}] &= \begin{bmatrix} 1 \\ -0.3846 \\ -0.3846 \end{bmatrix}
 \end{aligned}$$

The absolute relative approximate error in the eigenvalues is

$$\begin{aligned}
 |\varepsilon_a| &= \left| \frac{\lambda^{(2)} - \lambda^{(1)}}{\lambda^{(2)}} \right| \times 100 \\
 &= \left| \frac{1.3 - 1.5}{1.5} \right| \times 100 \\
 &= 92.307\%
 \end{aligned}$$

Conducting further iterations, the values of $\lambda^{(i)}$ and the corresponding eigenvectors is given in the table below

i	$\lambda^{(i)}$	$[X^{(i)}]$	$ \varepsilon_a $ (%)
1	2.5	$\begin{bmatrix} 1 \\ -0.2 \\ -0.2 \end{bmatrix}$	—
2	1.3	$\begin{bmatrix} 1 \\ -0.38462 \\ -0.38462 \end{bmatrix}$	92.307
3	1.1154	$\begin{bmatrix} 1 \\ -0.44827 \\ -0.44827 \end{bmatrix}$	16.552
4	1.0517	$\begin{bmatrix} 1 \\ -0.47541 \\ -0.47541 \end{bmatrix}$	6.0529

i	$\lambda^{(i)}$	$[X^{(i)}]$	$ \varepsilon_a $ (%)
5	1.02459	$\begin{bmatrix} 1 \\ -0.48800 \\ -0.48800 \end{bmatrix}$	1.2441

The exact value of the eigenvalue is $\lambda = 1$
and the corresponding eigenvector is

$$[X] = \begin{bmatrix} 1 \\ -0.5 \\ -0.5 \end{bmatrix}$$

Eigenvalues and Eigenvectors Quiz

? Quiz 1

The eigenvalues of

$$\begin{bmatrix} 5 & 6 & 17 \\ 0 & -19 & 23 \\ 0 & 0 & 37 \end{bmatrix}$$

are

- (A) $-19, 5, 37$
- (B) $19, -5, -37$
- (C) $2, -3, 7$
- (D) $3, -5, 37$

? Quiz 2

If $\begin{bmatrix} -4.5 \\ -4 \\ 1 \end{bmatrix}$ is an eigenvector of $\begin{bmatrix} 8 & -4 & 2 \\ 4 & 0 & 2 \\ 0 & -2 & -4 \end{bmatrix}$, the eigenvalue corresponding to the eigenvector is

- (A) 1
- (B) 4
- (C) -4.5
- (D) 6

? Quiz 3

The eigenvalues of the following matrix

$$\begin{bmatrix} 3 & 2 & 9 \\ 7 & 5 & 13 \\ 6 & 17 & 19 \end{bmatrix}$$

are given by solving the cubic equation

- (A) $\lambda^3 - 27\lambda^2 + 167\lambda - 285$
- (B) $\lambda^3 - 27\lambda^2 - 122\lambda - 313$

- (C) $\lambda^3 + 27\lambda^2 + 167\lambda + 285$
 (D) $\lambda^3 + 23.23\lambda^2 - 158.3\lambda + 313$

? Quiz 4

The eigenvalues of a 4×4 matrix $[A]$ are given as 2, -3, 13, and 7. The $|\det(A)|$ then is

- (A) 546
 (B) 19
 (C) 25
 (D) cannot be determined

? Quiz 5

If one of the eigenvalues of $[A]_{n \times n}$ is zero, it implies

- (A) The solution to $[A][X] = [C]$ system of equations is unique
 (B) The determinant of $[A]$ is zero
 (C) The solution to $[A][X] = [0]$ system of equations is trivial
 (D) The determinant of $[A]$ is nonzero

? Quiz 6

Given that matrix $[A] = \begin{bmatrix} 8 & -4 & 2 \\ 4 & 0 & 2 \\ 0 & -2 & -3 \end{bmatrix}$ has an eigenvalue value of 4 with the corresponding Eigenvectors of

$[x] = \begin{bmatrix} -4.5 \\ -4 \\ 1 \end{bmatrix}$, then $[A]^5 [X]$ is

- (A) $\begin{bmatrix} -18 \\ -16 \\ 4 \end{bmatrix}$
 (B) $\begin{bmatrix} -4.5 \\ -4 \\ 1 \end{bmatrix}$
 (C) $\begin{bmatrix} -4608 \\ -4096 \\ 1024 \end{bmatrix}$
 (D) $\begin{bmatrix} -0.004395 \\ -0.003906 \\ 0.0009766 \end{bmatrix}$

Eigenvalues and Eigenvectors Exercise

? Exercise 1

The eigenvalues λ of matrix $[A]$ are found by solving the equation(s)?

- A. $[A][X] = [I]$
 B. $[A][X] - \lambda[I] = 0$

- C. $|A| = 0$
 D. $|A - \lambda I| = 0$

Answer

D

? Exercise 2

Find the eigenvalues and eigenvectors of

$$[A] = \begin{bmatrix} 10 & 9 \\ 2 & 3 \end{bmatrix}$$

using the determinant method

Answer

$$(12, 1), \begin{bmatrix} 0.9762 \\ 0.2169 \end{bmatrix}, \begin{bmatrix} 0.8381 \\ -0.8381 \end{bmatrix}$$

? Exercise 3

Find the eigenvalues and eigenvectors of

$$[A] = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 0 & 1 \\ 2 & 0 & 1 \end{bmatrix}$$

using the determinant method

Answer

$$(0, 4, 5615, 0.43845) \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0.87193 \\ -0.27496 \\ 0.48963 \end{bmatrix}, \begin{bmatrix} -0.27816 \\ 3.5284 \\ 0.99068 \end{bmatrix}$$

? Exercise 4

Find the eigenvalues of these matrices by inspection

- A. $\begin{bmatrix} 2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 6 \end{bmatrix}$
 B. $\begin{bmatrix} 3 & 5 & 7 \\ 0 & -2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$
 C. $\begin{bmatrix} 2 & 0 & 0 \\ 3 & 5 & 0 \\ 2 & 1 & 6 \end{bmatrix}$

Answer

- A. 2,-3,6
 B. 3,-2,0
 C. 2,5,6

? Exercise 5

Find the largest eigenvalue in magnitude and its corresponding vector by using the power method

$$[A] = \begin{bmatrix} 4 & 0 & 1 \\ -2 & 0 & 1 \\ 2 & 0 & 1 \end{bmatrix}$$

Start with an initial guess of the eigenvector as

$$\begin{bmatrix} 1 \\ -0.5 \\ 0.5 \end{bmatrix}$$

Answer

$$4.5615, \begin{bmatrix} 1 \\ -0.31534 \\ 0.56154 \end{bmatrix} \text{ after 4 iterations}$$

? Exercise 6

Prove if λ is an eigenvalue of $[A]$, then $\frac{1}{\lambda}$ is an eigenvalue of $[A]^{-1}$.

? Exercise 7

Prove that square matrices $[A]$ and $[A]^T$ have the same eigenvalues.

? Exercise 8

Show that $|\det(A)|$ is the product of the absolute values of the eigenvalues of $[A]$.

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